

Strengthening Fiscal Policy and Management through Public Expenditure Reviews: Reference Guide on Approaches and New Frontiers

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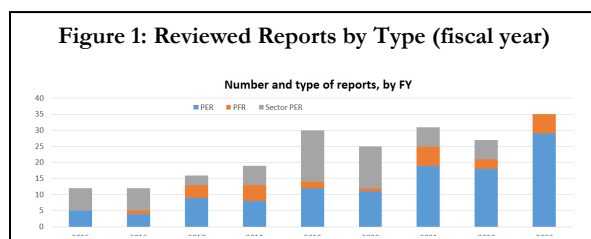
Acronyms

CCDR	Country Climate and Development Report
CEQ	Commitment to Equity
CIT	Corporate Income Tax
CPAT	Carbon Policy Assessment Tool
CPEIR	Climate Public Expenditure and Investment Review
DEA	Data Envelopment Analysis
DEMPA	Debt Management Performance Assessment
DPOs	Development Policy Operations
DRM	Domestic Revenue Mobilization
DSA	Debt Sustainability Analysis
HCI	Human Capital Index
HCP	Human Capital Project
HD	Human Development
HRM	Human Resource Management
IDA	International Development Association
IFMIS	Integrated Financial Management Information System
IMF	International Monetary Fund
INFRA SAP	Infrastructure Sector Assessment Program
IT	Information Technology
MAPS	Methodology for Assessing Procurement Systems
MFD	Maximizing Finance for Development
MRIO	Multi-Regional Input-Output
MTBF	Medium-Term Budget Framework
MTEF	Medium-Term Expenditure Framework
MTFF	Medium-Term Fiscal Framework
MTI	Macroeconomics, Trade and Investment Global Practice
OECD	Organization for Economic Co-operation and Development
OOP	Out of Pocket
O&M	Operations and Maintenance
PEA	Public Expenditure Analysis
PER	Public Expenditure Review
PEFA	Public Expenditure and Financial Accountability
PEIR	Public Expenditure and Institutional Review
PETS	Public Expenditure Tracking Surveys
PFM	Public Financial Management
PFR	Public Finance Review
PIM	Public Investment Management
PIMA	Public Investment Management Assessment
PIT	Personal Income Tax
PPP	Public-Private Partnership
PROST	Pension Reform Options Simulation Toolkit
SDGs	Sustainable Development Goals
SFA	Stochastic Frontier Analysis
SOEs	State-Owned Enterprises
SP	Social Protection Global Practice
TADAT	Tax Administration Diagnostic Assessment Tool
TCCRE	Typology of Climate Change Response Expenditure
TO	Thematically Oriented
TSA	Treasury Single Account
TTL	Task Team Leader
VAT	Value Added Tax
WBG	World Bank Group

Overview

Motivation

Public Expenditure Reviews (PERs) remain a central diagnostic instrument for the analysis of public expenditure and financial management systems in client countries and to inform engagements with client governments (including lending) as well as other stakeholders. PERs support policy dialogue around reforms covering the full public finance cycle and help mainstream related diagnostics in sectoral operations. They also help inform lending, such as Development Policy Operations (DPOs) and Program-for-Results financing. In FY21 and FY22, PERs (thematic/sectoral or fiscally centered) were undertaken in over 58 countries (Figure 1), a testament to the impact of PERs, even more salient now in helping clients navigate the economic fallout from the COVID-19 pandemic and accelerate green, resilient, and inclusive development. Flexibility and tailoring of policy options to country specific circumstances remains an attractive hallmark of PERs. It is worth noting that PERs are one of the World Bank Group's four core diagnostics (along with Poverty Assessments, Country Economic Memorandums, and Country Climate and Development Reports), and are to be undertaken with a periodicity of no more than 5 years in IDA countries.



Given their key role in underpinning policy dialogue and informing World Bank Group (WBG) operations (and the considerable resources expended preparing PERs), continual effort is warranted to improve the diagnostic instrument, while ensuring systematic support to task teams.¹ Despite the operational importance of PERs as a key pillar of wider public finance engagements, standard guidance or templates to inform the scope, content and structure of PERs are lacking, leading to great variation and missed opportunities for more effective cross fertilization, learning, innovation, and use of existing resources. While some flexibility and modularity are needed given the client driven nature of these diagnostics, it is also important that PERs build on best practices and as much available information as possible to boost consistency, quality, innovation, and impact. Task teams preparing PERs seek support tailored to country context and client demand, including broad guidance, analytical tools, data, and good practice examples.

The PER Programmatic Support Platform aims to help meet these needs, and this reference guide is a part of an integrated package of support.² The foundation of the effort is the stocktaking, with a comprehensive methodology and review of PERs from 2015 to 2021, building on the stocktake prepared in 2015. This reference guide is based on the findings, accompanied by a comprehensive repository, structured along several key dimensions to help inform PERs going forward and serve as a resource to task teams. The programmatic effort endeavors to continue regular reviews of PERs along with updates of the reference guide and repository, depending on the number of new PERs that are delivered and the need to capture emerging best practices and innovation. The Reference Guide is not meant to be prescriptive and should not be interpreted as an exhaustive list of areas that all PERs should cover. Rather, the Reference Guide is intended as a resource for PER teams with elements and approaches to be considered based on country context (including fragility, conflict and violence affected), fit (such as re-engagement) and feasibility (including data availability).

¹ Considering the 160 PERs undertaken between 2015 and 2021 that were reviewed as part of the stocktaking exercise, the cost / resources expended in preparation ranged from \$33,000 to \$1,096,000, with an average of \$276,000. Key drivers include the size of the country and whether the PER was undertaken as part of Reimbursable Advisory Services. The PERs reviewed were undertaken in 88 countries, with 17 countries having more than one FC PER undertaken in the review period, and 12 countries with multiple sector PERs.

² Additional pillars of the PER Programmatic Support Platform include: (a) providing timely access to granular fiscal data, (b) developing and disseminating analytical tools and targeted guidance notes in support of PER preparation and implementation, and (c) leveraging synergies across thematic / sectoral areas (such as climate change and the Human Capital Project) to facilitate enhanced coverage, cross-fertilization and inter-operability.

Methodological Framework

Extensive quantitative and qualitative indicators underpin the methodological framework of the stocktaking, complemented by task-team leader (TTL) feedback to gain a sense of impact.³ The framework retains some of the main features of the stock-take prepared in 2015 (facilitating comparisons for tracking progress over time as well as surfacing persistent issues), while significantly increasing the number of dimensions to over 200 indicators for tagging. The framework is structured around six inter-connected pillars:

1. **Macroeconomic context.** The macro-fiscal framework plays a crucial role in the budget cycle and drives the forecasts of revenue, expenditure, and financing. This pillar is concerned with assessing and reviewing the depth and breadth of macroeconomic analysis presented in PERs, highlighting best practices across various approaches that can help guide and enhance the analytical rigor of future endeavors.
2. **Fiscal policy (expenditure and revenue).** Broadly defined as government spending, revenue mobilization, and financing to achieve a range of economic and development outcomes, fiscal policy is the cornerstone of PERs. The methodological framework considers the extent to which PERs include indicators on key public expenditure goals of technical and allocative efficiency, as well as equity and effectiveness, while public revenues are generally analyzed with a view to increasing available public resources and ensuring appropriate incidence and improved efficiency. Financing is covered as a part of debt-sustainability analysis and related issues under the macro pillar.
3. **Quasi-fiscal activities/fiscal risks.** These are typically analyzed by the source of imbalances, which can stem from the banking system, the occurrence of unforeseen events (contingent liabilities and guarantees), financial performance from off-budget components of the public sector (such as state-owned enterprises), or the government's interactions with private sector agents (including public-private partnerships; PPPs). This pillar is designed to capture the extent to which PERs incorporate these dimensions in the overarching analysis along with emphasizing innovative approaches, either as a stand-alone chapter or within the broader macro-economic context.
4. **Budgetary institutions.** Integrated assessment of policy and institutions focuses on the quality of public finance management practices (such as budget planning, preparation and execution, accounting and reporting, and institutional accountability and transparency), as well as budget procedures and laws more broadly. For example, a discussion of Public Financial Management (PFM) challenges has been critical in helping understand institutional and process gaps hindering greater effectiveness and efficiency of fiscal policies, and more generally in identifying bottlenecks that undermine the quality of public spending.
5. **Thematic/sectoral areas.** While a large proportion of PERs are centered on key sectors, like education, health, infrastructure (INFRA SAPs), social protection, and water, areas of coverage are expanding, and will continue to grow in future. These include approaches to integrating climate change, human capital (as part of the Human Capital Project (HCP), gender equality, private capital mobilization (as part of Maximizing Financing for Development; MFD), political economy, and the Sustainable Development Goals (SDGs), among others. As only a small percentage of PERs currently cover these areas, monitoring progress and developing tools will be especially important to help task teams use PERs to engage with clients seeking to address related challenges. The salience of topics will evolve over time, and the team will endeavor to provide an updated list of potential topics and

³ While indicative data was collected as part of the stock-take, assessing impact more fully will require additional data to be collected and could be done separately at a later stage of the programmatic activity, potentially involving longer-term studies like those undertaken through the Development Impact Evaluation efforts of the World Bank. Though not specifically on PERs, the Impact Evaluation Group undertook a related evaluation:

<https://ieg.worldbankgroup.org/evaluations/addressing-country-level-fiscal-and-financial-sector-vulnerabilities>.

examples where related analyses have been undertaken, aiming to support task teams and country management in crafting the most relevant and impactful PER agenda and inform task team selection.

6. **Process and impact.** There is broad recognition that in many ways the process of PER implementation is as important as the content in supporting impactful client engagement. This pillar tracks key features, like the inclusiveness of the PER process, the extent to which PERs assess the impact of past efforts (reviewing previous recommendations and government follow-up as part of a wider engagement on public finance issues), and the documentation of timelines and projected social and fiscal impacts of proposed recommendations. In addition to regular tagging indicators, quantitative information is augmented with feedback from TTLs, as well as information on cost, duration and other key features.

Findings and Recommendations

Using the methodological framework outlined above, the stocktaking of PERs generated key insights that will help enhance the contribution of PERs towards strengthened fiscal policy and management in client countries. Over 160 reports spanning FY16-20 were reviewed, including both EFI and sector led PERs, Public Finance Reviews (PFR) and Public Expenditure and Institutional Reviews (PEIRs), along with a small set of other typologies.⁴ At the outset, it is important to note the wide range of PERs undertaken, with analyses that vary in breadth (such as thematic and institutional coverage) and depth (like flow of funds and impact). The framing of the PERs depends on the operational context, needs and circumstances of a given country, and the available budget allocated to the task, among other factors.

1. *Narrative*

Compelling PERs that underpin strong country engagements include an introduction / overview that provides a consistent narrative on key messages. Given the wealth of information and breadth of analysis in PERs, it is critical that a strong storyline be developed to contextualize the findings. Otherwise, PERs can come off as overly descriptive and lacking judgement on what it all means. More is needed to help the policy maker / audience navigate numerous priorities.

2. *Macroeconomic context*

The stocktaking revealed extensive variation in approaches adopted to undertake macroeconomic analysis. The coverage of macroeconomic dimensions might differ based on country policy contexts and performance, in many cases there are areas needing further attention to enable more depth and rigor in support of policy uptake and impact, including:

- a. **Expanding the coverage of budget credibility considerations,** which in most cases are addressed only at a basic level, limiting the policy impact for greater fiscal discipline.
- b. **Increasing the emphasis on issues of cyclicity, drivers of expenditure growth, and fiscal rigidity,** which are covered in a handful of PERs but can be more systematically integrated to generate insights into the capacity of policy makers to mobilize fiscal policy quickly and address macro imbalances.
- c. **Enhancing impact assessments of fiscal policy in terms of macro stability, fiscal space, fiscal adjustment, output gaps, growth and debt dynamics,** based on estimates of fiscal multipliers, customized debt sustainability analyses, econometrics, macroeconomic simulations/scenario analyses and other approaches. While many PERs include excellent analyses in these areas, the majority of PERs

⁴ These include just-in-time notes, self-standing policy notes, lighter versions of PERs, and other efforts to deliver analytical and advisory support of clients, which were reviewed but not included in the summary analysis presented in this reference guide.

could be strengthened, particularly via more systematic inclusion of fiscal risk from quasi-fiscal activities.

- d. **The impact of long-term structural issues on macroeconomic stability, such as demography and climate change, were rarely analyzed,** and there is scope for greater coverage of these megatrends as related tools are developed and disseminated.

3. Fiscal policy (expenditure and revenue)

Expenditure analysis is typically centered on effectiveness, efficiency, and equity, including assessing the incidence and composition of government expenditure, benchmarking countries vis-à-vis peers on spending levels and outcomes, determining the adequacy of public expenditure, and identifying opportunities for efficiency gains. Much variation is evident in cross-cutting thematic areas such as wage bill, adequacy, equity and efficiency analysis, with unequal and inconsistent coverage and levels of sophistication across PERs. More PERs may wish to consider in depth analyses on climate change, human capital, gender gaps, disaster risk management, operations and maintenance (O&M), inter-governmental transfers, and efficiency of public investment management and procurement systems.

Analysis of domestic revenue mobilization has been substantially strengthened since 2015, both in terms of coverage and the depth of analysis. Compared to the 2015 baseline, the review highlighted greater inclusion of domestic resource mobilization considerations in PERs and increased sophistication in analytical approaches. At the same time, there are a few areas where analyses could be further improved, including:

- a. **Increasing coverage of emerging new areas,** such as the digital economy, compliance and policy gaps, transfer pricing, and international taxation, among others.
- b. **Integrating core tax diagnostics,** particularly around the use of the Tax Administration Diagnostic Assessment Tool (TADAT) in discussions of tax administration challenges, and the Commitment to Equity (CEQ) methodology and new supplemental guidance for evaluating gender equality.
- c. **Expanding further the analysis of tax expenditures in PERs,** especially in complementing estimates of foregone revenue with broader assessments of fiscal management frameworks.
- d. **Deepening analytical sophistication of tax simulations and econometric approaches, as well as expanding the collection and use of granular administrative and survey data.** In the few PERs where these techniques were implemented, the analysis strengthened the understanding of tax effort and potential, tax expenditure, and the progressivity of tax systems, among other key issues.

4. Quasi-fiscal activities/fiscal risks

Fiscal risks from quasi-fiscal activities can be substantial and potentially destabilizing if left unchecked. Fiscal risks from quasi-fiscal activities are typically analyzed by the source of imbalances which can originate from the banking system, unforeseen events (contingent liabilities and guarantees), financial performance from off budget components of the public sector (such as state-owned enterprises; SOEs), or government interactions with private sector agents (through PPPs, for example). Fiscal risk analysis informs stakeholders of the potential budgetary impact of various risks, which may stem from SOEs, subnational governments, PPPs, the financial sector, debt/guarantees, quasi-fiscal activities/contingent liabilities, and natural disasters/pandemics. These issues have been addressed either in stand-alone chapters or within the broader macroeconomic context.

The stocktaking revealed both signs of improvement and some lingering challenges. Overall, the coverage of quasi-fiscal activities increased significantly vis-à-vis the 2015 baseline, with over 50 percent of reviewed reports discussing these issues explicitly. Within this group, the most prevalent topics included SOEs, arrears, tax expenditures, PPPs, guarantees and contingent liabilities. The depth of analyses differed

significantly, however, with only a small fraction of PERs presenting deep dives on any of the above thematic areas. Given the potential destabilizing effects associated with such off-budget balances, more PERs may wish to consider integration of fiscal risks emanating from quasi-fiscal activities within macroeconomic scenario analysis.

5. Budgetary institutions

Integration of budgetary institutions remains a challenge, even though coverage has increased overall since the last stocktaking in 2015. This is particularly evident in macroeconomic chapters that rarely analyze how weaknesses of budgetary systems and PFM help explain bottlenecks and impediments to greater fiscal discipline and efficiency. Despite wide availability, governance diagnostics such as Public Expenditure and Financial Accountability (PEFA) assessments remain under-utilized. More generally, stronger connections to expenditure analysis and more comprehensive and structured approaches to PFM could enhance the impact of PERs.

6. Thematic/sectoral areas

The treatment of thematic/sectoral areas is mixed, and increased coverage on emerging themes is worth considering. While the coverage of tax expenditures and sophistication of general tax analytics have expanded significantly, gaps remain:

- a. **Only 5 percent of reviewed reports have some coverage of climate considerations.** With the development and dissemination of new tools and analytical approaches, and the heightened importance of climate change on the international development agenda, treatment is expected to expand sharply.
- b. **Integrated HCP approaches are nascent in PERs** and are among the least represented areas in PERs.
- c. **Of the PERs reviewed, only three addressed gender equality extensively,** while eight had some coverage and five included a basic reference.
- d. **Discussion of MFD is also very limited,** with fewer than 5 percent of PERs addressing the topic.
- e. **Only about 10 percent of PERs include an in-depth analysis of political economy issues,** though they are central to making progress on policy and institutional reforms.
- f. **Coverage of SDGs was limited,** particularly among fiscally centered PERs.

7. Lessons learned and key considerations

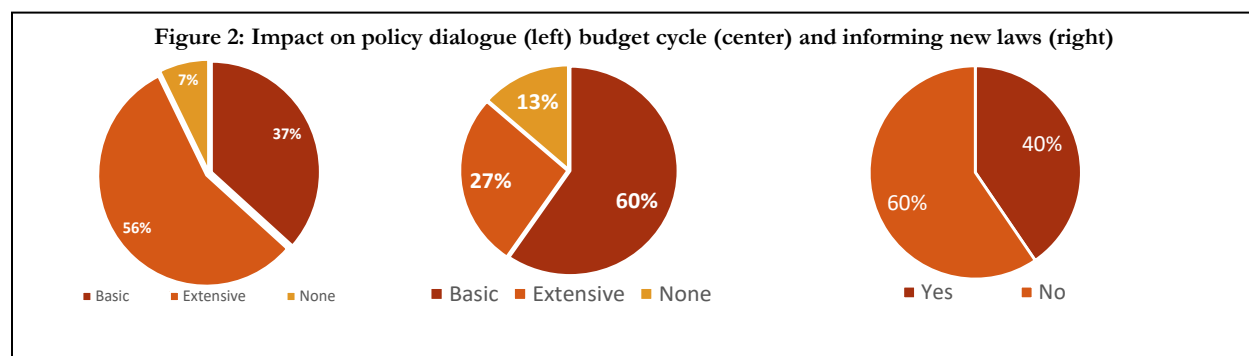
The stocktaking generated numerous lessons, highlighting opportunities for improving the consistency and depth of analysis in PERs, including scope for:

- a. Greater consistency in coverage of the macroeconomic context, highlighting key trends on growth, fiscal risks, debt, the impact of long-term structural issues (such as demography, climate change, and other megatrends), fiscal space considerations and scenario analysis.
- b. Sharper focus on cross-cutting thematic areas on the expenditure side, using the lens of efficiency, equity, and effectiveness.
- c. Better articulation of domestic resource mobilization issues, considering tax incidence and overall fiscal stance progressivity, tax expenditures, tax potential, and quality of tax administration, drawing more on existing diagnostics.

- d. Increased use of available diagnostic instruments to review the quality of budgetary institutions, noting the policy, regulatory and institutional framework, as well as budget preparation, execution, controls, audit, public investment management, and procurement systems.

Several additional considerations emerge from the stocktaking centered on the preparation process as part of a close overall engagement with the client, dissemination and training, and preparation of recommendations.

- a. **PERs prepared through an inclusive process that is closely integrated within a strong overall engagement with clients on public expenditure management are very impactful in underpinning policy dialogue and reforms.** How PERs are prepared is central to relevance and impact. This is especially the case during episodes of fiscal consolidation when PERs can help identify specific fiscal adjustment measures and estimate their potential gains, providing policymakers with policy options aimed at increasing the efficiency, effectiveness, equity and impact of public expenditures. PERs can support the development of proposals to minimize the negative impacts of expenditure adjustments in the face of limited or declining resource envelopes, enabling client countries to continue advancing their development objectives while maintaining fiscal discipline.
- b. **The operational relevance of PERs is enhanced by strong dissemination, training and integration of PER findings in lending operations.** Desk reviews and feedback from PER teams collected during the stocktaking exercise indicated a strong record of success in underpinning policy dialogue, particularly when PERs focus on the whole-of-government and are well integrated in broader operational contexts. More needs to be done to increase impact on the budget cycle and legislative initiatives. (Figure 2).



Source: World Bank

- **There is room to improve the process of PER preparation and implementation.** For example, facilitating public/stakeholder participation in discussions during the analysis stage can help enable testing of some hypotheses, raise public awareness of issues with objective data, and build public consensus around reforms. In addition, the way in which policy recommendations are presented is central to how they are received and their potential for impact. The majority of PERs do not present timelines, sequencing and estimated fiscal and social impacts of proposed recommendations. Only a handful of PERs refer to prior PERs, including assessing progress in implementing previous recommendations. Similarly, PERs undertaken by the Equitable Growth, Finance and Institutions Practice Group (EFI) and those completed by sectors often appear to be siloed products, and there is scope to leverage synergies across thematic / sectoral areas, including in sequencing and coordination.

Using this Reference Guide

The audience for this Reference Guide is mainly internal to the WBG, including task teams preparing PERs and country teams in their efforts to advance the public finance policy dialogue and deepen the provision of

integrated analytical and advisory services. The text below provides an extensive analysis and lessons learned across the five pillars noted above (Sections 2–6). For each area and sub-topic, the discussion notes: (a) key issues and analytical relevance, (b) detailed findings and best practices, (c) recommendations and key considerations, and (d) list of resources, such as key references in the literature, as well as best practice examples of PERs and data sources, in addition to the repository. As noted above, these are not intended as an exhaustive list relevant and needed in all contexts, but rather to be used separately, depending on specific interest, country context, and client demand. Section 7 presents lessons learned and key considerations for enhancing the consistency of PERs (in terms of scope and breadth of analysis), as well as key tables.

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1. Introduction

Public Expenditure Reviews (PERs) remain a critical analytical tool to engage on the quality of public expenditure and financial management systems with client countries. They serve as the main instrument guiding and influencing policy dialogue around reforms covering the full public finance cycle and help mainstream public finance diagnostics in sectoral operations. Over the past few years, formal expenditure reviews (whether thematic, sectoral or fiscally centered) have increased substantially, reaching almost 50 PERs led by the Equitable Growth, Finance, and Institutions Practice Group (EFI) in FY22. In addition, it is worth noting that PERs are one of EFI's three core diagnostics (along with Country Economic Memorandums and Poverty Assessments) that are to be undertaken with a periodicity of no more than 5 years in IDA countries.

PERs are instrumental in supporting fiscal consolidation efforts and strengthening policy dialogue with client counterparts. This is particularly true for those PERs that identify specific fiscal adjustment measures and estimate their potential gains, providing policymakers with options to enhance the efficiency, equity and impact of public expenditures in the context of fiscal reforms and consolidation efforts. PERs can also help minimize negative impacts of expenditure adjustments by identifying actions that maximize the efficiency and effectiveness of a limited or declining resource envelope that may enable clients to continue advancing their development objectives while maintaining fiscal discipline.

Despite their operational relevance, however, standardized guidance to inform the scope, content and structure of these core diagnostics has been limited or absent. This leads to missed opportunities for mutual learning, as well as fragmentation of analytical undertakings. PERs vary significantly in scope, quality, and methodology, with mixed results in incorporating innovations, reflecting lessons learned, and taking advantage of existing resources. While some flexibility and modularity are needed given the client driven nature of PERs, it is also important that they are built on best practices and up-to-date information to ensure high quality, consistency, and innovation for maximum relevance and development impact. In this context, the PER Programmatic Support Platform was established to support task teams and help strengthen and mainstream data, knowledge and innovation for greater quality, consistency and effectiveness of PERs.

A stocktaking of the universe of PERs implemented during FY16-21 offered important insights into both existing process and knowledge gaps, as well as surfacing best practice approaches. This Reference Guide provides a detailed analysis of the main findings from the stocktaking along with lessons learned and steps that could improve their design and implementation. Augmented by perspectives and inputs shared by PER task team leaders, the Reference Guide compiles best practices in terms of analysis, innovation, and frontier areas, such as climate change informed PERs and integration human capital considerations. It also suggests measures to improve the overall process of PER preparation and implementation through stronger synergies with other diagnostics, best practices on PER recommendations and deepened collaboration with sectors. The final scope and structure of any PER will be determined by country context and circumstances, reflecting the policy dialogue and reform needs. Hence, the reference guide is not intended to be prescriptive, but rather to offer a comprehensive framework and tools to support impactful PERs going forward.

The rest of the reference guide is structured as follows: Section 2, discusses emerging patterns and best practices related to the analysis of the macroeconomic context, highlighting key approaches and thematic areas. Section 3 presents findings on revenue and expenditure analysis. Section 4 discusses approaches to integrate analysis of quasi-fiscal activities, and Section 5 presents findings on the coverage of budgetary institutions in PERs. Section 6 provides links to some of the key resources at sector level and frontier areas, such as mainstreaming climate change in core diagnostics, amid a growing recognition about the central role of fiscal policy on the challenging climate change agenda. Finally, Section 7 notes lessons learned and key considerations towards a standardized core.

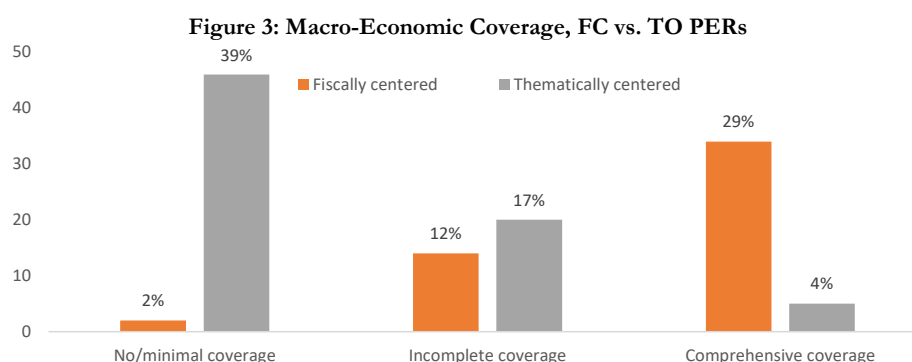
2. Macroeconomic Context

Analyzing the macroeconomic context is a critical pillar of robust PER diagnostics and central to developing and implementing sound policy interventions. This typically includes a medium-term fiscal framework with sectoral projections (real, external, fiscal, and monetary sectors), debt sustainability analyses, and references to fiscal rules or targets as applicable. It provides the context under which fiscal targets are set, policy choices are determined, and realistic revenue and expenditure projections are prepared. Hence, PERs need to be placed in a macro-fiscal framework to ensure both consistency and feasibility of findings and recommendations.

Well-crafted coverage of the macroeconomic context enables assessment of key challenges, including:

1. Identifying macroeconomic issues and determining whether they are the result of structural or temporary imbalances, as well as how they might best be addressed.
2. Assessing the scope for government interventions, and if so, what instrument (monetary, fiscal, or a combination of both) should be used.
3. Analyzing whether the level and composition of public expenditures and revenues are in line with the country's long-run financing needs and capacity.
4. Considering the drivers of public debt dynamics and the exposure to currency, rollover and interest rate risk, as well as whether debt levels are sustainable and pose manageable fiscal risks.
5. Assessing whether the redistributive structure of spending meets government priorities and objectives.
6. Reviewing vulnerability to external shocks and the capacity of fiscal policy to respond adequately, looking at issues of rigidity, cyclical, credibility and fiscal space, among others.
7. Identifying opportunities for efficiency gains to generate fiscal space or support fiscal consolidation.

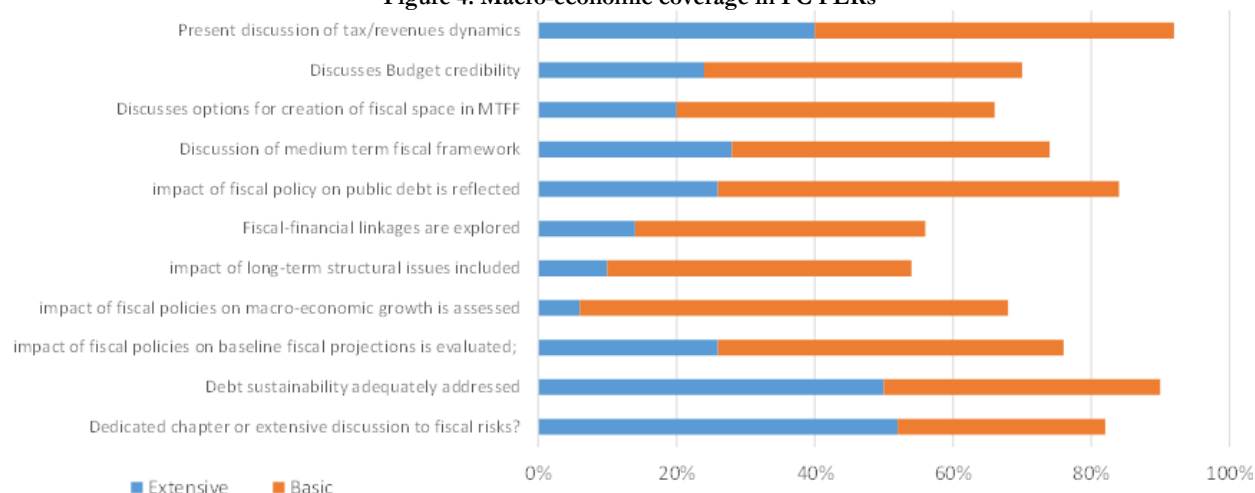
The PER stocktaking surfaced the range of methodological approaches adopted to undertake macro-fiscal analysis. The coverage and level of sophistication of macroeconomic analysis is uneven across the reports reviewed. Fiscally centered (FC) PERs typically include more in-depth coverage of macro-fiscal topics than thematically oriented (TO) PERs (Figure 3), but even among FC PERs there is substantial variation in the depth of analysis (Figure 4)⁵.



Source: 2020 Stocktaking; World Bank calculations.

⁵ To determine boundaries between comprehensive and incomplete coverage, the team undertook qualitative assessments for each of the individual macro dimensions to assess both their presence and sufficient depth of analysis, contingent on country context.

Figure 4: Macro-economic coverage in FC PERs



Source: 2020 Stocktaking, World Bank calculations.

To assess the breadth and depth of analysis in the discussion of the macroeconomic context, the stocktaking noted several key elements, including whether:

1. Current and projected aggregate levels of general government expenditures and revenues are consistent with the macroeconomic framework and any fiscal rules in place.
2. A scenario analysis is presented to inform options for generating of fiscal space, estimating needed fiscal consolidation, and gauging the impact of fiscal policy on public debt.
3. Public debt dynamics are adequately addressed given the country context.
4. Cyclical policy of fiscal policy is analyzed along with other dimensions, such as rigidity and impact on growth, to assess capacity to mobilize fiscal policy to address macroeconomic imbalances and respond to shocks.
5. Fiscal-financial linkages are explored, comparing fiscal and monetary measures to achieve policy aims.
6. Budget credibility is assessed.
7. Long-term structural issues, such as demographics and climate change, are analyzed and assessed.
8. Public financial management practices that shape fiscal discipline and macroeconomic stability are examined.

These elements are presented in more detail below, highlighting intended goals and methodological approaches. The omission of some of these should not necessarily be viewed as a weakness, to the extent that such analysis may not be relevant in a specific country context. The purpose of this discussion is rather to provide an overarching framework that can help inform future PERs by listing the most salient approaches and surfacing best practice examples on coverage of the macroeconomic context. The priority is to ensure that relevance and applicability to specific country contexts drives the decision about what elements to feature and the appropriate depth of analysis of each individual approach.

2.1 Profile, trend and benchmarking analysis of key macro-fiscal indicators

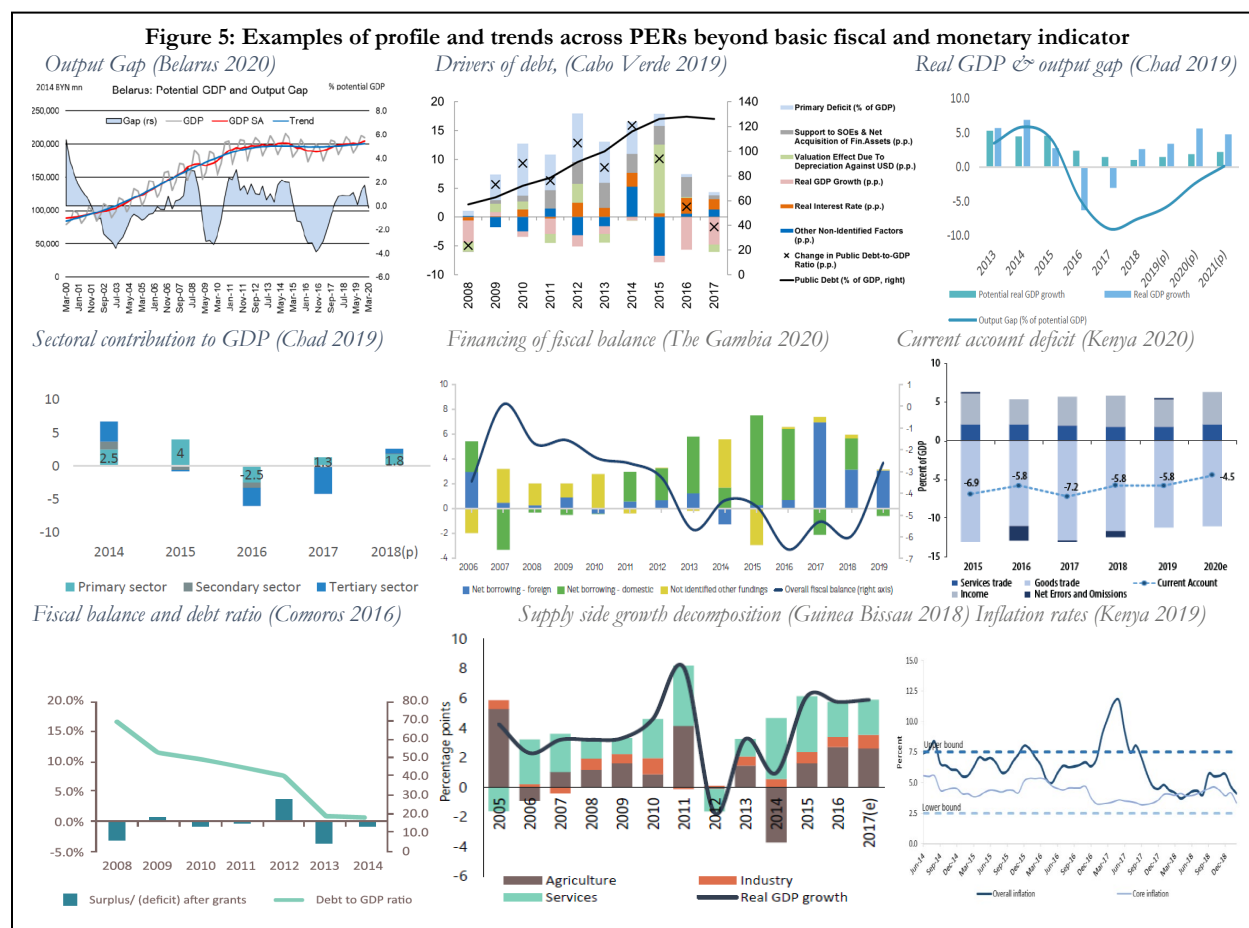
Almost every PER presents the macro-fiscal context, but at different levels of granularity and sophistication. At a minimum, any macro chapter should include basic profile and trends analysis for revenues, expenditures, and fiscal balances, ideally complemented by monetary indicators to explore fiscal-monetary linkages (Table 1 shows an example from the Montenegro 2019 PFR). Based on context, however, PERs might also seek to complement this by displaying trends of additional indicators, such as reserves, de-seasonalized trends in gross domestic product (GDP), output gap, projected financing needs, debt composition, cyclically adjusted balances and sectoral contribution to GDP, among others. Figure 5 provides some examples of macro trends presented across reviewed PERs.

Table 1: Main macro fiscal and monetary indicators, Montenegro (2019)

	2010	2011	2012	2013	2014	2015	2016	2017	2018*
Real sector, inflation and exchange rate									
GDP growth (%)	2.7	3.2	-2.7	3.5	1.8	3.4	2.9	4.7	4.9
Domestic demand (% growth)	-2.2	-2.3	-1.6	1.3	2.6	3.1	9.3	8.2	6.0
Unemployment rate (period average)	19.6	19.7	19.8	19.5	18.0	17.6	17.7	16.1	15.2
Inflation (period average)	0.5	3.1	4.1	2.2	-0.7	1.5	-0.3	2.4	2.6
Broad money (% growth)	3.5	2.3	8.4	4.4	9.1	11.3	9.5	13.7	1.9
Private sector credit (% growth)	-5.1	-9.3	-0.1	1.0	-0.1	1.5	1.0	9.7	8.3
Government finance (% of GDP)									
Revenues	42.2	39.5	40.9	42.3	44.6	41.5	42.5	41.4	42.4
Expenditures	46.7	44.6	46.2	47.6	47.7	48.8	45.3	47.0	46.2
Fiscal deficit	-4.6	-5.1	-5.3	-5.2	-3.1	-7.3	-2.8	-5.6	-3.8
Primary deficit	-3.6	-3.7	-3.4	-3.1	-0.8	-4.9	-0.7	-3.2	-1.8
Public debt and guarantees	50.7	57.2	65.4	65.9	67.1	73.7	71.4	70.0	75.7
External sector (% of GDP)									
Current account balance	-20.3	-14.8	-15.3	-11.4	-12.4	-11.0	-16.2	-16.1	-17.2
Net FDI	17.7	11.9	14.5	9.6	10.2	16.9	9.4	11.3	7.1
Gross external debt	142.6	145.4	156.4	156.3	162.3	171.0	162.5	160.5	168.3

Source: Montenegro (2019).

Many PERs benchmark with comparisons to global, regional, aspirational and/or structural peers. These analyses ranged from international comparisons of basic indicators (such as total revenue and expenditures levels, or decompositions such as the ratio of tax to GDP or capital spending), to more nuanced and sophisticated approaches on both tax (tax gaps, efficiency, value added tax (VAT) performance), expenditure (efficiency indicators, unit costs) and financing (debt stocks). Existing country scan tools offer an easy-to-use interface to identify meaningful income and behavior peers for such benchmarking exercises and there is room for increasing their use in PERs.



The discussion of revenue dynamics in the macroeconomic chapters typically include overall trends as share to GDP and its main components. Virtually all of the FC PERs presented this type of analysis in their macroeconomic chapters, often accompanied by a narrative on current reforms and outstanding gaps and challenges. These representations could be complemented – either in the macroeconomic section or in a dedicated chapter - by deep dives within sub-categories to assess trends and performance of key features, such as VAT performance (The Gambia 2020), personal income tax (PIT) progressivity (Liberia 2019, Mongolia 2018), international benchmarking (Cabo Verde 2019, Ukraine 2018), efficiency metrics (The Gambia 2020), policy and compliance gaps (Cabo Verde 2019), and inclusion of tax expenditures (Ethiopia 2016, Costa Rica 2018, Tanzania 2018, Ghana 2017, The Gambia 2020, Uzbekistan 2020, Lesotho 2019 and Georgia 2017, among others). The main features and analytical approaches on each of these thematic areas are discussed in more detail in Section 3.1.

Expenditure analysis commonly includes a description of basic trends by economic category, and administrative units, as well as function and programs of interest, as available. A more elaborate approach would seek to complement this basic discussion with a range of additional analysis that could help understand vulnerability to shocks and the ability of fiscal policy to respond, including analysis of fiscal rigidity, force and momentum analysis (Kenya 2020), execution rates (Comoros 2017), cyclical policy of expenditure (Costa Rica 2018), and fiscal multipliers (Timor-Leste 2019), as well as a more granular assessment of the composition of spending across sectors and economic categories to identify potential for efficiency gains. These approaches are described in detail below.

Discussion of main fiscal and monetary indicators can be complemented by an analysis on the role and effectiveness of fiscal rules in supporting enhanced fiscal discipline. Several PERs discuss fiscal rules in their macro chapter either in an attempt to provide options for their introduction or in assessing their effectiveness. Kazakhstan (2018) conducts an analysis of alternative fiscal rules, such as establishing a non-oil deficit target, and weighing between the nonoil fiscal balance (defined as the difference between public expenditure and nonoil fiscal revenue) and structural fiscal balance (based on budget formulation using smoothed oil prices over time). The Costa Rica (2018) runs scenario analyses to document how implementing the new fiscal rule fully is critical to put public debt on a downward, sustainable path. North Macedonia (2015) considers alternative fiscal consolidation measures and a proposed fiscal rule through a constitutional amendment establishing a binding ceiling on public debt at 60 percent of GDP and on the fiscal deficit of 3 percent of GDP, which could play a significant role in strengthening the path of public finances. Mexico (2016) highlights the importance of limiting the use of the escape clause to the Federal level balanced budget rule, as well as maintaining fiscal rules to ensure fiscal discipline at the sub-national level. Finally, Romania (2016) notes that while recommendations from the PER of 2010 regarding the establishment of fiscal rules and an independent fiscal council were adopted, challenges remain in implementing strong fiscal rules, which have been circumvented in the past. Taken together, these examples show how PERs have focused on fiscal rules as a core part of the analysis and development of policy options.

2.2 Medium-term fiscal framework and debt sustainability

Discussion of the medium-term fiscal framework (MTFF) is foundational in PERs. An MTFF provides the context under which fiscal targets are set, policy choices and fiscal adjustments are determined, and realistic revenue and expenditure projections are prepared. A sound MTFF assesses the sustainability of existing fiscal projections and notes deviations from the current baseline as a result of changes in fiscal policy stance⁶ or materialization of negative external shocks, and how these elements affect debt sustainability.

At a minimum, PERs should include projections of debt dynamics, highlight potential risks for long-term debt distress, and note mitigation measures. A good practice approach on debt dynamics is to use

⁶ A change in fiscal policy stance can have important ramifications in terms of fiscal space, growth and debt dynamics. For instance, higher spending needs – when not sufficiently countered by an equivalent increase in revenue collection - may require more borrowing and hence more public debt. A change in fiscal policy will also directly affect economic activity and indirectly impact the economy through consumption (taxes, expectations), investment (interest rates, confidence), and net exports (demand for imported goods, effects on exchange rate).

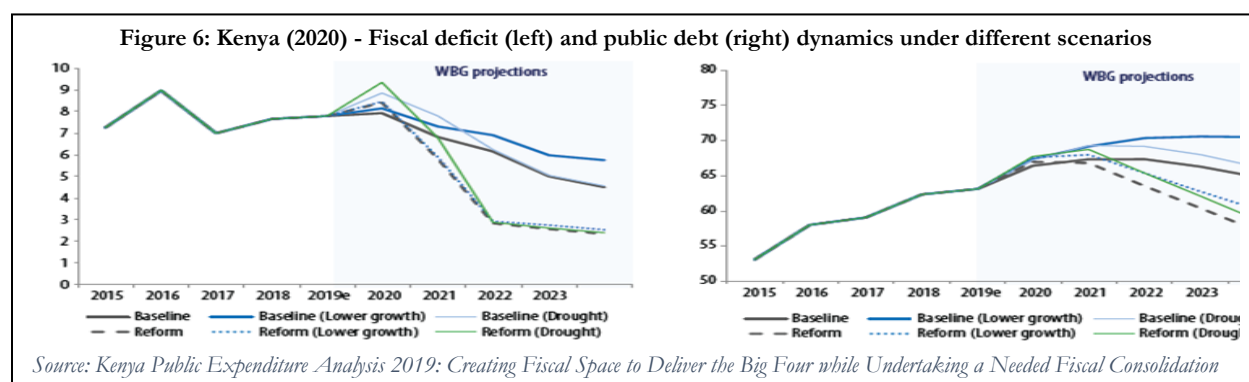
the established debt sustainability analysis (DSA) framework.⁷ A DSA helps understand and identify vulnerabilities in the debt structure or policy frameworks via a series of sensitivity tests applied to the baseline scenario, helping guide borrowing decisions and identify needed policy reforms to ensure long-term debt sustainability. Over 40 DSAs are typically conducted each year, providing a timely and widely available resource to underpin PERs.⁸

In order to undertake this analysis, PERs may:

1. Assess the impact of current (“stay the course scenario”) or proposed (“reform scenario”) fiscal policies on baseline fiscal projections and on debt dynamics through DSAs and related customized approaches.
2. Consider a range of potential external shocks and broader fiscal risks to assess overall vulnerability and challenges to macro framework stability.
3. Review long-term structural issues, such as demographics and climate change, and their expected effects on the macroeconomic framework and stability (discussed in more detail below in Section 2.3).
4. Identify/estimate debt thresholds and likelihood of debt distress.
5. Provide international benchmarking.
6. Complement the elements above with analysis of debt management practices, borrowing strategies, transparency, reporting, and domestic debt market development.

Several PERs stood out in implementing macroeconomic scenario analysis, identifying reform options, and estimating the size of fiscal adjustment needed in order to bring debt levels under control.

For instance, Kenya (2020) presented a compelling scenario analysis to assess the scope to generate fiscal space and restore public debt sustainability after the economic fallout of the COVID-19 pandemic, showing how continuing current policies would adversely affect Kenya’s fiscal and debt sustainability, documenting threshold breaches under various macroeconomic shocks, and identifying fiscal consolidation measures that could restore public debt sustainability and create fiscal space to finance COVID-19 recovery (Figure 6 and Table 2). Similar analysis was undertaken in Dominican Republic (2018), noting the size of the fiscal adjustment needed to stabilize the debt stock while acknowledging vulnerability to climate induced risks, contingent liabilities and sensitivity to growth rate assumptions). In Chile (2017), scenario analyses identified efficiency measures to support the needed macro-fiscal adjustment process. In Cambodia (2018), impact assessments of reform scenarios were prepared as part of fiscal projections, and Ukraine (2017) estimated the fiscal impact of additional reforms relative to the baseline.⁹



⁷ It should be emphasized that the level of depth of analysis of DSA should always be weighed against specific country context. For countries with low and sustainable debt levels, providing brief references to the size of public debt would be sufficient.

⁸ A list of DSA is available at <https://www.worldbank.org/en/programs/debt-toolkit/dsa> with links to additional resources.

⁹ Appendix C presents the full list of PERs for a range of thematic areas.

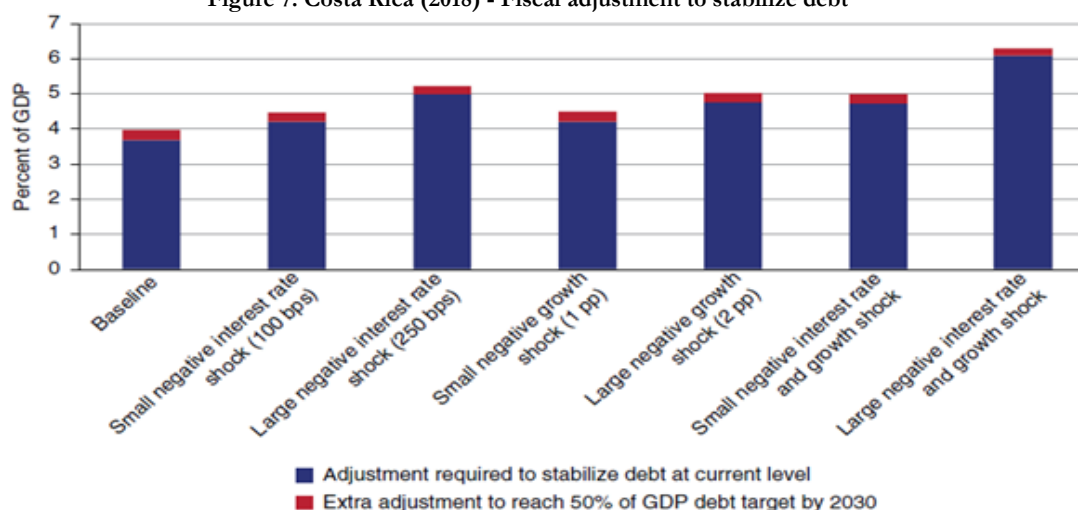
Table 2: Kenya (2020) - Scenario analysis

	In percent of GDP, unless otherwise stated									
	Historical					Projections				
	FY2015	FY2016	FY2017	FY2018	FY2019	FY2020	FY2021	FY2022	FY2023	FY2024
GDP at current prices (KSh billion)	6,654	7,594	8,529	9,316	10,240	11,287	12,550	13,982	15,562	17,321
GDP at constant prices, annual growth rate (%)	5.7	5.9	5.6	5.9	3.5	3.3	5.8	5.9	5.8	5.8
GDP deflator, annual growth rate (%)	10.0	8.1	5.0	3.7	6.2	6.7	5.1	5.2	5.2	5.2
Exchange Rate (KSh/USD)	102.4	102.9	102.5	101.5	104.3	108.7	111.3	114.2	117.2	120.2
Real Exchange Rate, Index 2015=100 (%)	100.0	94.0	90.7	88.6	87.7	87.5	87.1	86.6	86.2	85.7
Revenue	19.1	19.1	18.2	18.5	18.9	17.6	17.8	17.9	18.2	18.2
Direct Taxes (Tax on Income & Profits)	8.4	8.1	7.5	7.4	7.2	6.7	6.9	7.1	7.2	7.2
Indirect Taxes (Taxes on Goods and Int'l Trade)	7.7	7.7	7.3	7.7	6.9	7.0	7.3	7.4	7.6	7.6
Grants (2)	0.4	0.4	0.3	0.2	0.3	0.4	0.4	0.3	0.3	0.3
Non-Tax Revenue (3)	2.6	2.9	3.1	3.2	4.4	3.5	3.2	3.1	3.1	3.1
Expenditure	26.9	27.5	25.2	26.2	27.2	25.5	24.6	24.1	23.2	22.7
Wages and Salaries	4.6	4.4	4.6	4.5	4.7	4.5	4.2	3.9	3.7	3.7
Goods and Services (4)	8.0	7.3	7.5	7.9	8.4	8.0	7.3	6.9	6.6	6.6
Transfers to Counties	4.1	4.0	3.8	3.9	3.3	3.9	3.6	3.4	3.2	3.2
Other Expenses	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Interest Payments	2.9	3.5	3.8	4.0	4.3	4.1	4.3	4.4	4.4	4.3
Development Expenditure (5)	7.3	8.4	5.5	5.8	6.6	5.0	5.2	5.5	5.3	4.9
Primary Balance	-4.2	-5.1	-3.6	-3.6	-4.0	-3.8	-2.5	-1.8	-0.6	-0.2
Overall Balance	-7.1	-8.6	-7.4	-7.7	-8.3	-7.9	-6.8	-6.2	-5.0	-4.5
Gross Borrowing Requirements	19.9	21.5	21.2	21.6	19.2	19.4	18.7	16.7	15.1	15.0
Overall Balance (+ indicates deficit)	7.1	8.6	7.4	7.7	8.3	7.9	6.8	6.2	5.0	4.5
Amortizations Payments	11.4	11.9	13.3	14.1	13.1	11.4	11.9	10.5	10.1	10.5
Other Funding Needs	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Net Acquisition of Fin Assets (e.g., Deposits) (6)	1.5	1.0	0.5	0.2	-2.2	0.0	0.0	0.0	0.0	0.0
Borrowing Sources	19.9	21.5	21.2	21.6	19.2	19.4	18.7	16.7	15.1	15.0
Issuance of Domestic Debt (% share) (7)				80.0	80.0	80.0	80.0	80.0	80.0	80.0
Issuance of External Debt (% share)				20.0	20.0	20.0	20.0	20.0	20.0	20.0
Public Debt	53.1	58.0	59.1	62.4	63.1	66.4	67.3	67.3	66.2	64.8
Domestic Debt				31.6	30.4	33.2	34.5	35.2	34.9	35.3
External Debt				30.7	32.8	33.2	32.8	32.1	31.3	29.4

Source: Kenya 2020 PER.

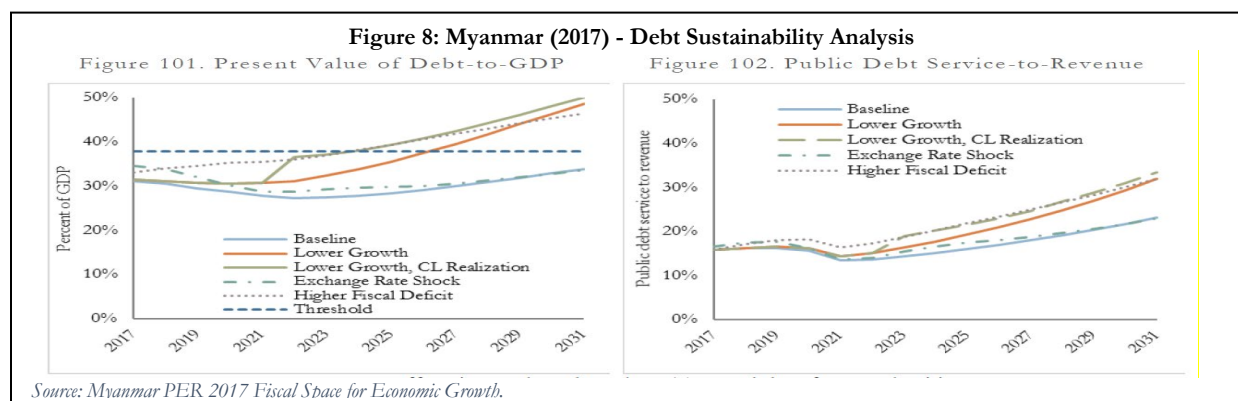
Another strong example of scenario analysis was implemented in the Costa Rica PFR (2018). The report used scenario analysis to estimate the size of the fiscal adjustment required to stabilize public debt and come back towards a target of 50 percent of GDP by 2030 (Figure 7), based on a commonly used methodology accounting for differing parameters and sensitivity to underlying assumptions. The study concluded that the recently introduced fiscal policy package – comprising both revenue and expenditure measures – would be enough to bring debt levels back to the desired threshold.

Figure 7: Costa Rica (2018) - Fiscal adjustment to stabilize debt



Source: Costa Rica Public Finance Review 2018.

Several other PERs applied scenario analyses with DSAs or through customized approaches to consider paths to debt sustainability. Myanmar (2017) presented a customized DSA to track the robustness and sensitivity of the baseline scenario to potential external shocks and reform scenarios (Figure 8). The report concludes that while the current risk of debt distress is low, external vulnerabilities (such as exchange rate depreciation, commodity dependence, and vulnerability to disasters), along with high sensitivity to lower growth scenarios, could pose heightened risk for fiscal slippages and long-term debt sustainability challenges. These can be mitigated through the adoption of proactive measures to enhance fiscal space. The study also looked at global trends in debt sustainability to place Myanmar's experience within a broader international context.



In one of the most comprehensive analyses, Cabo Verde (2019) discussed historical drivers of public debt and the impact of recent fiscal consolidation measures introduced to contain fiscal imbalances. Motivated by the rapid build-up of public debt fueled by persistent fiscal imbalances, scenario analysis was used to formulate medium-term fiscal trajectories under current policies compared with reform scenarios along with stochastic simulations to estimate effects of macroeconomic risks using a Vector Auto-Regressive model. Projections of debt levels according to different scenarios were overlapped with identification of debt thresholds (reflecting prudent, safe levels of public debt) using probit modeling to estimate a 20 percent probability of debt distress at current levels (Table 3). The analyses showed that further fiscal consolidation efforts were needed, in conjunction with structural reforms, particularly on tax side.

Table 3: Cabo Verde (2019) - Debt threshold and debt distress probability

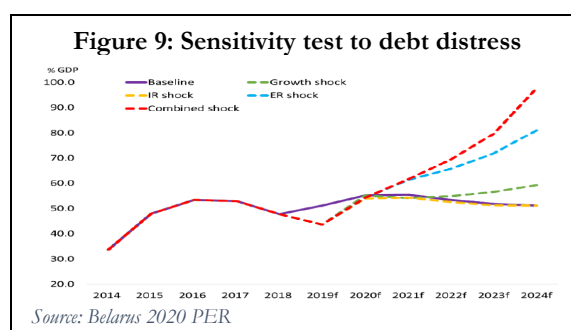
Explanatory variables	Coef.	Estimated value	Signif.	Probability of debt distress (%)	Thresholds of public debt-to-GDP ratio (%)
Public debt-to-GDP indicator (%)	β_1	1.0080	***	10	77
MIC dummy * Public debt-to-GDP indicator (%)	β_2	0.0214		15	101
CPIA score (1 to 6)	β_3	-0.5260	***	20	120
Real GDP growth (%)	β_4	-6.1500	***	25	136
Constant	β_0	0.3360		30	151
No. of observ.		597		Assuming: CPIA score of 4 Real GDP growth of 5 percent Cape Verde MIC status	
Pseudo R-squared		0.138			
Sign. *** p<0.01, ** p<0.05, and * p<0.10					

Source: IMF-IDA (2012)

Source: World Bank estimates based on IMF-IDA (2012)

Source: Cabo Verde 2019 PER: "Revisiting the Efficiency of Public Spending to Reduce Debt and Improve Education and Health Outcomes."

Analysis of the effects of reform policies and vulnerability of public debt dynamics to fiscal risks was well presented in several other PERs. Sierra Leone (2021) included a customized scenario analysis built upon a recently implemented DSA integrating COVID-19 related fiscal risks. Belarus (2021) conducted sensitivity analysis to macroeconomic shocks, such as swings in the exchange rate (Figure 9). Malawi (2018) ran fiscal and debt sustainability projections under baseline, shocks and strong reform scenarios, while Sri Lanka (2019) implemented sensitivity analysis against exchange rate depreciation and tax reform scenarios. Other examples include Bangladesh (2015) (updating the DSA to account for a worst-case scenario), The Gambia (2020) (presenting deviations from the baseline due to positive reforms or negative shocks based on the realization of adverse SOE contingent liabilities and policy reversals), Comoros (2016) (showing alternative DSA scenarios surfacing ongoing risk of debt distress), and Egypt (2019) (disentangling contributions of main debt drivers).



Finally, debt sustainability analysis can be complemented by additional considerations around debt management practices which can help further strengthen the overall narrative. Key approaches include assessments of the debt management framework, medium-term debt strategies, and domestic debt market development. One of the most comprehensive discussions was presented in Sri Lanka (2019), which linked the success of bringing debt to a sustainable path by strengthening the medium-term debt strategy in terms of reducing fragmentation of debt management functions, making structural improvements in domestic security markets, establishing a single database with real time information on government debt, and creating an autonomous debt management office. Other examples included Myanmar (2017), which assesses the impact of various borrowing scenarios on cost/risk tradeoffs of the debt portfolio and options to strengthen the domestic debt market. Kazakhstan (2018) looks at options to limit SOE debt exposure. Laos (2018) and Mexico (2016) discuss the findings of a debt management performance assessment (DEMPA), while Tanzania (2017) highlights a DEMPA report to formulate options for improving debt management performance.

2.3 Impact of long-term structural issues on fiscal stability

Discussion of the macroeconomic and fiscal impact of long-term structural issues, such as demographics and climate change, is generally limited. While several PERs consider the fiscal sustainability of pension obligations, few analyze the effect of changing demographics on health systems and financing. More concerning is the virtual absence of risk assessments to estimate vulnerability to climate change-induced shocks.

Pension issues have typically been covered in countries where related pressures are salient given population dynamics. The Pension Reform Options Simulation Toolkit (PROST) has been the instrument of choice for simulations. One example is the Belarus PER (2021), which discusses pension challenges stemming from demographic changes and presents scenario analysis based on PROST. The Angola social protection PER (2018) also uses PROST to develop projections of pension related fiscal expenditures and benefit levels, showing the long-term unsustainability under the current rules and parameters. The Argentina PER (2020) offers an in-depth examination through PROST of the adequacy, sustainability, coverage, and equity of pensions, surfacing inequitable distribution across provinces and sustainability challenges, compounded by fragmented institutional arrangements that hamper parametric adjustments. Other examples include Montenegro (2019), documenting the trade-off between sustainability and adequacy; Kosovo (2015), noting how pressures to grant pension increases have pushed projected pension spending onto a higher albeit sustainable trajectory given assumptions about growth rates and demographics; Costa Rica (2018), highlighting weaknesses of contributory system in ensuring the sustainability of pension obligations; Moldova (2017) documenting the low replacement rate, as well as the inequity and unsustainability pension system; and Egypt (2019), preparing financial projections of various scenarios (adjusting the base wage, early retirement provisions, retirement age, indexation method and survivor benefit policy) to identify reform options while modernizing the pension system and ensuring equity and sustainability.

The impact of demographics on health systems and related fiscal pressures is noted in a few PERs. The analysis typically concentrates on fiscal pressures stemming from the need to extend social insurance coverage, meet the rising cost of procuring drugs, and pay elevated preventive care expenses. Sri Lanka (2019) documents the challenge for the national health system of transitioning into a more complex and costly health care system needed to deal with a growing burden of non-communicable diseases, a rapidly aging and urbanizing population, and rising public expectations. St. Maarten (2020) presented extensive parametric estimation of financial sustainability of health spending due to the aging population.

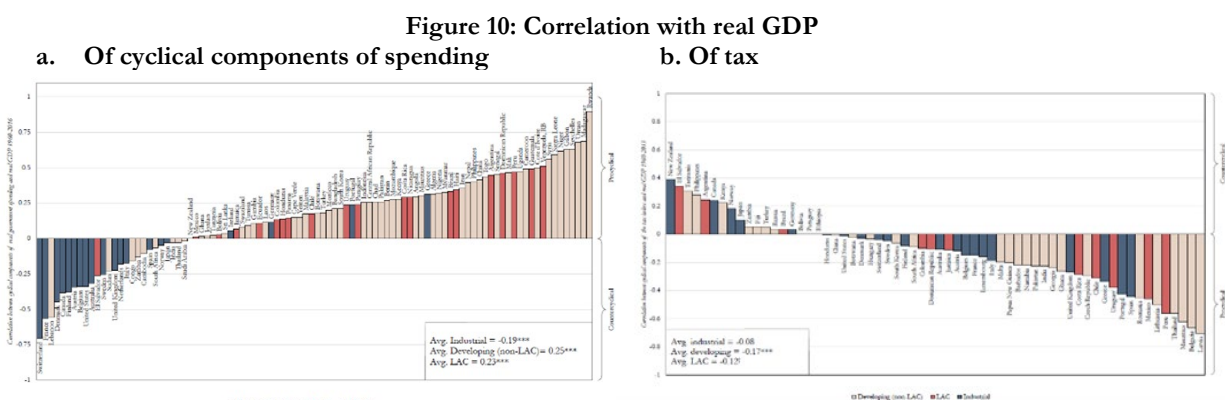
As discussed in more detail below (Section 6.2), very few PERs address climate change issues. Fiscal risk assessments of vulnerability to macroeconomic shocks related to climate change are included in only a handful of FC PERs, with only limited analysis. Dominican Republic (2019) recognizes the need to generate fiscal space to counter vulnerability to natural disasters (hurricanes) that have wreaked damages equivalent to an average of 0.7 percent of GDP. Climate change considerations and their impact on macroeconomic stability is featured in Malawi (2019), running scenario analysis to show debt vulnerability to a series of fiscal risks including climate change induced natural disasters, while St. Maarten (2020) documents the need for a disaster

risk management strategy. Other PERs include Mali Social Protection PER (2018), Morocco (2020), Indonesia (2017), Vietnam (2017), and Uganda (2020). The development of new tools, like the Carbon Policy Assessment Tool (CPAT) provides new options to analyze systematically these key policy issues in PERs.

2.4 Cyclicality and multiplier effects of fiscal policy

Estimating the cyclicality of the fiscal position is an important step in assessing vulnerability and/or resilience to negative shocks. Procyclical spending amplifies the business cycle, with expansionary fiscal policy during economic upswings, and cuts during economic contractions, as well as shifts in the composition of spending due to inertia in current spending, while capital expenditures typically bear the brunt of adjustment with negative long-term growth repercussions.

Several regional studies have looked at the issue of fiscal policy cyclicalities. The most common technique is to estimate cyclical components of real spending (or revenues) and GDP using the Hodrick-Prescott filter.¹⁰ As an example, Vegh et al (2017), analyze the main forces that shape fiscal policy in developing countries (with focus on Latin America and Caribbean countries), and conclude that in most cases, the fiscal stance has been procyclical, reflecting both limited access to finance during downturns (surfacing a link between fiscal procyclicality and debt sustainability) and political economic forces pushing for a rise in expenditures during upturns (Figure 10). A similar approach was deployed in Abdih et al (2010), looking at the cyclical properties of fiscal policy in the Middle East and North Africa region during the past decades and their implications in terms of fiscal space and ability to respond to downturns.¹¹ The study considered different response levels, showing that procyclicality is larger during good times (greater rise in expenditure) than bad times. This approach has been used as foundation for several PERs, such as Saudi Arabia (2019) and Belarus (2020).



Source: Carlos Vegh, Daniel Lederman, and Federico R. Bennett, "Leaning against the Wind: Fiscal Policy in Latin America and the Caribbean in a Historical Perspective" (LAC Semiannual Report, World Bank, Washington, DC, April 2017).

Table 4: Correlation between real spending and GDP, goods vs. bad times

	Correlation	
Middle East, North Africa and Central Asia	good times	0.82**
	bad times	0.28
MENA Oil exporters	good times	0.51*
	bad times	0.27
MENA Oil importers	good times	0.14
	bad times	-0.05
Caucasus and Central Asia	good times	0.97**
	bad times	0.23

Source: Yasser Abdih, Pablo Lopez-Murphy, Agustin Roitman, and Ratna Sahay, "The Cyclicalities of Fiscal Policy in the Middle East and Central Asia: Is Current Crisis Different?" (IMF working paper, International Monetary Fund, Washington, DC, 2010).

¹⁰ The Hodrick-Prescott filter refers to a data-smoothing technique applied to remove short-term fluctuations associated with the business cycle to detect longer-term cycles. An alternative technique is the use of the Kalman filter.

¹¹ Yasser Abdih, Pablo Lopez-Murphy, Agustin Roitman, and Ratna Sahay The cyclicalities of Fiscal policy in the Middle East and Central Asia: Is current crisis different? IMF working paper, 2010

Several PERs include robust assessments of the cyclicity of fiscal policy. Building on the methodological approach in Vegh et al, Costa Rica (2018) presents an in-depth analysis and cross-country comparison of cyclicity of different components of government expenditures, demonstrating that while Costa Rica has shifted towards a counter-cyclical fiscal stance on average, an investigation of its components reveals a more fragmented picture of procyclical investments and counter-cyclical wage policy (Figure 11). Saudi Arabia (2019) follows the analytical approach in Abdi et al (2010), providing international benchmarking (Figure 12) and confirming the highly procyclical nature of fiscal policy. The analysis underscores the need to shift to a rules-based fiscal framework to further support macro-stability and de-link spending from oil price volatility, though it recognizes that even fiscal rules can take a long time to establish and are generally not recommended for countries where off-budget fiscal operations are large.

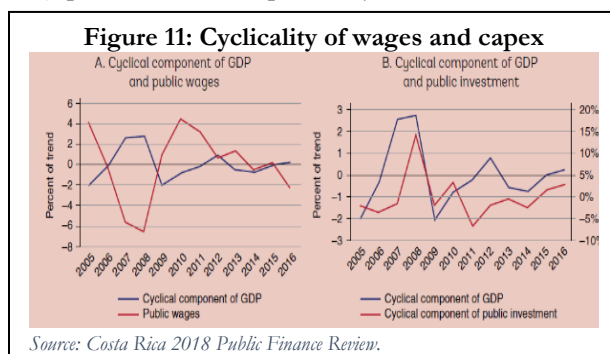
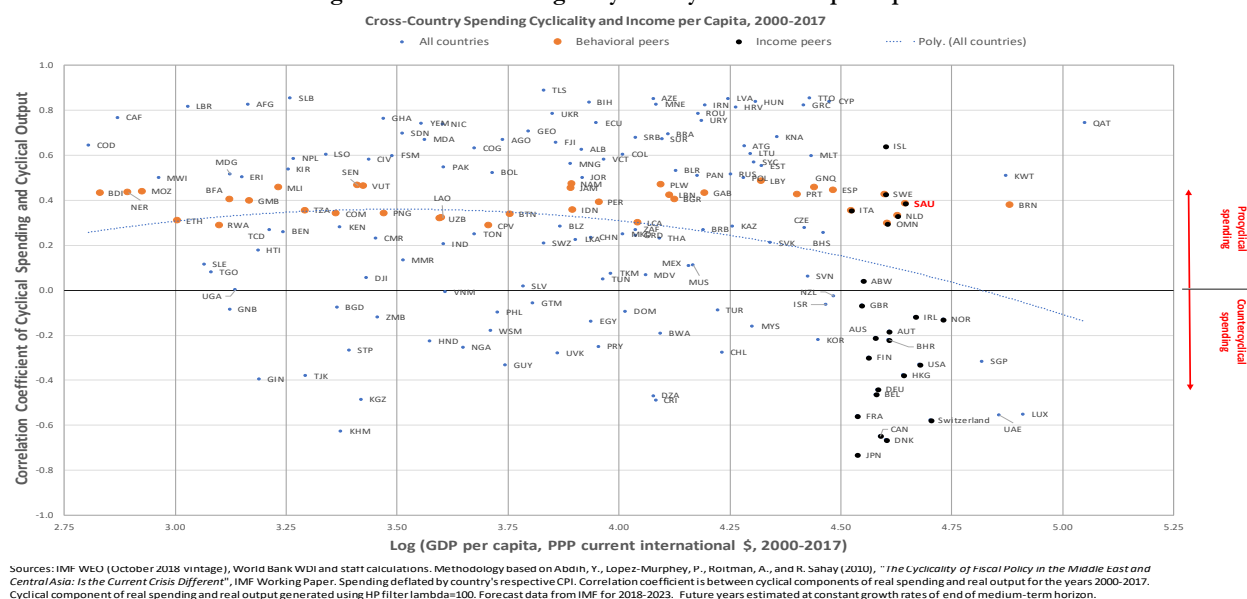


Figure 12: Benchmarking of cyclicity and income per capita



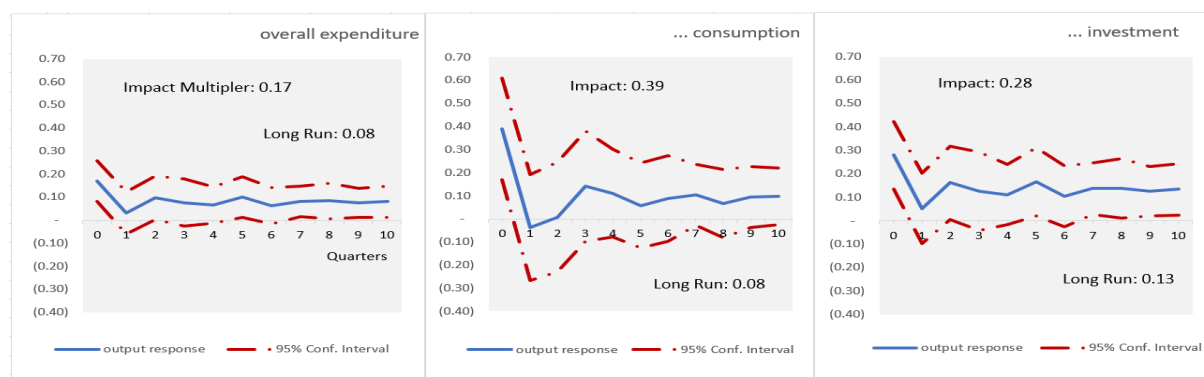
Several other PERs present similar analyses in their macro-chapters, including estimation and benchmarking of cyclicity of fiscal policy, adjustment and output multipliers. Argentina (2020) examines spending procyclicality at the provincial level using a standard error correction model to highlight scope for potential mitigation through risk sharing that could reduce revenue buoyancy. Belarus (2020) documents the procyclical nature of capital outlays compared with more rigid recurrent expenditures. Kenya (2020) documents the procyclicality of the fiscal stance by relating structural fiscal balances with the output gap. Chad (2019) emphasizes high procyclicality of spending due to the connection of revenues to oil prices, a common trend among most oil rich countries. Dominican Republic (2019) notes a moderate shift in procyclicality in recent years. Other examples include Gabon (2018), The Gambia (2020), Ukraine (2017), Mexico (2015), Uganda (2020), Romania (2018), and Namibia (2016).

Estimating fiscal multipliers (the change in GDP induced by a unit change in public spending) is key to assessing the impact of fiscal policies on output.¹² Timor Leste (2019), Cabo Verde (2019) estimate

¹² A multiplier larger than 1 suggests that public expenditure stimulates GDP by more than the initial increase in spending. A multiplier lower than 1 indicates that public spending is counteracted by effects that reduce its effectiveness – such as a “crowding out” of private sector activity and/or import “leakage”.

fiscal multipliers to assess the link between spending and growth.¹³ The impulse responses of the output variable to the fiscal policy shocks are estimated using a Structural Vector Autoregressive model. As expected, output responds positively to total and capital expenditure shocks at both the point of impact and over longer time horizons, with gains tapering off quickly (Figure 13). However, the long run multiplier for capital expenditure was low, raising concerns over the effectiveness of capital spending. Similar analysis was implemented in Costa Rica (2018).

Figure 13: Timor-Leste (2019) Impulse response outputs

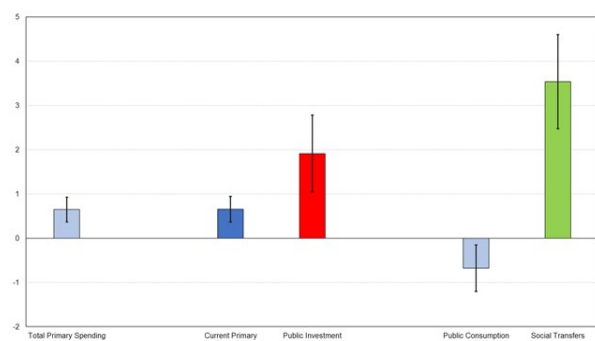


Source: Timor-Leste, "Public Expenditure Analysis: A Joint Ministry of Finance and World Bank Analysis on Public Spending," 2008–2016.

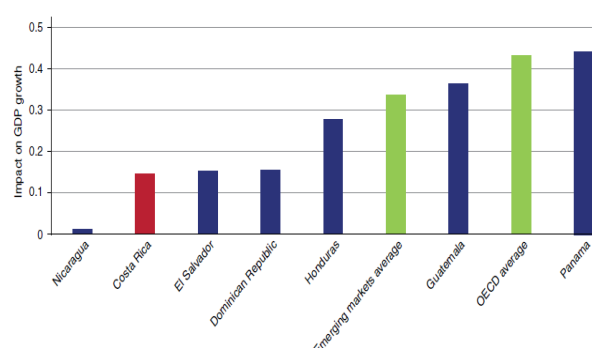
Several PERs discuss output multipliers by drawing on existing studies. Dominican Republic (2018) and Costa Rica (2018) reference Vegh et al (2018) and make slight modifications based on country specific context (Figure 14).¹⁴ Estevao and Samake (2013)¹⁵ and Ilzetzki et al (2013)¹⁶ provide further regional and global analyses of output multipliers that are referenced in PERs, such as Costa Rica (2018), in documenting different impacts on GDP growth from expenditure vs. revenue consolidation efforts.

Figure 14: Output multiplier analysis in PERs in Latin America and the Caribbean

Dominican Republic (2019) - Output multipliers by category



Costa Rica (2019) – Output loss due to spending adjustments



Source: "Vegh, Carlos; Lederman, Daniel; Bennett, Federico R.. 2017. *Leaning Against the Wind: Fiscal Policy in Latin America and the Caribbean in a Historical Perspective*. LAC Semiannual Report; April 2017. World Bank, Washington, DC.

2.5 Fiscal-financial linkages

A change in fiscal policy can impact inflation, foreign exchange reserves, interest rates and exchange rates. Achieving policy objectives requires coordinating fiscal policies with monetary, exchange rate, and structural policies. Another key policy consideration is the current account deficit, as persistent fiscal deficits

¹³ Others however – like Costa Rica 2019 PFR – presented discussion of fiscal multipliers – drawn from existing literature – to document the estimated impact of fiscal consolidation on GDP growth.

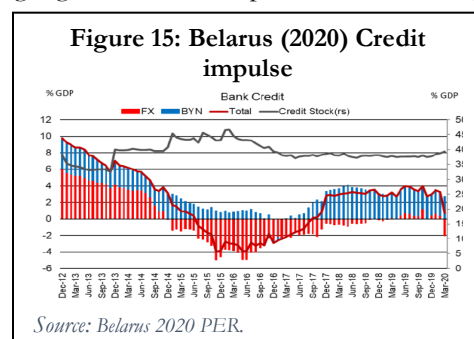
¹⁴ Vegh, Carlos; Lederman, Daniel; Bennett, Federico R.. 2017. *Leaning Against the Wind: Fiscal Policy in Latin America and the Caribbean in a Historical Perspective*. LAC Semiannual Report; April 2017. World Bank, Washington, DC.

¹⁵ Estevao, M and Samake, I. 2013. "The Economic Effects of Fiscal Consolidation with Debt Feedback", IMF WP 13/136, DC.

¹⁶ Ilzetzki, E, Mendoza E.G., Vegh, C.A. 2013. "How Big (Small?) Are Fiscal Multipliers?" JME, Vol. 60, pp. 239-54.

often spill into the external accounts, and fiscal tightening and exchange rate measures may be needed to narrow the current account deficit. Similarly, higher budget deficits lead to more borrowing, with financial sector implications, depending on the source of financing. For instance, financing from the central bank can increase inflationary pressure, while borrowing from the domestic market can crowd out private sector investment by bidding up interest rates. Foreign borrowing can lead to rising debt and balance of payment issues. Finally, as witnessed during the 2008-9 global financial crisis, episodes of excessive lending can lead to crises, which can trigger expansionary spending policy responses, weakening debt sustainability.

These transmission channels, and their implications for growth, fiscal sustainability, and financial stability, need to be considered. Belarus (2021) (Figure 15) highlights how fiscal pressures caused by expansionary fiscal policy (driven mainly by the wage bill), inflexible exchange rates, and limited scope for interest rate policy (given high dollarization) led to procyclical monetary policy impulses. In underscoring the urgency of fiscal adjustment, Dominican Republic (2018) documents the effect of growing debt issued by the central bank in a context of relatively undeveloped financial markets in driving up interest rates and crowding out private sector investment. The Gambia (2020) relates chronic imbalances due to low revenue mobilization, rigid expenditure, and pressures from SOEs to crowding out of private investment (among lowest in the region) as government domestic borrowing pushed up interest rates. Montenegro (2019) describes the boom-bust cycle of large capital inflows and unsustainable credit expansions, which led to non-performing loans in the wake of the financial crisis and efforts to support growth with expansionary fiscal policy and resultant elevated debt.



Other PERs include similar analyses. Kazakhstan (2017) discusses the effect of expansionary fiscal policy on exchange rate appreciation, while Chad (2019) documents the effect of oil price shocks and procyclical fiscal policy on international reserves, leading to credit crunches as the central bank tightens monetary policy to ensure they remain adequate. Kenya (2020) highlights the linkages between widening fiscal deficits, increased borrowing, and the imposition of interest rate caps, and a subsequent slowdown in overall credit growth. Lesotho (2018) emphasizes that the lack of a Treasury Single Account (TSA) complicates monetary policy by preventing timely monitoring of available reserves for short-term cash management.

2.6 Budget credibility

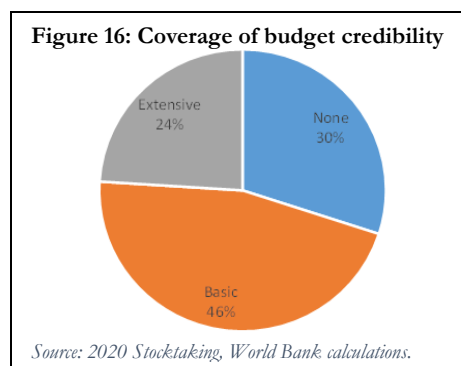
The deviation of outturns from the original (and potentially revised) budgets has important implications for fiscal discipline. Budget credibility analysis enables an assessment of the level of legitimacy and reliability of the budget process by documenting the extent to which original allocations are adequately executed and/or whether systematic re-allocation of resources occurs during implementation.

Despite the analytical usefulness and the straightforward nature of the analysis, coverage in PERs is limited. As shown in Figure 16, less than 25 percent of all FC PERs presented a disaggregated analysis of budget credibility. The granularity of data platforms like BOOST – available in the majority of developing countries – make it possible to routinely monitor the budget credibility by systematically assessing the extent of budget deviations – execution versus original budget - across agencies, functions and economic classification, including at highly disaggregated levels (such as procurement items, wage allowances, and the like). In addition,

despite the broad availability of PEFA reports, their findings and overall narratives have rarely been integrated into PERs as part of broader budget credibility analysis.¹⁷

Credibility analyses are at their best when:

1. Drilling down across economic categories to identify which items are most likely to be cut during implementation, highlighting the treatment of capital spending as a residual.
2. Looking across ministries and functions to detect any systematic pattern of re-allocation within ministries/functions (indicating power relations at play during execution) as opposed to common patterns (cuts or increases) across the board, more likely to be the result of commitment controls or other PFM deficiencies.
3. Considering line-item data to perform deep dives in particular areas, for instance procurement indicators or capital investments.
4. Connecting revenue and expenditure to explore linkages between poor revenue forecasting and expenditure re-prioritization (cuts across the board vs. targeted reduction in few sectors).
5. Including budget deviation analysis with in-depth assessments of underlying PFM weaknesses undermining overall budget credibility (references to PEFA can help strengthen the narrative).



Several PERs present a fully disaggregated budget credibility analysis. Chad (2019) replicated the PEFA methodology by comparing data from the budget laws to those reported by the treasury, along with a benchmarking exercise for revenues and expenditures vis-a-vis relevant peers. This exercise was enhanced further by referring to PEFA identified weaknesses in PFM systems that reduce credibility. Tanzania (2018) looks at under-execution of revenues, expenditures and borrowing plans and their relation to PFM deficiencies and accumulation of arrears. Drawing from budget and execution reports, Comoros (2017) (Figure 17) examines both revenue and expenditure under-execution, concluding that revenue execution has been largely determined by out-turns of non-fiscal revenues. On the expenditure side the report notes that while on average under-execution has been relatively modest, there are significant differences with larger under-execution rates in specific islands. The PER also documented extensive manifestations of re-allocation within ministries and within economic categories, undermining the credibility of the budget process. Malawi (2019) dissected budget credibility by economic and functional classification to demonstrate consistent patterns of over and underperformance driven by poor revenue forecasting, leading to the “budget being reformulated as it is released.” Tanzania (2017) documents under-execution in maintenance spending and its implications in terms of weaker value for money of underlying investments.

Figure 17: Comoros (2017) - Budget deviation



Source: Comoros (2017) Public expenditure and fiscal management review.

¹⁷ PEFA contains a full pillar on budget credibility, assessing the magnitude of change at aggregate and sectoral level for both expenditures and revenues.

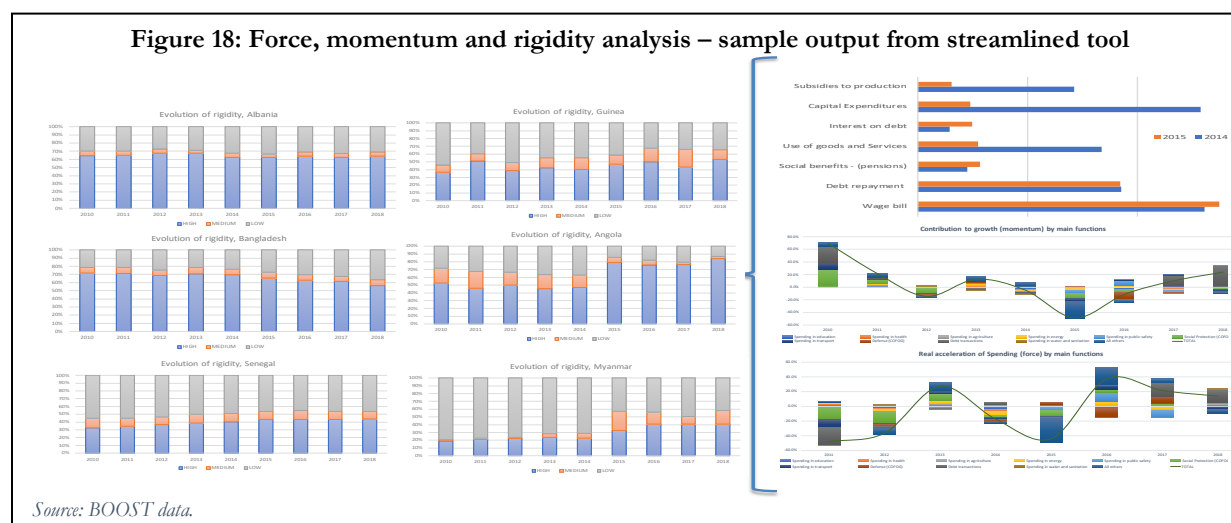
2.7 Fiscal risks from quasi-fiscal activities and off-budget balances

Fiscal risks are broadly defined as the possibility of short- to medium-term deviations in fiscal variables as compared to what was anticipated in the MTF or annual budget (IMF, 2008). Fiscal risks from quasi-fiscal activities are typically analyzed by the source of imbalances, which can originate from the banking system, debt and debt guarantees, unforeseen events (contingent liabilities), off-budget components of the public sector (such as SOEs), subnational governments, interactions with private sector agents (such as PPPs), and natural disasters/pandemics. These issues can be discussed either in a stand-alone chapter or within the broader macroeconomic context.

The treatment of fiscal risks has improved, but challenges remain. Overall, the coverage of quasi-fiscal activities increased significantly compared with the 2015 stocktaking, with about 50 percent of PERs now discussing these issues explicitly. Within this group, the most prevalent topics including SOEs, arrears, tax expenditures, PPPs, guarantees, and contingent liabilities. The depth of analysis, however, varies significantly, with only a small fraction of PERs presenting deep dives on any of these key issues. Furthermore, integration of fiscal risks from quasi-fiscal activities within macro scenario analysis was limited, despite the potential destabilizing effects linked to off budget imbalances. The issue of fiscal risk from quasi fiscal activities is covered in more detail in Section 4 below.

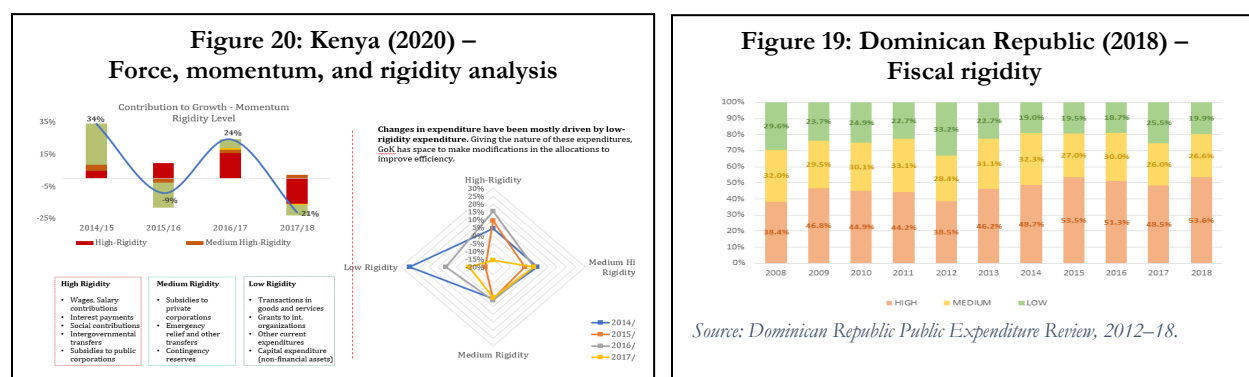
2.8 Fiscal rigidity and drivers of expenditure growth

Force and momentum analysis of public spending is a relatively new approach to help identify drivers of expenditure growth. Measuring force and momentum across the entire budget allows practitioners to monitor and decompose the micro drivers of public spending pressure, recognize rapidly expanding spending lines, and identify priorities for reform before these pressures lead to macro-fiscal problems. When combined with assessments of rigidity, applying such measures can improve fiscal monitoring and inform fiscal consolidation by helping surface pressing fiscal issues that could undermine fiscal space.¹⁸ The larger the size of spending and momentum, the bigger the deceleration that will be needed to slow down its relative contribution to expenditure growth. The quality/potential of adjustments/deceleration might be undermined by rigidity of particular spending categories. A tool streamlining the application of force, momentum and rigidity analysis at the country level, complemented by international benchmarking, is available on the [PER Programmatic Support Platform page](#). The tool uses BOOST to highlight the evolution of rigidity over the past 15 years by country, as well as basic force and momentum tables and charts as shown in Figure 18.



¹⁸ Momentum illustrates relative contribution to growth of specific items computed a product of relative size and growth rate (“velocity”). Force captures the acceleration/deceleration of such contribution computed as the difference between momentum at period t and t-1. For a thorough discussion, see Merotto et al, Dismal Science, Accounting and Newton’s Second Law Identifying Force and Rigidity in Public Expenditure Analysis, World Bank 2015.

Several PERs analyzed force, momentum and rigidity in their macro-chapters. Dominican Republic (2018) documented how rigidity – modeled after Cetrangolo et al. (2010) – increased in recent years driven by remuneration, transfers, and interest payments (Figure 20). Saudi Arabia (2019) notes the role of high rigidity of spending in amplifying procyclical fiscal policy, where significant inertia in current spending leads to capital expenditure bearing the brunt of adjustments during downswings, underscoring the need to shift to a rules-based fiscal framework to support macro-stability and delink spending from oil price volatility. Kenya (2019) (Figure 19) demonstrates how changes in expenditure have been mostly driven by low-rigidity expenditure, concluding that the government has space to make modifications in the allocations to improve efficiency. Chile (2017) includes an estimate of fiscal rigidity and its impact on fiscal consolidation, finding that broadening the range of expenditures that can be reduced (lowering overall rigidity) would increase the efficiency of fiscal consolidations efforts. Using Alier’s taxonomy (2006), Argentina (2020) highlights how rigidity at both federal and local levels complicate future fiscal consolidation efforts.¹⁹ Other PERs presenting similar analyses include Cabo Verde (2019), Arab Republic of Egypt (2019), The Gambia (2020), and Costa Rica (2018).



2.9 PFM considerations in shaping macro and fiscal outcomes

Integrating PFM aspects in macro-fiscal chapters continues to be a challenge (as covered in more detail in Section 5 below). While both FC and TO PERs generally focus on expenditure analysis seeking to improve the efficiency and effectiveness of sectoral spending, they often fail to integrate the analysis in the broader PFM context. In macro-economic chapters, less than 10 percent of PERs include explicit references to PFM considerations. There is substantial scope to expand the coverage of PFM in their analyses, including recalibrating the discussion of the macroeconomic context with greater PFM coverage. This could include:

1. Reviewing legislative, policy, institutional and regulatory frameworks.
2. Analyzing general government accounting practices and their consistency with budget classification.
3. Reviewing global cross-cutting issues, such as procurement systems, public investment management, corruption and budget processes.
4. Integrating findings from existing governance diagnostics, such as PEFA, Public Investment Management Assessments (PIMA) and TADATs, as relevant and applicable.

Several PERs stand out for their structured discussion of PFM issues in their macroeconomic chapters. Lesotho (2018) presented a comprehensive integration of PFM considerations in the macro chapter, detailing how better expenditure controls and revenue management can help reduce macro volatility by curbing the procyclical nature of fiscal policy. The report also discusses how improved macro-fiscal management can be achieved by ensuring more effective and transparent PFM practices (including the establishment of a TSA), enhancing alignment between the projections of the Integrated Financial Management information System (IFMIS) and the Medium-Term Expenditure Framework (MTEF), and taking action to improve inadequacies in the PFM systems as identified in PEFAs related to budget credibility, fiscal discipline, transparency, financial accountability and fiscal risk monitoring, among others. Malawi (2019) also presented direct linkages between select weak PFM practices and a lack of fiscal discipline. Similar analysis was also offered in Cambodia (2018).

¹⁹ Alier M. “Measuring Budget Rigidities in Latin America, IMF WP, Washington DC 2006

3. Fiscal policy

Fiscal policy—defined as the use of government spending, revenues and financing to influence the economy—is the foundation of the analytical framework encompassing a PER. Analysis of public revenues in PERs is generally analyzed by unbundling headlines figures across a range of dimensions including tax types, taxpayer segments and collecting agencies to study the evolution of levels and composition of revenues, predictability and tax gaps. At a more micro level it aims to assess distributional implications, efficiency metrics, effective tax rates, taxpayer behavioral responses, and tax expenditure analysis to enrich the understanding of the mechanisms at play behind headline numbers. On the other hand, public expenditures analysis is conducted with an objective to improve the technical and operational efficiency of public expenditure and strike an appropriate balance between poverty reduction – that is assessing how the costs and benefits arising from public policies are distributed across society - and growth promotion objectives of spending policies.

The stocktaking revealed encouraging positive trends related to coverage of domestic resource mobilization thematic areas. The review highlighted a marked improvement from the previous round with greater inclusion of domestic resource mobilization considerations and greater sophistication in analytical approaches including use of survey and administrative tax data to explore tax incidence, tax evasion and tax expenditures, along with regression analysis and other techniques to estimate tax effort, tax capacity, tax gaps and efficiency metrics.

At the same time, the evidence emerging from the stocktaking on coverage and analysis of revenue side within fiscal policy pillar suggests that there are a few areas where analysis of the revenue side could be improved. These areas include:

1. Coverage of emerging new areas such as digital economy, transfer pricing, international taxation, tobacco taxation and tax arrears.
2. Use and integration of tax diagnostics.
3. Further in-depth analysis of both foregone revenue estimates and broader assessments of tax expenditure fiscal management frameworks.
4. Mainstreaming of sophisticated analysis including tax simulations, econometric approaches, collection of granular administrative data, and use of survey data.
5. Analysis of progressivity of tax policy instruments continue to be under-represented in PERs.
6. Diagnosis and use of tools to assess gender biases in fiscal policy.

On the expenditure side, there is room for improvement in integrating more sophisticated approaches to equity and efficiency analysis, including broadening coverage to under-represented topics. Much variation is detected in cross cutting thematic areas such as wage bill, adequacy, equity, and efficiency analysis, with unequal and inconsistent coverage and levels of sophistication across PERs. Furthermore., absent from the majority of PERs were analyses to address questions around mainstreaming climate change, gender equality objectives, and disaster risk management in government expenditure policies along with insufficient coverage of thematic issues such as size, targeting and value for money of subsidies, adequate funding of operation and maintenance, inter-governmental transfers, and efficiency analysis in public investment management and procurement systems and practices.

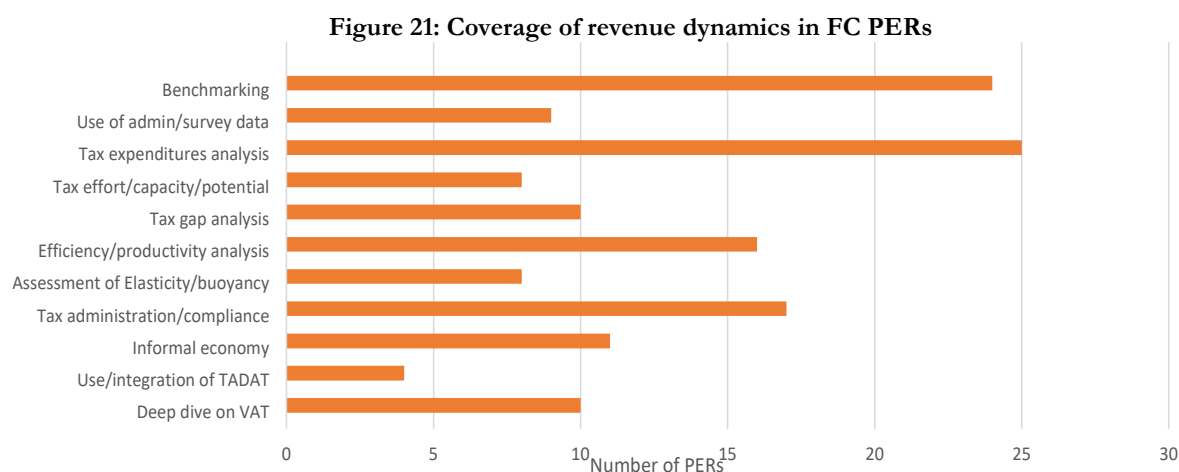
3.1 Domestic resource mobilization

The 2015 Addis Tax Initiative brought domestic resource mobilization (DRM) to the forefront of the development agenda. It identified effective DRM as crucial for financing the public investments that are required to end extreme poverty and boost shared prosperity, while reducing aid dependence, a commitment also reaffirmed by the SDG and International Development Association (IDA) frameworks. Since effective DRM is not just about raising more revenue but also about how it is done, strengthening DRM requires a multipronged approach that seeks not only to maximize revenues but also to inform the strategic design of tax systems to encourage transparency, good governance, equity, and efficiency while maintaining opportunities for economic growth.

The renewed focus on DRM is reflected in its increased coverage and the level of sophistication of analysis in PERs. Virtually every FC PER includes explicit references to revenue dynamics with more than half of the reviewed sample complemented by deep dives, typically within dedicated chapters. Effective DRM analysis requires unbundling headline figures across a range of dimensions, including tax types, taxpayer segments, and collecting agencies, in order to study at a macro level the composition and adequacy of revenues, predictability, and tax gaps, and to assess at a micro level the distributional implications, efficiency metrics, effective tax rates, taxpayer behavioral responses, and tax expenditures. Such DRM analysis is central to understanding the mechanisms at play behind headline numbers.

As reported in Figure 21, beyond basic profile and trend analysis, revenue policy is typically explored through the following lenses and analytical approaches:

1. Benchmark revenue performance vis-a-vis a global, regional, aspirational, and structural set of peers.
2. Use granular administrative or household survey data to undertake microsimulations and/or analyze tax incidence and overall fiscal stance progressivity through the Commitment to Equity (CEQ) methodology.
3. Conduct tax expenditure analysis focusing on the estimation of foregone revenues.
4. Undertake regression analysis to estimate tax potential, effort, capacity, and other constraints, and use instruments such as supply-use tables to estimate policy and compliance gaps.
5. Compile efficiency metrics, buoyancy, and elasticity to assess the overall productivity and responsiveness of various tax types.
6. Review the quality of tax administration and/or estimate the level of informality, tax evasion, and compliance—including through the use of existing tax diagnostics and assessment frameworks such as TADAT and DIAMOND.
7. Undertake deep dives (particularly in VAT) consistent with the tax policy assessment framework.
8. Address other issues such as volatility and vulnerability to economic cycles, arrears, digital economy, tobacco taxation, transfer pricing, and international taxation.

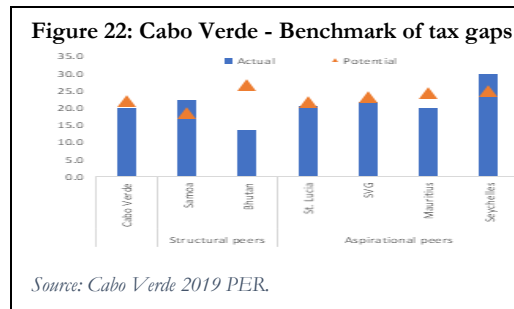


Source: 2020 Stocktaking, World Bank calculations.

a. Revenue benchmarking

International benchmarking can help explain the urgency of action in particular areas and, through peer learning, provide motivation for reform. These comparisons are all the more compelling when the analysis is sufficiently granular and the policy implication actionable. At the same time, it is important to recognize that comparisons of basic tax performance measures may be skewed by differing tax potentials, which are contingent on a variety of institutional, geographic, economic, and demographic factors, among others, so it is important to identify appropriate peer countries.²⁰ Of the FC PERs reviewed, the majority provided some benchmarking (typically centered around basic revenue performance), but only a fraction did so against carefully identified structural or aspirational peers, as opposed to comparing with global, regional, or income groups.

Several PERs provided a series of innovative benchmarking analyses, beyond simple performance statistics. Cabo Verde (2019) presented benchmarking of tax gaps regarding structural peers using a stochastic frontier analysis (SFA) to determine the potential for further improvement in revenue performance (Figure 22). The PER also included additional benchmarking for efficiency of corporate income tax (CIT)/PIT, extent of tax expenditures, and tax volatility. Serbia (2016) compiled a series of benchmarked efficiency indicators, including comparison of cost per employee (calculated by dividing the administrative budget by the number of employees) to show that on average, Serbia spends less per employee than EU countries. Other practical examples include:



- Sri Lanka (2019) compares the country tax effort across countries to document how its performance remains below that of aspirational comparator countries.
- Lesotho (2018) presents a regional comparison of VAT productivity and C-efficiency, as well as the cost of revenue collection, showing that Lesotho compares favorably to the broad set of Sub-Saharan Africa countries, but below other Southern Africa Customs Union countries.
- Egypt (2019) presents a cross-country comparison of tax buoyancy indices to show that the country's tax buoyancy has underperformed that of peers, particularly on CIT.
- Ukraine (2018) provides benchmarking of tobacco taxation to highlight the scope for further increases in excise taxation.
- Chad (2019) presents international comparisons of statutory CIT rates and non-oil revenues to highlight the rate is the highest in Central African Economic and Monetary Community, making Chad less competitive in the region.
- Bangladesh (2015) presents international comparisons on levels of tax effort to show the country has the lowest levels in the region.
- The Gambia (2020) presents benchmarking of VAT/CIT and PIT performance and changes in tax to GDP ratios, highlighting the margin for improvement.

b. Using administrative and/or survey data for incidence analysis and micro-simulations

In order to understand the determinants of revenue performance, it is often imperative to go beyond basic revenue data and consider administrative or survey data. This enables the modelling and documentation of behavioral responses to changes in the tax system—allowing for an in-depth investigation of dimensions such as the concentration of taxpayers at thresholds where tax rates change—evaluate responses to tax policy changes, estimate tax expenditures, and calculate effective rates, among other features. Household poverty surveys can help assess the distributional consequences of tax policy, including the combined effect of

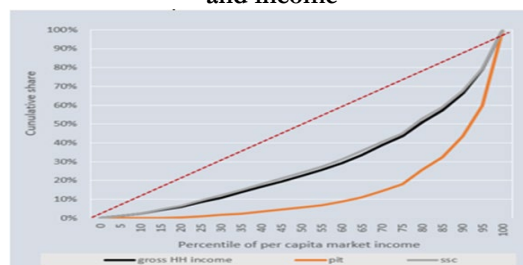
²⁰ Widely available tools, like Find My Friends and Country Economic Memorandum country scans can be used to identify suitable structural and aspiration peers.

expenditure and revenue policy on the progressivity of fiscal stances through the CEQ methodology. Administrative or household budget expenditure data can be used to analyze the gendered incidence of taxes and transfers, for instance on individuals or on different types of households.

Accessing and using granular administrative data can help deepen the understanding of how tax policy changes, such as adjustments in tax bracket thresholds, affect the tax base, revenue collection, and overall progressivity.

Drawing from Brockmeyer and Hernandez (2017),²¹ Costa Rica (2018) reports evidence of tax avoidance/evasion by documenting significant bunching at tax bracket thresholds. Liberia (2018) presents a robust analysis using household survey data to assess the distributional consequences of government taxation on employment income in Liberia,²² revealing that the poorest 40 percent of the population in terms of per capita gross income account for 17 percent of “market income” but only 3 percent of personal income taxes, confirming the intended redistributive design of the PIT system (Figure 23). A similar analysis was prepared in Mongolia (2018), which used PIT administrative data to conduct a series of simulations to inform the adoption of a more progressive PIT system after showing that the impact of the proposed 10 percent flat personal income tax and associated exemptions was only slightly more progressive in relative terms.

Figure 23: Liberia (2018) - Lorenz curve—PIT and income



Source: Liberia 2018 PER.

A few PERs have internalized the CEQ methodology²³ to assess the incidence of tax (and expenditure) and enable efforts to improve the overall progressivity of fiscal policies. The Gambia (2020) (Table 5), for instance, references The Gambia Fiscal Incidence Analysis, which was CEQ-informed and documents the progressivity of the PIT and VAT. In perhaps the best application, Georgia (2015) presents comparative perspectives of CEQ analysis to illustrate the tax system’s relative regressivity compared to regional peers, driven by indirect taxes and offset by large spending programs with negative concentration coefficients, all leading to an overall regressive fiscal stance. Montenegro (2019) notes the analysis from the 2015 Household Budget Survey to chart the distributional impact of fiscal policies broken down by various components and concludes that the net effect is progressive (Figure 24).

Table 5: The Gambia (2020) - Redistributive effects of the fiscal stance

	Quintiles					All
	Poorest	II	III	IV	Richest	
Number of People	734,948	735,013	735,050	734,317	735,748	3,675,077
Market Income Per Capita	425	1,055	1,822	2,926	6,924	2,631
All Taxes (-)	159	193	299	512	1,234	480
Indirect Taxes	157	177	230	297	495	272
Direct Taxes	2	16	69	215	738	208
All Transfers (+)	693	630	533	488	378	545
Direct Transfers	546	492	400	363	270	414
In-kind Transfers	147	138	133	125	108	130
Indirect Subsidies (+)	1	3	5	10	24	9
Final Income Per Capita	965	1,495	2,061	2,913	6,092	2,706

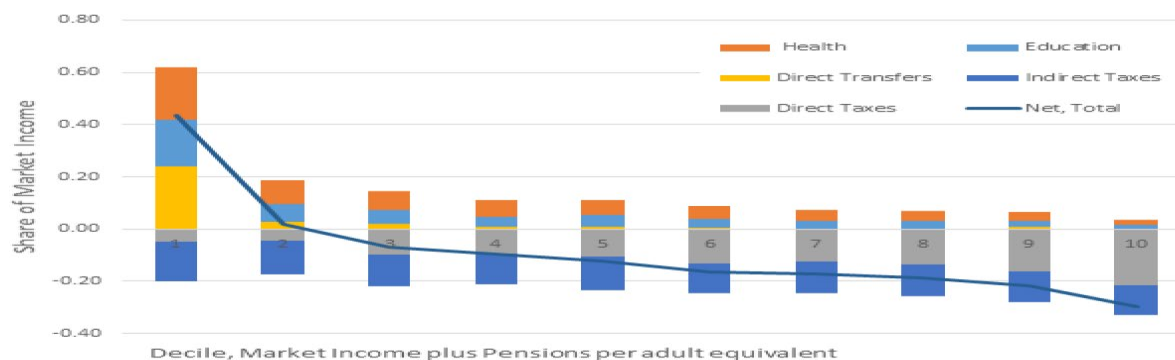
Source: The Gambia 2020 PER.

²¹ Anne Brockmeyer and Marco Hernandez. 2018. Taxation, Information and Withholding: Evidence from Costa Rica, World Bank Working Paper 7600.

²² Personal income tax and social security contributions were allocated to individuals in the survey based on the assumptions about formal-sector employment and report of wages/salaries net of taxes.

²³ The CEQ methodology aims to support fiscal policy decisions to improve inequality and poverty through comprehensive and rigorous tax and benefit incidence analysis, and active engagement with the policy community. It was developed by the CEQ Institute at Tulane University. Tax and benefit incidence studies using CEQ have been completed in numerous low- and middle-income countries in all regions of the world. For more information on CEQ, see Lustig (2016), “Commitment to Equity Handbook. A Guide to Estimating the Impact of Fiscal Policy on Inequality and Poverty,” Tulane University or visit <https://commitmenttoequity.org/>.

Figure 24: Montenegro (2019) - Impact of taxes and social programs



Source: Montenegro 2019 PFR.

Finally, a few PERs include tax simulations to look at a variety of tax policy issues. Haiti (2015) uses the Tariff Reform Impact Simulation Tool to run simulations on four scenarios of tariff reform to estimate impact in terms of increased revenues and indirect effects on imports. Mongolia (2018) uses PIT administrative data to conduct a series of simulations to inform the adoption of a more progressive PIT System. Georgia (2015) simulates the distributional impact of lowering the threshold for property taxes while compensating for removal of select regressive VAT exemptions. This results in enhanced revenue collections with preserved progressivity. Sri Lanka (2019) simulates the distributional impact of the removal of VAT exemptions by using household poverty data, highlighting the need to use these resources to finance better targeted social assistance programs to keep the overall effect progressive.

c. Tax expenditure analysis

The stocktaking reveals an increased focus of PERs on tax expenditures, both in terms of coverage and detail. More PERs include an extensive coverage of tax expenditures since the last round typically through the revenue foregone methodology with greater emphasis on CIT and VAT exemptions. Examples of PERs that look at foregone revenues include:

- Cabo Verde (2019) provides estimates of key tax expenditure items (import duties, CIT, VAT, special consumption tax) over time, while recognizing the paucity of data available for accurate measurements.
- Cambodia (2018) assesses the cost of investment incentives and options for rationing.
- Chad (2019) estimates that tax expenditures are equivalent to 1 percent of GDP and notes that this estimate is a lower bound given the paucity of data and inadequacy of monitoring systems.
- Costa Rica (2018) presents estimates of foregone revenues for VAT and PIT in a regional perspective, as well as a decomposition to identify main drivers.
- Ecuador (2020) provides an estimate of revenue losses for various tax sources in a comparative regional perspective.
- The Gambia (2020) and Tanzania (2020) estimate VAT foregone revenues—a major impediment for efficiency—as well as foregone revenues in border trade.
- Georgia (2017) estimates foregone revenues at 3.2 percent of GDP.
- Ghana (2019) includes detailed estimates of VAT, trade-related, and tax expenditures along with a discussion of the limited effectiveness of tax incentives.
- Guinea-Bissau (2018) discusses generous tax exemptions and tax liabilities that account for a large portion of foregone revenues, estimated at 2.7 percent of GDP.
- Haiti (2015) references a recent International Monetary Fund (IMF) study to document the high share of corporate income tax revenues lost because of exemptions.
- Kazakhstan (2018) estimates tax expenditures by tax type (such as CIT versus VAT) and by the number of taxpayers.
- Kenya (2019 and 2020) document the extent of tax expenditures and discuss recent (and future) reforms of the fiscal management framework.

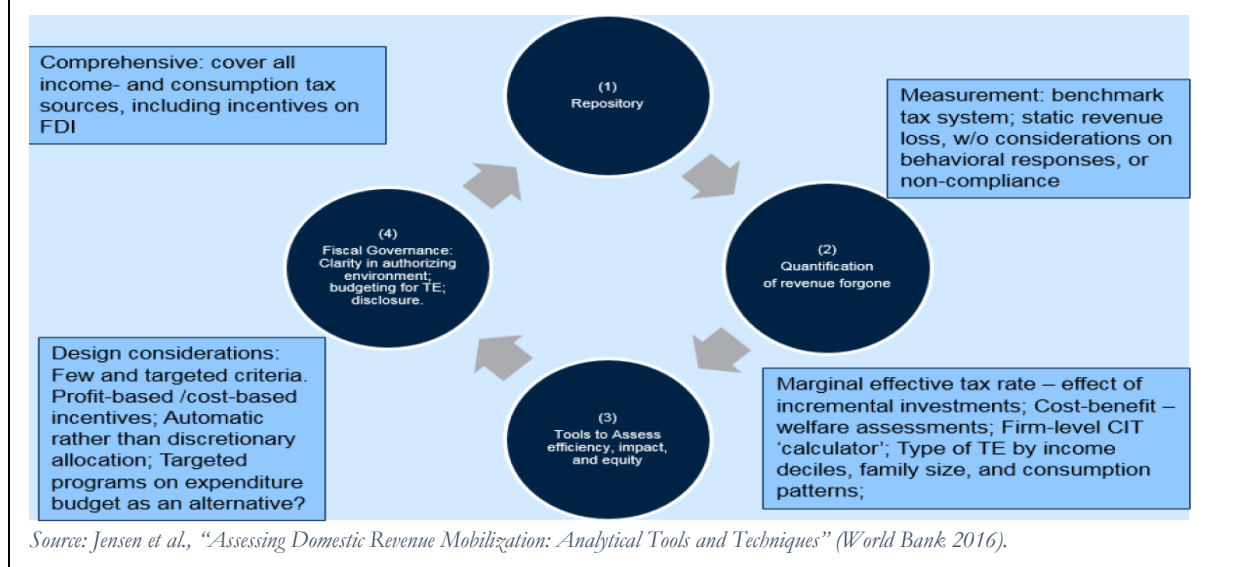
- Lesotho (2019) estimates fiscal losses due to VAT exemptions (including reduced and zero ratings), emphasizing both the extent of the foregone revenues (over 6 percent of total revenues) and their poor progressivity. Similar estimations were conducted for the CIT exemptions.
- Moldova (2017) presents a list of exemptions in VAT and CIT with estimates of foregone revenues in agriculture.
- Mongolia (2019) looks at tax expenditures by each type (CIT, PIT, VAT, customs, and excises) over time.
- Myanmar (2017) looks at tax incentives and estimates CIT tax expenditures under different scenarios.
- Ukraine (2017) estimates the cost of key VAT exemptions and reduced withholding rates.
- Uzbekistan (2020) provides a comparative perspective on the use of tax incentives along with estimates of revenues foregone by sector and an evaluation of the tax incentive regime.

Several PERs also looked at the impact and profile of tax expenditures. For instance, Gabon (2019), assesses the limited impact of tax expenditures on increasing foreign direct investment (FDI), as the tax expenditures are unable to compensate for a relatively poor investment climate. Moldova (2015) quantifies the economic and social impact of eliminating tax expenditures on the national budget, while Georgia (2017) and Sri Lanka (2020) argue that most tax expenditures are broadly efficient and devoted toward social equity objectives (such as reduced VAT rates on education, health, and food), leaving limited room for further rationalization. North Macedonia (2019) discusses investment incentives and other exemptions narrowing the tax base, while Uzbekistan (2020) assesses the ineffectiveness of current tax incentives in attracting more FDI and argues for more targeted measures, such as investment-linked credits or allowances as opposed to existing measures like tax holidays.

A few PERs went beyond basic estimates of foregone revenues and looked at the quality of fiscal management of tax expenditures. Chad (2019), for example, offers a good discussion of the weaknesses in the institutional framework, such as the lack of monitoring of compliance, the existence of multiple granting authorities, and the extent of manual procedures involved in recording, which hampers accounting on the full universe of tax expenditures. Uzbekistan (2020) identifies fiscal management weaknesses, such as incentives offered outside tax laws, that lead to discretion and potential for abuse, while Myanmar (2017) emphasizes poor information sharing between incentive-granting agencies and tax authorities. In a couple of PERs, a four-step diagnostic framework (Figure 25) was applied to look at a comprehensive set of fiscal management challenges. As documented in Ethiopia (2016) and Tanzania (2018), for instance, many countries lack a basic definition of tax expenditures, possess incomplete repositories of tax sources, or fail to introduce a comprehensive national fiscal reference against which revenues foregone could be assessed. As a result, estimates of foregone revenues is often carried out on an ad hoc basis, without adequately defining deviations from the national benchmark. Furthermore, because of data limitations and capacity constraints in tax policy units, tax expenditures are rarely the subject of systematic oversight and ex ante/ex post evaluations to assess their effectiveness. In all pilots, the integrated fiscal governance set-up was found to be poorly defined, with decision-making fragmented across many stakeholders and with blurred criteria for granting tax expenditures, leaving a great deal of discretion for specific awards.

Despite the increase in coverage of tax expenditures in PERs, there is scope for further improvement. Estimation of foregone revenues continues to be implemented on an inconsistent basis and for a limited range of tax sources and types. The scope of the analysis could be expanded into broader assessments of fiscal management frameworks that underpin selection, awarding, and monitoring of tax expenditures. In particular, none of the PERs reviewed in the stocktaking assessed the effectiveness of tax expenditures in terms of achieving its goals, as well as whether those objectives would have been achieved regardless, or whether it would have been more efficient to address via expenditure policy.

Figure 25: Four-step diagnostic framework at a glance



d. Undertaking econometric analysis to estimate tax effort/potential

A number of PERs undertake more sophisticated regression analysis to explore the drivers of revenue dynamics. Econometric analysis is particularly relevant to estimate measures of tax potential, effort, and capacity, among others. While the number of PERs undertaking these types of analysis remains small—indicating potential for further improvement—these early pilots can serve as inspiration for future endeavors.

- **Cabo Verde (2019) uses a panel econometric study to assess tax potential.** First, a regression based on a standard random-effect GLS regression is conducted to estimate tax potential. Then the analysis is complemented by SFA, as in Battese and Coelli (1995), to compare with the maximum achieved by peers with similar characteristics (Box 1).²⁴
- **A different parametric approach is undertaken in The Gambia (2020) to estimate the tax gap and tax potential.** Using panel analysis following Khwaja and Lyer 2014, the PER estimates tax potential at 17 percent of GDP, well above current levels of 11 percent. Similar analysis is conducted in Guinea Bissau 2018.²⁵
- **Chad (2019) develops an econometric model to estimate tax effort relative to other countries.** Using panel fixed effects regression, it estimates tax effort and compares Chad with other resource-rich Sub-Saharan Africa countries, differentiating between oil and non-oil revenues (Box 2). The results show that tax effort is high for oil revenues—signaling reliance on easy-to-collect taxes—but low for non-oil revenues. Similar analysis was undertaken in Sri Lanka (2019) following Le et al (2012).²⁶

²⁴ Battese G.E. and Coelli, T.J. (1988): “Prediction of Firm-level Technical Efficiencies: With a Generalized Frontier Production Function and Panel Data”, *Journal of Econometrics*, 38, pp. 387-399.

²⁵ M. Khwaja, and I. Lyer, “Revenue Potential, Tax Space and Tax Gap—A Comparative Analysis” (WPS6868, World Bank, 2014).

²⁶ Le, Tuan Minh; Moreno-Dodson, Blanca; Bayraktar, Nihal. 2012. Tax Capacity and Tax Effort : Extended Cross-Country Analysis from 1994 to 2009. Policy Research Working Paper; No. 6252. World Bank, Washington, DC.

Box 1: Cabo Verde (2019, Annex 6) - Estimating the stochastic frontier for tax potential

Mathematically, the stochastic frontier analysis panel model is written as follows:

$$Y_{it} = \alpha + \beta X_{it} + V_{it} - U_{it}$$

Y_{it} represents the tax-to-GDP ratio in country i at time t . X_{it} is the matrix of independent variables. V_{it} is a random error term that reflects such random factors as trade or weather shocks. U_{it} is the inefficiency term that captures the gap between actual tax collection and maximum revenue potential. It follows a non-negative truncated normal distribution.

Following the economic literature, the independent variables are (1) GDP per capita in purchasing power parity terms; (2) the contribution of agriculture to GDP; (3) trade openness as measured by the import-to-GDP ratio; (4) control of corruption; (5) enrollment in secondary education; (6) the old-age dependency ratio; (7) broad money as a percent of GDP (a measure of financial deepening); and (8) a policy variable, such as the VAT rate, and dummy variables for small island countries and fuel exporters.

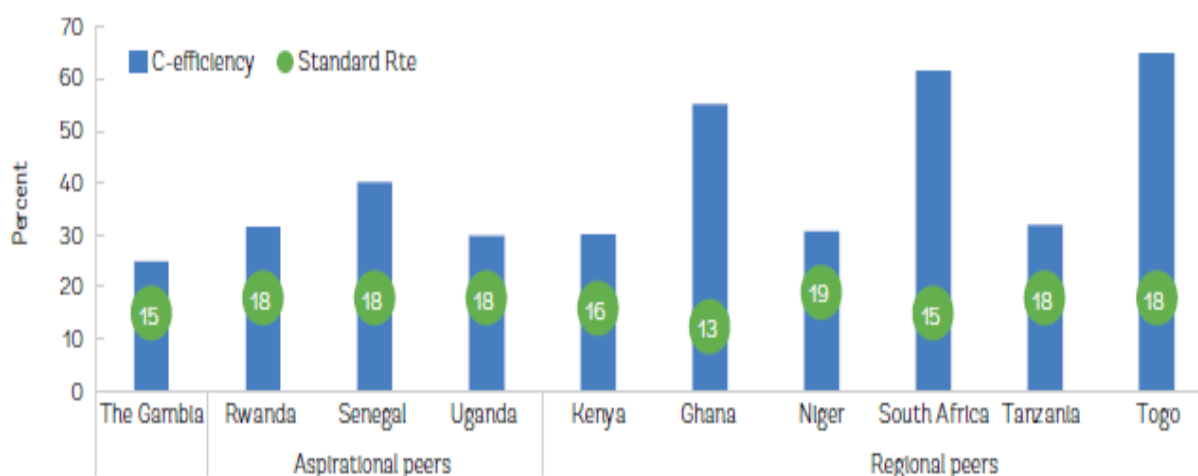
Data comes from numerous sources: (1) tax-to-GDP ratios are from OECD statistics, IMF's Government Finance Statistics, and the Asian Development Bank statistical database; (2) secondary education enrollment, GDP per capita in purchasing power parity terms, share of agriculture in GDP, old-age dependency ratios, and broad money come from the World Bank's World Development Indicators; (3) VAT rates come from private tax consultants such as PwC, E&Y, KPMG, and Deloitte; (4) institutional variables, like control of corruption, come from the World Governance Indicators database. The sample consisted of 160 countries from 1990 to 2016.

e. Revenue efficiency metrics

Assessments of efficiency was included in several PERs. One of the most common ways of measuring revenue efficiency is to look at VAT productivity by calculating C-efficiency ratios, defined as the ratio of VAT revenues to the product of the standard VAT rate and household consumption, or as the ratio between VAT collection (as a share of GDP) and the statutory VAT rate. A good application of these approaches is presented in Lesotho (2018).

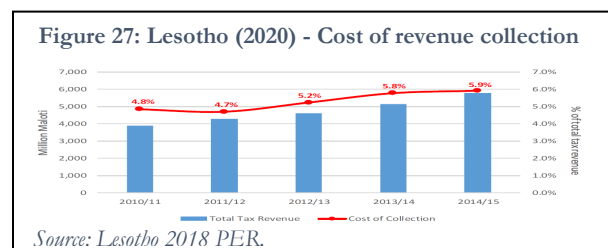
Similar analysis is implemented in The Gambia (2020). The PER looks at productivity of all main revenue types. CIT productivity is defined as the ratio of revenue collected over the corporate tax rate. VAT productivity is proxied through VAT efficiency, C-efficiency, and gross compliance ratios, which show improvement over time but are still below the levels of aspirational and structural peers given the prevalence of tax exemptions and zero-rated goods (Figure 26). Similar analysis was undertaken in Sri Lanka (2019).

Figure 26: The Gambia (2020) - Benchmarking VAT C-efficiency and standard rate



Source: The Gambia 2020 PER.

Alternative measures of efficiency have been explored in other PERs. Serbia (2016) looks at a range of efficiency metrics, which include cost per tax administration employee versus the number of taxpayers per tax employee to document under-resourcing of this function, particularly in analytical capacity and information technology (IT) systems vis-à-vis its peers. Lesotho (2018) looks at the unit cost of revenue collection (calculated as the ratio of operating costs and revenue collected) and documents how the cost of collection grew faster than tax revenues (Figure 27). Cambodia (2018) looks at the ratio of tax administration staff and active taxpayers to document weaker performance compared with peer countries.



f. Coverage of issues of tax administration

An in-depth analysis of the quality of tax administration can offer important insights into determinants of efficiency in revenue collection. This includes delving into issues of compliance costs, institutional and capacity constraints, data recording and IT systems, auditing and Human Resource Management (HRM) considerations along with a broader range of quality dimensions. Of the reviewed FC PERs, 16 carried out an in-depth analysis of tax administration, indicating room for improvement.

Tax administration has been tackled in different ways. Some PERs, such as Moldova (2017), analyze trends in compliance transaction costs through the Doing Business database or other enterprise surveys, or more by describing ongoing reforms and remaining gaps (Myanmar 2017). Few PERs have taken advantage of the existing tax framework or diagnostics, such as TADAT or DIAMOND,²⁷ to provide a more integrated analysis of tax administration. One of the best examples is offered in The Gambia (2020), which presents findings from the TADAT report and monitors reform plans based on its recommended actions. Similar analysis was presented in Costa Rica (2018), Laos (2018), and Tanzania (2018).

Few PERs offer an in-depth analysis of tax administration features that undermine efficiency. Serbia (2016) presents a strong overview of current weaknesses in tax administration that undermine compliance and efficiency, focusing on VAT audit performance and the need for a more efficient risk model to improve audit targeting and hit rates. Similarly, Moldova (2017) documents improvements that stem from enhanced risk-based auditing, while flagging continuing administrative challenges that hinder revenue collection. Other PERs, such as Tanzania (2020), document various administration challenges that hinder better efficiency in VAT revenue collection.

Use of firm surveys can also offer insights into compliance costs and constraints to business, but this was rarely implemented. Beyond the now-retired Doing Business, this information can be extracted from other available sources, including the World Bank Enterprise Surveys to identify the extent to which tax administration (and tax rates) affect business, and whether small firms are particularly hit (as evidenced in Serbia 2016). Other internationally administered surveys, such as the Global Competitiveness Survey, offer similar insights, including on the extent of informality and tax evasion.

Few PERs study the issues associated with the informal economy. Serbia (2016) looked at strategies to combat the shadow economy—estimated at 30 percent of GDP—by first documenting the compliance burden for smaller businesses and the lack of proactive taxpayer services, resulting in incentives for tax evasion via cash transactions. Reform measures focused on improving audit capacity, streamlining taxpayer services, and producing more responsive tax administration, along with IT innovations essential for curbing tax evasion. Chad (2019) estimates that the shadow economy comprises 42 percent of GDP by using predicted mean

²⁷ TADAT refers to the Tax Administration Diagnostic Assessment Tool, designed to provide an objective health assessment of a tax administration. DIAMOND is a toolset to support implementation and monitoring of tax reforms.

matching and multiple indicators, as well as using multiple cause methods presented in Medina et al. (2017).²⁸ The report emphasizes the need for greater use of withholding taxes to collect taxes from the informal economy and discusses examples from peer countries. Tanzania (2020) identifies specific steps to reduce the tax gap that results from informal sector agents amid increased pressures for further informality brought about by the digital economy. Other PERs with coverage of the informal economy include Kenya (2019), Kosovo (2015), and Ukraine (2017).

g. Deep dives on specific tax issues and outstanding gaps

A number of PERs have implemented deep dives on VAT. Consistent with the IMF's Tax Policy Assessment Framework, VAT is typically examined through overall performance, efficiency, and productivity measures, as well as the role of tax exemptions (policy gaps), the compliance gap, and VAT thresholds. The Gambia (2020) provides comprehensive coverage of all those areas, and highlights the potential for improved efficiency by reducing both policy and compliance gaps through rationalization of tax expenditures and raising VAT threshold levels and statutory rates. Kazakhstan (2018) discusses the determinants of low VAT productivity, including narrow base, high thresholds, generous zero-rating provisions in special economic zones, inefficient cascading effects of widespread exemptions, and high levels of arrears in VAT refunds. Costa Rica (2018) discusses moving to a full-fledged VAT from sales tax and estimates the increase in revenues and distributional impact. Lesotho (2018) discusses the policy determinants of the narrow VAT base due to exemptions and high thresholds, the complexity of the rate structure that hinders efficiency and productivity, and the limited progressivity embedded in various exemption schemes (more than 20 percent of potential VAT revenues are lost to exemptions). The analysis provided important insights into the scope for efficiency gains from unifying the statutory rate structure and removing or streamlining exemptions. Tanzania (2020) documents the improvement of VAT productivity and buoyancy following the implementation of reforms, but notes still subpar efficiency due to the proliferation of exemptions; it also documents challenges in VAT administration, such as the VAT refund procedures and the filing compliance / audit processes, drawing from TADAT. Sri Lanka (2019) discusses key features of the VAT, including exemptions, efficiency, incidence, gap analysis, determinants of low collection, and reforms to increase collection and reduce distortionary impacts.²⁹

A few PERs provided deep dives on customs alongside options to improve revenue collection and mitigate governance challenges. Tanzania (2019), for instance, presented an analysis using the standard “mirror statistics” approach to examine trade discrepancies, estimate potential revenue losses, help pinpoint high-risk sectors, and mitigate tax evasion. Other PERs included Moldova (2017) and Haiti (2015), which run simulations on four scenarios of tariff reform to estimate the impact in terms of increased revenues and indirect effects on imports.

Finally, a few other DRM-related areas have been covered sparingly. For instance, Tanzania (2020) is the only PER that has looked into the issue of capturing the digital economy in the VAT to minimize disruption to the tax base due to growing e-commerce sales. Lesotho (2018) and Ukraine (2019) look into the issue of tobacco taxation, whereas Tanzania (2020) explores options to improve excise taxation. The analysis of the extent and impact of transfer pricing and international taxation was seldomly undertaken, with brief discussions presented only in Myanmar (2017). Finally, the issue of tax arrears was rarely analyzed. Mongolia (2018) discusses tax arrears by first documenting its nature and size (about 25 percent of total revenue collected) and then identifying bottlenecks from the lack of a tax arrears management strategy, performance targets, a dedicated debt collection function supported by adequate information technology, and improved risk assessment strategies. Serbia (2016) documents arrears related to over-time trends, short-term impact of amnesty initiatives, and the inefficiency of tax administration in improving compliance. Guinea Bissau (2018) estimates tax arrears at 1.4 percent of GDP in 2016, mostly caused by SOEs, while Tanzania (2020) highlights VAT refund arrears.

²⁸ Leandro Medina, Andrew Jonelis, and Mehmet Cangul, *The Informal Economy in Sub-Saharan Africa: Size and Determinants*, IMF Working Paper 17/156, July 2017.

²⁹ For a more comprehensive discussion, see Michael Keen, “[The Anatomy of the VAT](#)” (WP/13/111, International Monetary Fund, Washington, DC, 2013). Also see the VAT module at <https://www.imf.org/en/Data/TPAF>.

3.2 Analysis of public spending

The issue of expenditure policy was generally approached by focusing on efficiency, effectiveness, and equity considerations. Expenditure policy questions are generally analyzed in the following ways:

1. Preparing an overall expenditure policy diagnostic by analyzing the profile, trends, and composition of the main expenditure aggregates and by drilling down for a comprehensive expenditure analysis.
2. Assessing where a country stands vis-a-vis peers in terms of spending and outcomes by undertaking a variety of benchmark analyses with global, regional, aspiration, and structural sets of peer countries.
3. Considering the adequacy of public spending by comparing actual expenditures with identified needs.
4. Reviewing the efficiency of spending and options to generate savings and greater value for money by looking at allocative and technical efficiency issues and by analyzing the extent to which budget planning and execution are in line with government priorities and policy objectives.
5. Assessing how well the budget is executed by undertaking budget deviation analysis in sectors.
6. Evaluating the effectiveness and efficiency of capital investments and their implications for growth through capital expenditure modules and estimation of output multipliers, as well as analysis of public investment management practices.
7. Considering the equity of spending by undertaking benefit incidence analysis, CEQ reviews, and various quality of targeting assessments, particularly within social assistance programs.
8. Assessing the wage bill in terms of sustainability, productivity, equity, and efficiency, including undertaking micro-level decomposition analysis (such as looking at the the private/public wage gap).
9. Assessing sustainability, rationality, and value-for-money of public subsidies, particularly for SOEs.
10. Analyzing the scope for expenditure policy versus private sector financing.
11. Mainstreaming climate change, gender, and disaster risk management considerations in government expenditure policies.

The recent stocktaking highlighted a number of issues worth exploring. While efficiency and equity analysis are among the most commonly adopted approaches, such analyses vary considerably across PERs, ranging from basic descriptive approaches to more sophisticated techniques, such as data envelope analysis, SFA, and other regression analysis. Much variation also is detected in the treatment of cross-cutting thematic areas, such as wage bill analysis, adequacy analysis, and the mainstreaming climate change, gender, and disaster risk management. Only a few PERs address issues related to subsidies (including value for money considerations), O&M (Timor Leste 2015), inter-governmental transfers (Argentina 2020), or efficiency of public investment management and procurement systems.

a. Expenditure benchmarking

A basic level of benchmarking entails comparisons across total expenditures. However, even when applied with carefully selected peers, this approach does not carry much analytical potential at an aggregate level, since many factors can contribute to variations and little is gained in terms of key drivers and implications. More nuanced comparisons are needed in order to surface operationally relevant analysis and policy recommendations.

broken down geographically. As documented in Haiti 2015, coverage of social assistance programs is unevenly distributed with inadequate targeting and capability in several poor regions.

Alternatively, PERs might assess country performance against a set of recommended benchmarks that have been established in sectors such as education, health, water and sanitation, and agriculture to detect instances of funding gaps.³⁰ For instance, Tanzania (2017) documents inadequate funding in agriculture (with 2 percent of the budget allocated versus the benchmark of 10 percent of total expenditures) and in road maintenance by comparing actual allocations against established benchmarks for Sub-Saharan Africa.³¹ Similarly, Haiti 2015 documents low spending on health care (5 percent of the budget) versus the established benchmark (15 percent). It is important to recognize, however, that standard benchmarks always need to be considered in specific country contexts.

A more precise adequacy analysis draws from actual estimation and costing of needs. The analysis entails calculating funding gaps between costed needs in specific sectors or economic items and actual allocations. This was the approach implemented, for instance, in Angola (2020), Tanzania (2017), and Vietnam (2017), which estimate road maintenance needs and costs and document misalignment with allocated resources. Similarly, Timor-Leste (2015 INFRA PER and 2016 PER) estimate operation and maintenance (O&M) resources needed in each infrastructure subsector while Myanmar (2020 Water PER) estimates a funding gap in the current resource allocations for water and sanitation needed to meet the SDGs.

Other PERs have tackled the issue of adequacy of particular spending categories, such as public wages and social protection, often weighed against fiscal sustainability. As discussed in the wage analysis section, several PERs look at the wage premium to assess the adequacy and competitiveness of public sector wages. Other PERs have attempted to estimate the adequacy of benefits levels. Using household poverty data, Belarus (2021), for instance, documents that social assistance accounts for about 23 percent of total household expenditures in the bottom quintile of the income distribution, while Argentina (2020) presents an in-depth examination of the adequacy, sustainability, coverage, and equity of pensions.

c. Efficiency analysis

Efficiency analysis is both the most common theme within expenditure policy and the least consistent in terms of methodological approaches across reviewed PERs. This reflects differing country contexts and sector coverage, as well as intrinsic challenges with appropriately measuring efficiency. The purpose of this section is to document common approaches and particularly innovative approaches with a view toward informing future efforts.³²

One approach entails benchmarking country performance across a range of indicators that relate expenditures to specific outputs or outcomes in order to identify efficiency gaps. This is the preferred approach in the majority of PERs, which typically look at cross-country variations in terms of unit costs that relate the cost of inputs and the size of outputs (such as the ratio between spending on education and the number of pupils or completion rates or learning outcomes) or linkages between outputs and inputs (such as the ratio of the number of teachers to pupils, or of hospitalizations to the number of hospital beds). More nuanced relationships have included examining the cost per hospital stay and average length of stays (Costa Rica 2018) and variations in unit cost per kilometer in road maintenance across regions (Tanzania 2017). Others

³⁰ In education the basis to assess adequacy is the international benchmark introduced by Global Partnership for Education in July 2015 calling for at least 4 to 6 percent of GDP and at least 15 to 20 percent of total public expenditure to be allocated to the education sector. For agriculture, the Comprehensive Africa Agriculture Development Program estimated the optimal size in select African countries to be around 10 percent of total spending. A recent UNICEF study quantified annual costs needed to reach SDG goals in access to water and sanitation to amount to US\$114 billion, equivalent to 3–4 percent of GDP. Finally, the Abuja declaration calls for at least 15 percent of budgets to be spent on health care.

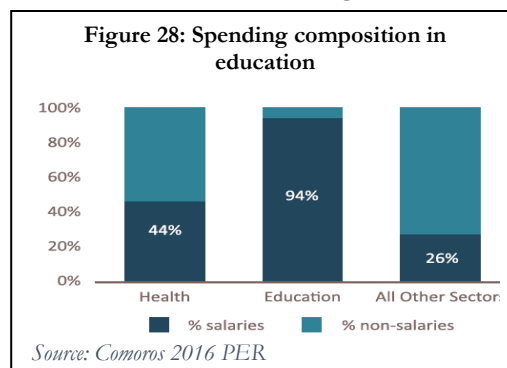
³¹ As documented in the PER, estimates (www.infrastructureafrica) suggest that Sub-Saharan African countries need to spend 15 percent of GDP on infrastructure with 5 percent committed to O&M and 10 percent to development.

³² A note on efficiency analysis that delves more deeply into these issues is available on the [PER Programmatic Support Platform website](#).

entailed comparing spending levels with international benchmarks, such as those established by Global Partnership Education that advocate for at least 20 percent of total spending, as discussed in Chad (2019). While intuitive and insightful, these approaches often fail to isolate country fixed effects in explaining differences in performance, requiring additional levels of analysis to strengthen the narrative.

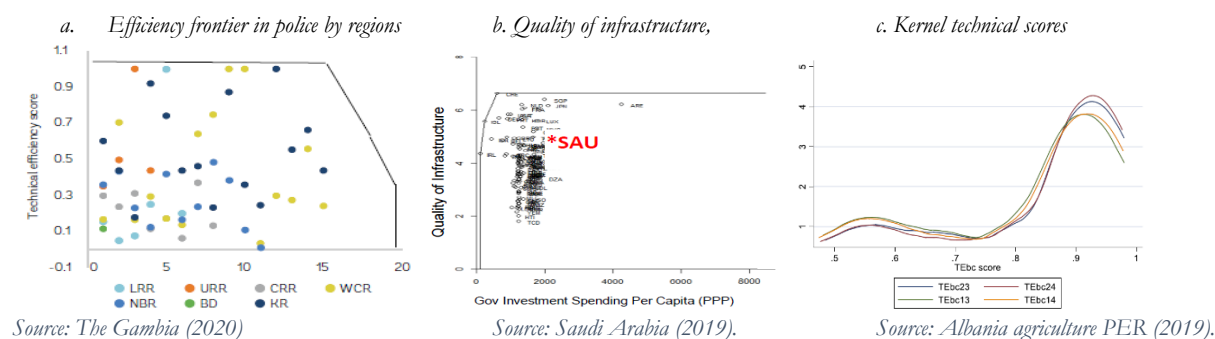
An alternative approach undertaken in select PERs has been to drill down on the composition of public spending to identify inefficient allocations of resources across economic categories.³³ This

includes looking at ratios of maintenance versus capital to detect evidence of underspending in maintenance that could lead to faster depreciation of assets and reduce the overall efficiency of public investment. It might also include analysis of the ratio of wages to goods and services (as a whole as well as within a sector) to determine the extent to which the wage bill crowds out spending on other items. Within goods and services, drilling down further enables assessment of the extent to which such expenses are not related to service delivery outcomes (like teaching versus nonteaching materials) or are not used efficiently.³⁴ For instance, in The Gambia (2020), the majority of the education budget in goods and services is found to be allocated for expenses not related to learning outcomes, while Bangladesh (2015) highlights low spending levels in maintenance in infrastructure. Comoros (2016) documents crowding out of goods and services by large spending in wages, with over 90 percent of spending in education due to expansive recruitment policies (**Figure 28**) leading to efficiency losses and over-reliance on out-of-pocket (OOP) payments. Sierra Leone (2020) develops a set of efficiency indicators.



Technical efficiency – defined as the effectiveness with which a given set of inputs is used to produce an output – can be examined through frontier analysis. Methodological approaches such as data envelopment analysis (DEA) (19 reports) and SFA (4 reports) have been applied to measure efficiency gaps either within or across countries. SFA was used in Costa Rica (2018) looking at academic achievement in secondary schools, accounting for a range of school characteristics, and found a lower efficiency than private schools. A two-step approach was used in Mali 2019 to estimate a subnational efficiency score and its determinants. While this approach typically involved analysis of education and health, a few PERs have applied similar techniques to other sectors, including Saudi Arabia (2019) for roads; The Gambia (2020) for the security sector, Albania (2019) for agriculture (Figure 29), and Bulgaria on policing and firefighting.

Figure 29: Example of DEAs across nonsocial sectors



Investigation of these relationships can yield important insights, especially when conducted at the facility level. Belarus (2021) undertakes facility-level unit-cost analysis to inform optimization of the hospital

³³ There is a growing literature on this topic. See for instance Cobb-Clark and Jha, 2016 on why budget allocation across spending categories matters more for student achievement.

³⁴ For instance, Costa Rica (2018) documents that increases in teacher salaries have not translated into better outcomes because of an inefficient salary structure that is delinked from educational outcomes.

network in order to generate technical efficiency gains. The results of the spatial analysis of hospitals that are included in the sample along with comparison of the unit costs shows a dense network of facilities with duplicative functions and wide variation in efficiency. Analysis of key variables, such as low rates of hospitalization and utilization, as well as length of stays, number of beds, and bed occupancy rates, can also help identify low economies of scale that hinder operational efficiency, as documented in Georgia (2017) and Moldova (2017). Similar analysis was undertaken for the education sectors in several other PERs—such as Costa Rica (2018)—highlighting that in order to achieve high levels of efficiency, schools need to reach certain minimum thresholds in terms of resources and the number of students.

Efficiency can also be assessed by looking at the functional composition of public spending, balancing internal and external efficiency considerations.³⁵ The distribution of spending within the same sector can be used to help evaluate the extent to which allocations between subsectors are balanced and how spending affects both the internal and external efficiencies of the sector, as the process of producing public services varies between subsectors, leading to different cost structures. Examining spending efficiency in subsectors can help identify rationalization measures. The internal efficiency of a subsector may be high, but with relatively high cost and low economic returns, reducing both the internal and external efficiency of the whole sector. In health and education, for example, internal efficiencies might be high for tertiary levels, but low for the sectors overall, as tertiary service provision typically has high-cost structures, while social returns from investment are higher at lower levels.³⁶

Several PERs have looked into these internal efficiency issues. For instance, The Gambia (2020) and Georgia (2017) document allocative bias toward expensive curative services and then recommend rationalizing the allocation of the health budget in favor of primary care toward a long-term aim of providing universal health coverage. Chad (2019) identifies gaps in resource allocations to pre-primary education that might carry long-term effects on human capital accumulation. It also includes an assessment of the internal efficiency of education, determined by looking at repetition and dropout rates and by quantifying the cost of these efficiency losses, mostly driven by gender and location disparities.

Another approach undertaken in some PERs entails looking at cross-cutting areas such as procurement and public investment management, to identify potential for efficiency gains. Very few PERs (6 reports) look at the issue of efficiency savings in procurement, despite being one of the biggest-ticket items in government budgets. The analysis of efficiency gains in procurement can be done using strategic sourcing methodology that assesses the potential for fiscal savings from customized strategies. Great examples of this approach are presented in Costa Rica (2018), Ecuador (2018), and Kenya (2020) (Box 2). Numerous PERs look into the issue of the quality of public investment management at some level of analysis, but only a handful conduct deep dives, either by dissecting execution rates by sectors and geographically, or by seeking to link up project-level disbursement data with administrative fields to estimate cost overruns and time delays, as well as broader portfolio analysis considerations. Two of the best examples are Kenya (2020) and Myanmar (2017), in which individual project-level data were collected to identify portfolio-wide issues in terms of cost overruns, time delays, and dormant projects, and highlighting the need for rationalization and prioritization. Another good example is presented in Bangladesh (2015) (Figure 32), which looks at a portfolio of capital investments and documents the prevalence of cost and time overruns, as well as other operational inefficiencies.

³⁵ Internal efficiency relates to the effectiveness of converting inputs into outputs (such as completion rates in education), while external efficiency measures effectiveness of converting inputs and outputs into outcomes (such as quality of learning).

³⁶ For instance, there is growing rigorous evidence (Gertler et al., 2015; and S. Berlinski, S. Galiani, and P. Gertler, “The Effect of Pre-Primary Education on Primary School Performance,” *Journal of Public Economics* 93 (2009): 219–34) on the causal link between early years interventions in education and long-term achievement.

Box 2: Kenya (2020)—Strategic sourcing methodology

Strategic sourcing refers to seeking institutional arrangements to maximize a purchasing entity's buying power. The goal of the analysis is to understand and explain price differences across the whole public sector procurement process and to identify customized strategies to maximize fiscal savings. In Kenya, this was pursued through framework agreements or consolidated procurements in which all needs are centralized and purchased centrally, a process that is particularly helpful for homogeneous products.

The approach adopted in the PER entailed looking at two purchase categories, motor vehicles and IT assets, in order to identify potential for fiscal savings. Despite the categories being quite homogenous, the unit price dispersion for motor vehicles was quite high, an outcome which suggests scope for efficiency savings. Savings were quantified by calculating differences between actual prices and median prices for each vehicle category equivalent to 1 percent of total procurement spending. The case for making efficiency gains by lowering costs through economies of scale, lower price dispersion, and reduced transaction costs was even more obvious for IT equipment, for which the PER documented that procurement expenses were almost three times higher before the introduction of a centralized procurement policy. The PER concludes that applying a more centralized procurement framework to other items could further improve cost efficiency, while taking into consideration potential pitfalls in terms of capacity and the challenges of ensuring competitiveness, including maintaining opportunities for small and medium firms.

Figure 3.10: Unit Price Dispersion for Vehicles, FY2018/19

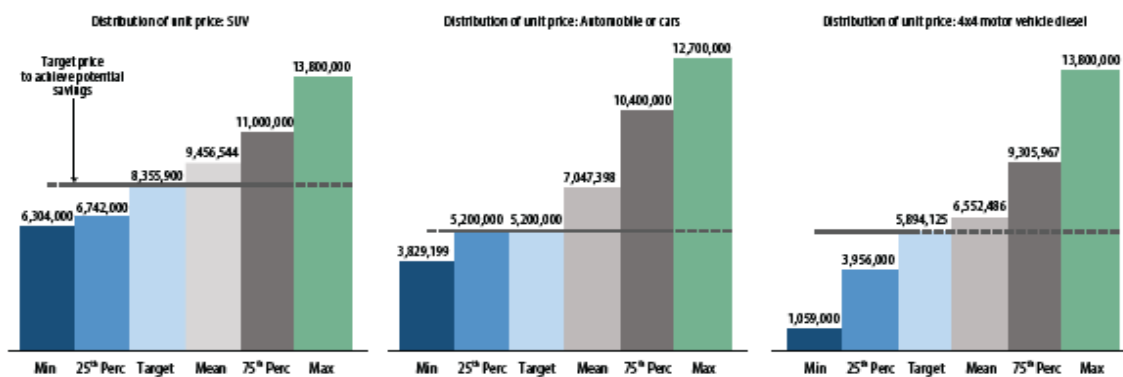
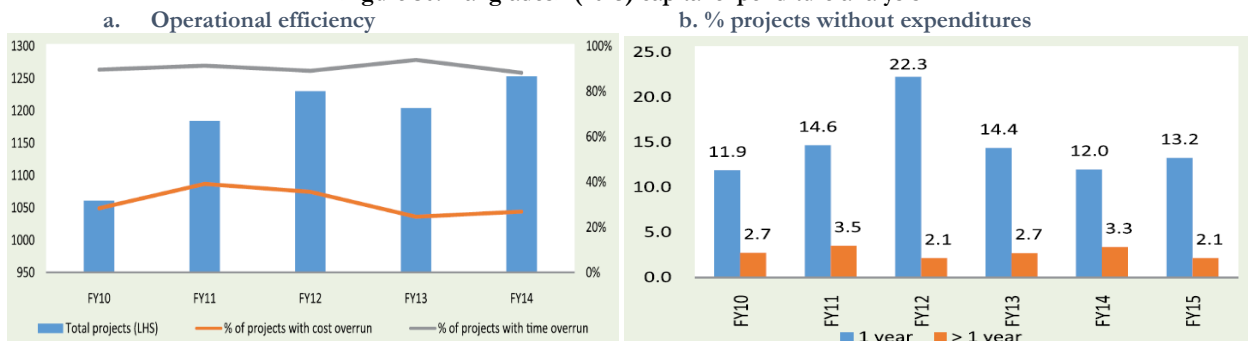


Figure 30: Bangladesh (2015) capital expenditure analysis



A starting point for improving efficiency is to assess the reasons behind low efficiency. A simple process entails looking at execution rates to assess overall budget execution rates, which is particularly damaging in the case of capital projects because they extend over years (requiring multiyear funding allocations) and adherence to strict implementation schedules to avoid cost overruns. More generally, analysis of PFM practices can shed light on inefficient outcomes of public expenditures, as discussed in more detail in Section 4.

Other approaches to addressing efficiency issues include discussing the operational efficiency of SOEs and their subsidies (Indonesia 2020 for water utilities), undertaking regression analysis to identify the determinants of spending efficiency/learning outcomes at the school/regional level (Papua New Guinea 2015) and estimating of the economic rate of return of infrastructure projects, including maintenance needs to support long-term efficiency (Timor-Leste 2015 INFRA PER). Appendix C provides several examples of policy recommendations to increase efficiency built on the spectrum of analytical approaches discussed here.

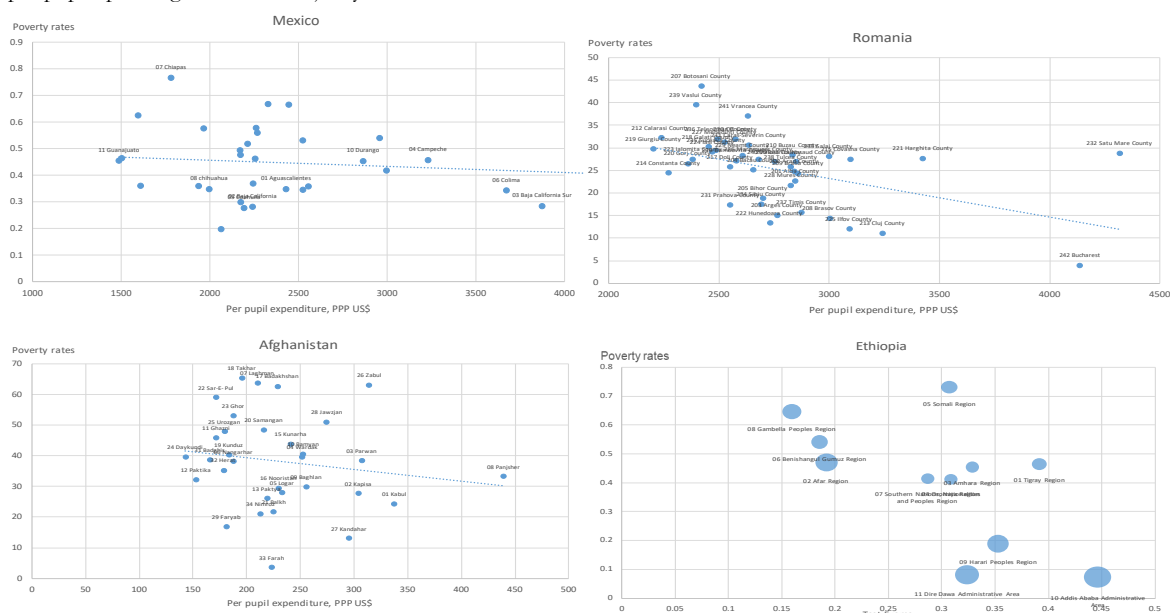
d. Equity analysis

Equity analysis can be undertaken in a variety of ways. A typical approach entails the use of household poverty data to estimate the incidence of spending on poor households, as implemented in benefit incidence analyses and the CEQ methodology (discussed above). Benefit incidence analysis has been integrated in over 25 PERs, either by referencing existing studies or by conducting the analysis using newly available household poverty data. For instance, Cameroon (2018) conducts a benefit incidence analysis using poverty survey data and demonstrates how public spending on health is generally progressive, but not pro-poor. Guinea-Bissau (2018) highlights the regressivity of public subsidies to the education sector as a whole. More generally, poverty surveys can help estimate OOP payments, which at high levels can distort incentives as documented in Georgia (2017). Similar analyses were undertaken in the Democratic Republic of Congo (2015), El Salvador (2015), Ethiopia (2015), Ghana (2016), North Macedonia (2018), Mali (2019), Moldova (2018), Myanmar (2017), Paraguay (2017), and Ukraine (2017), among others.

A second approach looks at the distribution of resources at the subnational level to detect evidence of misalignment between budget allocations and economic needs. Such analysis can reveal a significant disconnect between resource allocation and economic needs, surfacing room for more efficient allocation to improve overall effectiveness. Allocating resources in a way that is not well aligned with needs can result in both inequitable and inefficient distribution of spending. To measure allocative efficiency at this level requires first and foremost access to subnational spending data or at least fully deconcentrated central spending data that are broken down by geographic area, along with a series of nonfinancial data, including sector outputs/outcomes (for example, in the case of education, variables such as completion rates, literacy rates, PISA performance, and so on) and socioeconomic indicators (such as poverty rates, GDP, and the like). Most of the evidence (Box 3) emerging from the limited number of analyses performed in this arena suggests that the distribution of resources across subnational units is often disconnected from economic needs, and instead is driven by incrementalism or political economy considerations. The findings suggest that considerable efficiency could be gained through a better allocation of resources.

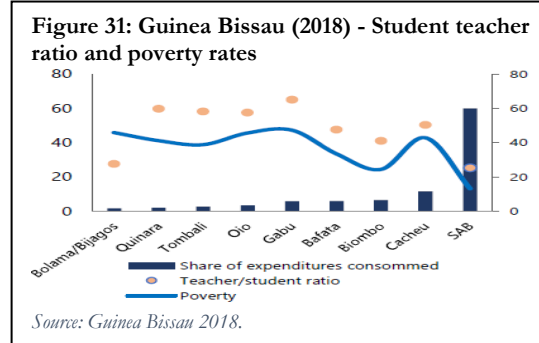
Box 3: Assessing allocative efficiency at the subnational level

Combining subnational spending with outcome indicators on poverty and literacy, for example, provides an excellent approach to assessing equity and efficiency of spending in the public sector. In education, BOOST data are used to connect annual subnational spending flows (2009 onward) with pupils, teachers, poverty rate, completion, and literacy rates for 30 developing countries with decentralized education delivery. The analysis suggests a prevalent disconnect between needs (poverty/completion/test scores) and per pupil spending across the majority of countries examined.



Source: BOOST, World Bank calculations.

Despite the compelling nature of these findings, however, only a limited number of PERs reviewed during the stocktaking present a similar analysis. Guinea Bissau (2018) looks at per pupil expenditures and student-teacher ratios versus poverty rates at the regional level and highlights a significant concentration of resources around the capital (Figure 31), while Cambodia (2018) reveals that provincial variations in expenditures per student are not associated with the incidence of poverty. Cabo Verde (2019) provides analysis of the geographical distribution of needs but does not explore linkages with resource allocation, while the Albania (2019) subnational PER documents political economy drivers underlying specific grant allocations. One of the best discussions is presented in Argentina (2020), which notes differences in expenditure levels and composition across provinces and highlights the scope for better use of equalization grants to match need and fiscal capacity.



A third approach entails dissecting specific functions—especially within social assistance—to determine the quality of targeting across various subprograms. For instance, Belarus (2021) discusses the impact of social assistance on poverty to highlight that ineffective targeting of its schemes leads to inefficient welfare outcomes (Figure 32). The PER dissects individual components of social assistance expenditures to determine the quality of targeting across subprograms, including child allowance benefits, energy subsidies, and unemployment, and cross-references expenditures against available information on the characteristics of poor households and other sources of information. It concludes that current schemes favor predominantly wealthier households and undermine both the efficiency and equity goals of these programs. Table 6 presents a list of other PERs with similar analyses of the targeting of social protection programs.

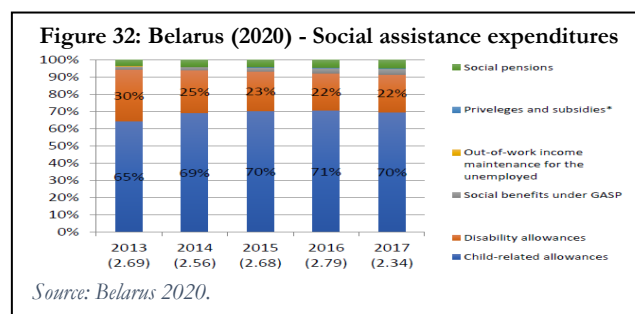


Table 6: Other PERs that review the quality of targeting of social assistance schemes

PER	Analysis
Cameroon (2018)	Documents that targeting of most social assistance programs is not progressive.
Costa Rica (2018)	Reviews targeting of social protection programs.
Haiti (2015)	Notes the inadequate and poorly targeted nature of social protection schemes.
Kosovo (2015)	Presents analysis of the social protection schemes driven by political choices rather than by economic needs.
North Macedonia (2018)	Analyzes the targeting of social assistance programs.
Malawi (2020)	Examines the nature, targeting, and sustainability of social protection programs.
Moldova (2017)	Highlights the poor targeting of social assistance benefits.
Montenegro (2019)	Discusses the weak targeting of social assistance programs using household budget survey data.
Trinidad and Tobago (2019)	Analyzes the poor targeting of social assistance programs using a poverty survey.
Ukraine (2017)	Reviews poor targeting of social programs and the need to transition from universal to means-tested programs.

e. Wage bill analysis

An in-depth wage bill analysis can reveal important insights about the sustainability, efficiency, and productivity of public spending. Examples of such analysis include: (1) examining the composition of wages (base versus allowances), (2) highlighting fiscal sustainability perspectives, (3) considering crowding out of other

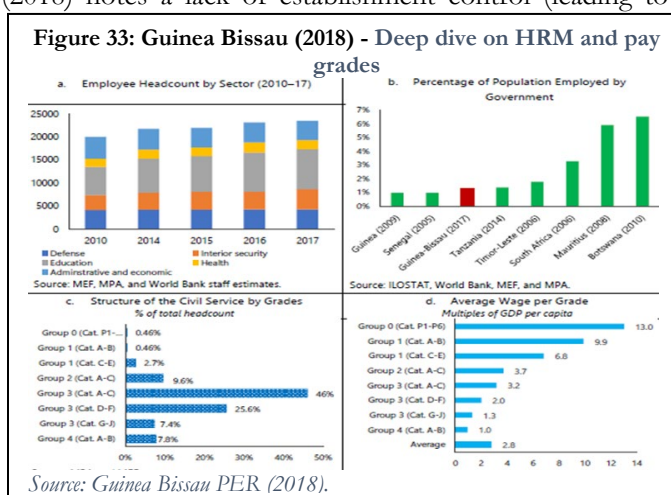
expenditures, (4) analyzing wage premiums, pay grades, and per worker productivity, and (5) placing emphasis on human resource management and related governance considerations. Given the relevance of this topic, a full guidance note has been developed to support teams undertaking comprehensive wage bill analysis in PERs (the note is available on the PER Programmatic Support Platform website).

Despite the broad relevance of wage bill analysis, the stocktaking reveals that coverage is uneven. Whereas the wage bill is analyzed in most PERs, the examination is typically conducted only in basic form through a discussion of recent trends, profiles, and relative shares of public spending, complemented by short narratives, leaving much scope for improvement. Several PERs include a deeper analysis along the following thematic areas and approaches: drivers of wage bill growth, reforms to promote wage bill sustainability, human resource management and pay grades, and adequacy of public wages.

Disaggregated analysis of the wage bill can help identify drivers of growth and areas where operational efficiency can be improved. Kenya (2020) offers an extensive discussion of the main types of allowances, along with fiscal savings that could be gained from their rationalization. Cabo Verde (2019) emphasizes how contractual and temporary workers comprise a significant share of the government workforce. Guinea Bissau (2018) presents a wage decomposition to show large nonwage expenses, virtually doubling the wage bill and creating a large budget deviation across its components. Lesotho (2018) identifies growth in compensation schemes—often not linked to performance—as the main cause of a spike in the wage bill, which is already among the highest in the region. Pay relative to GDP per capita and compared with South Africa reveals large differences across all pay grades. Malawi (2019) compares the country’s surge in public employment with its low teacher-student ratios to reveal that most of the employment was for administrative functions rather than for service delivery.

Some PERs look at the fiscal impact of proposed reforms to promote wage bill sustainability. For instance, Costa Rica (2018) identifies short- and medium-term measures—from hiring and employment freezes to administrative measures and structural reforms (such as offering a single salary and strengthening performance pay systems)—and estimates the associated fiscal and efficiency gains. Guinea Bissau (2018) simulates the fiscal impact of alternative wage bill policies—reducing public employment through attrition, cutting bonuses, and reforming the security sector and structural pay—meant to create fiscal space and help address the infrastructure financing gap. Lesotho (2018) makes wage spending projections that would continue to be a source of fiscal pressure unless compensation policies are changed.

Very few PERs include deep dives on HRM and pay grades.³⁷ By documenting the misalignment between projections and actual staff numbers, Kenya (2020) argues for the need to improve establishment structures for greater operational efficiency. Similarly, Lesotho (2018) notes a lack of establishment control (leading to uncontrolled hiring in low grade categories) and flaws in current compensation policies and suggests that a gradual shift toward performance-linked pay would help generate fiscal relief. Costa Rica (2018) documents the institutional fragmentation of the civil service that results in a proliferation of indexation and compensation systems, with high inequalities and loss of control over the wage bill. Guinea Bissau (2018) analyzes the civil service structures and pay grades and finds significant bunching in the middle (Figure 33), while Ghana (2017) documents progress towards establishing a unified pay structure that would contain wage bill growth and reduce disparities across occupation types and grades.

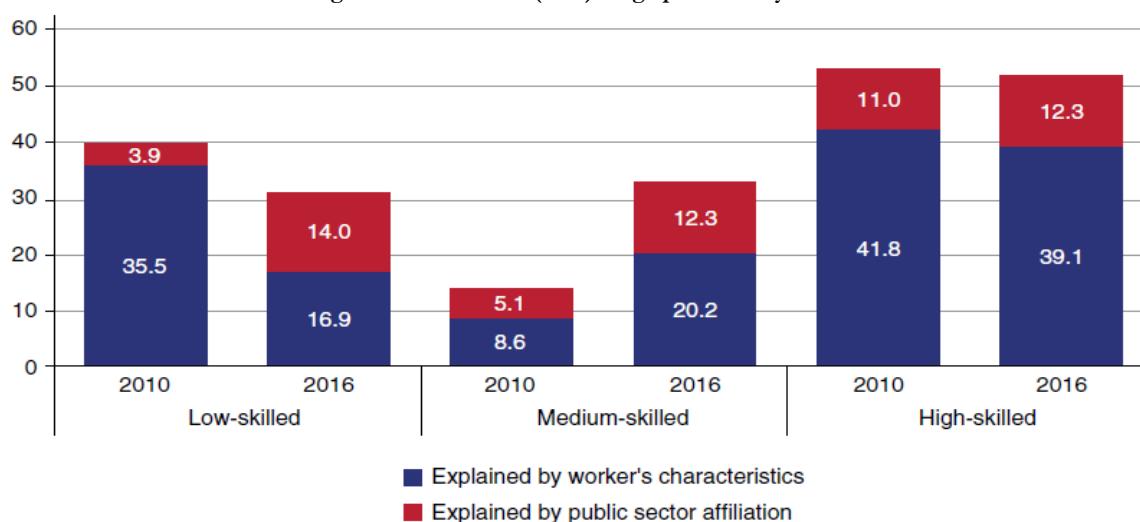


³⁷ World Bank, *Public Sector Employment and Compensation: An Assessment Framework* (Equitable Growth, Finance and Institutions Insight, Washington, DC: World Bank, 2021).

Other examples include Cameroon (2018), which discusses the complexity of the pay structure because the provision of allowances is not standardized, as well as Timor-Leste (2018) and Somalia (2020).

The adequacy of public wages is analyzed in several PERs, mainly through the lens of wage premiums. Kenya (2020) uses poverty survey data to compare average consumption expenditures with levels of public wages to show that pay is adequate across all grades and provide evidence that there is a significant public sector wage premium. Costa Rica (2018), Ecuador (2018), Ghana (2017), and Uzbekistan (2018)—mostly using the Worldwide Bureaucracy Indicators dataset—document high public sector wage premiums and conclude in each case that wages in the public sector are often misaligned with those in the private sector, even after accounting for differences in education, occupational groups, skills, regions, hours worked, and other characteristics. Using the Oaxaca-Blinder decomposition, Costa Rica (2018) is also able to differentiate wage premiums across skills levels (Figure 34). Lesotho (2018) compares ratios of public sector pay versus GDP per capita across all pay grades and against South Africa, an exercise which reveals that compensation levels are more generous than for its neighbor, a main determinant of a surging wage bill. Other PERs, such as Mongolia (2018), estimate the level of public wages to be below those in the private sector for similar grades.

Figure 34: Costa Rica (2018) Wage premium by skill levels



Source: Costa Rica 2018.

Finally, political economy analysis of public sector pay reform is rarely undertaken in PERs. Understanding the prevailing institutional, legal, and regulatory constraints is critical to advising on an appropriate sequencing of wage bill reforms. Only Ghana (2017) and Somalia (2020) present a discussion of political economy considerations—mainly strong labor movements and institutional fragmentation in wage setting—impeding ambitious wage bill reform.

f. Areas with insufficient coverage warranting further analysis

Several additional expenditure policy-related areas that do not receive adequate coverage are worth noting. For instance, gender considerations are rarely incorporated in PERs. Only a handful of PERs analyze gender, including Chad (2019; in the context of explaining efficiency losses in education) and Georgia (2017; in a discussion of gender differences in access to health care services). Similarly, the issue of donor financing receives inconsistent coverage (except for Cameroon 2018, Ethiopia 2015, The Gambia 2018, and Gabon 2018) with room for further discussion on its size and relevance, particularly in highly aid-dependent economies.

Assessments of the impact of structural issues, such as demographics and climate change, on expenditure policies is very limited. Integration of climate change considerations in expenditure policies is addressed in only a handful of PERs, often in countries with significant exposure to natural disasters, including St. Maarten (2020), which highlights the need for better disaster risk management strategies. Sri Lanka (2020)

and Ukraine (2019) are among the few addressing the fiscal pressures of aging on health systems and highlighting the need to focus on preventive health measures and shift from input-based to output-based financing. A few PERs in Europe and Central Asia focus on school optimization analysis to mitigate fiscal pressures in an environment of shrinking student populations and the reluctance to reduce the employment of teachers.

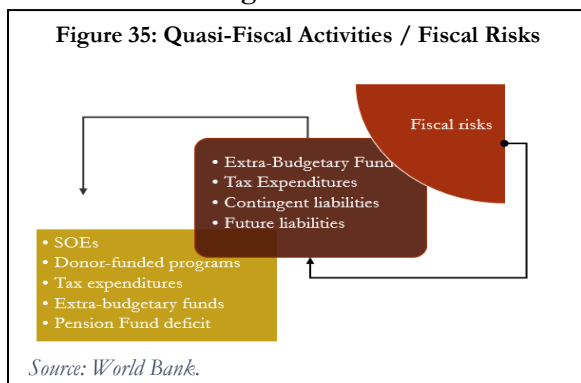
The regressivity and limited value for money of subsidies is rarely analyzed. Uzbekistan (2020) presents an extensive discussion of fiscal support (on and off budget) to SOEs, whereas Timor-Leste (2016) notes the fiscal drag from electricity subsidies and the need for transition toward cost recovery in public asset management to reduce fiscal pressures. Similar discussions are presented in Bangladesh (2015), Kazakhstan (2018), Kosovo (2015), and Myanmar (2017). Some PERs address the regressivity of universal subsidies compared with more targeted social protection measures, highlighting how energy subsidies benefit mostly richer households at a great fiscal cost. These PERs include Belarus (2021), Cameroon (2018), and Kosovo (2015). It is important for PERs to strike a balance between the need to replace universal subsidies with targeted, means-tested transfers while focusing on informing the transition toward cost recovery to reduce fiscal pressures, as presented in Timor Leste (2015) and Comoros (2016).

Assessments of the adequacy of O&M are also rarely undertaken. This is an important area to monitor because underspending in O&M can reduce the lifespan of capital assets, undermining overall efficiency of capital spending. Angola (2020) and Tanzania (2017) include in-depth assessments of the inadequacy of spending on maintenance in the road sector and provide options for reducing the funding gap. Timor-Leste (2016) computes the funding gap for maintenance of capital stock and documents that maintenance spending has not kept pace with capital spending, which is further compounded by under-execution of budgeted allocations.

Finally, the issue of whether spending is driven by incrementalism or by economic need considerations is also rarely covered. This question would be addressed by analyzing how resources are distributed at the subnational level to address inequities. A good example is presented in Argentina (2020), which applies fiscal gap methodology to improve the design of equalization grants with the potential to improve equity in the distribution of resources across provinces (discussed in Section 4). Although a few other PERs offer similar discussions, this is an area that continues to be significantly underrepresented in PERs.

4. Fiscal Risks, Quasi-fiscal Activities and Off-Budget Balances

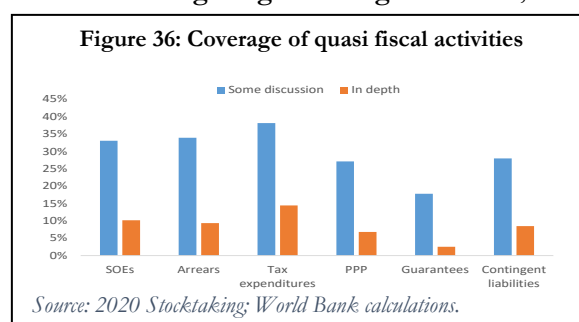
Fiscal risks are broadly defined as the possibility of short- to medium-term deviations in fiscal variables compared with what is anticipated in the MTFF or annual budget.³⁸ Deviations can arise due to numerous factors, as: (1) budgets may be based on assumptions that do not materialize; (2) external shocks may induce revenue shortfalls or surpluses; (3) implementation capacity may be limited; or (4) off-budget or off-balance sheet fiscal operations, government guarantees, and other contingent liabilities may impact government revenues, expenses, financial assets, or liabilities more than expected (Figure 35). Fiscal risk analysis informs stakeholders of the potential budgetary impact of various risks, which may originate from SOEs, subnational governments, PPPs, the financial sector, debt and debt guarantees, quasi-fiscal activities and contingent liabilities, as well as and natural disasters or pandemics.



In order to undertake the analysis, PERs may, among other steps:

1. Analyze the differences between expected and actual fiscal outcomes for key aggregates, such as the budget balance and public debt.
2. Estimate the levels and stock or arrears along with identification of key institutional challenges.
3. Review the budgetary impact of contingent liabilities that may arise for the general government when it implicitly guarantees losses.
4. Estimate the budgetary impact of quasi-fiscal activities (operations that result in a net transfer of public resources to the private or public sector through various nonbudget channels).
5. Simulate alternative scenarios that illustrate the implications of the differing probabilities of the realization of contingent liabilities.
6. Reference direct, indirect, and contingent liabilities related to a specific sector.
7. Use World Bank standard fiscal risk assessment tools, such as the PPP Fiscal Risk Assessment Model and DSAs.

The stocktaking revealed both signs of improvement and some lingering challenges. Overall, the coverage of quasi-fiscal activities increased significantly after 2015, with over 50 percent of reviewed reports discussing these issues explicitly. Within this group, the most prevalent topics include SOEs, arrears, tax expenditures (discussed separately in section 4), PPPs, guarantees, and contingent liabilities. The depth of analysis, however, varies significantly and only a small fraction of PERs present a deep dive on any of the aforementioned thematic areas (Figure 36).



A growing number of PERs analyze the role of SOEs. Coverage and methodologies include:

1. Looking at the financial performance of SOEs—mostly within sectoral analysis and INFRA-SAPs.
2. Discussing subsidies and cost recovery for water utilities (Indonesia 2020 and Zimbabwe 2017).
3. Reviewing public costs and fiscal risks associated with SOEs (Cameroon 2018).
4. Analyzing the public sector accounts with full integration of SOEs (provided only by Myanmar 2017).

³⁸ IMF, 2008.

5. Assessing the importance of SOEs at the macro level in terms of employment, output, debt, arrears, and nonperforming loans, as well as considering the impact of SOE liabilities amid eroding profitability on public debt (Belarus 2021).
6. Reviewing the fiscal risks associated with SOEs (The Gambia 2020).
7. Considering SOE governance issues and constraints with a focus on institutional frameworks, fiscal flows, oversight, and performance data (Bangladesh 2020, The Gambia 2020, Myanmar 2017, and Uzbekistan 2020).

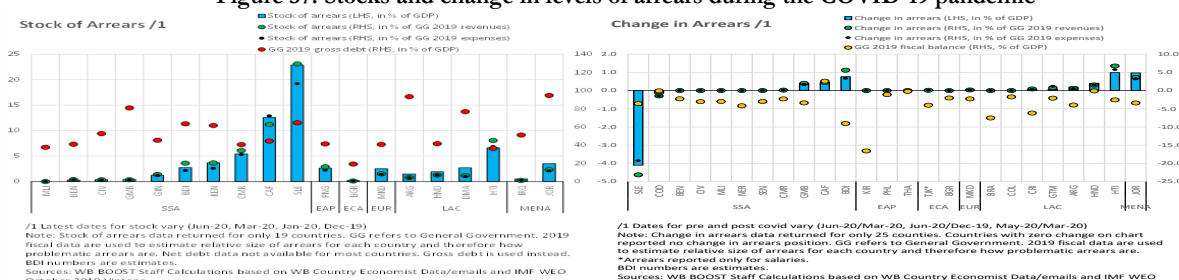
The stocktaking exercise surfaces several common challenges. Fiscal risks continue to be underrepresented in PERs, both in terms of coverage and depth of analysis. Whereas about half of the reports note the topic explicitly, treatment is typically mostly descriptive. The use of existing fiscal risk assessment tools is limited, with only a few PERs leveraging related tools for a more systematic analysis. The stocktaking also captures the low coverage of quasi-fiscal activities, such as contingent liabilities and PPPs, likely a reflection of underrepresentation of infrastructure sectors in FC PERs. Furthermore, despite the size, opacity, and potentially distortionary nature of subsidies to SOEs, none of the FC PERs assess the value-for-money of such subsidies, cost recovery considerations, or reform policy scenarios to improve fiscal balances—with two exception: (1) Comoros (2017), which introduces a model to simulate the impact of proposed reforms to improve operational efficiency and phase out subsidies to the electricity utility, and (2) Uzbekistan (2020), which estimates quasi-fiscal deficits. Finally, very few PERs (Lesotho 2018 is one) look at fully integrated public sector balances when assessing the macroeconomic context, which likely underestimates the potential risks to fiscal stability. Uzbekistan (2020) presents one of the most extensive discussions documenting large off-budget spending that hinders the comprehensive analysis of the budget, including breaking down individual components (Table 7).

Table 7: Uzbekistan (2020) - Consolidated public spending

	Budget	Targeted funds	EBFs	Off-budget accounts	Foreign financed investment	Policy lending	Consolidated spending without quasi-fiscal losses	Quasi-fiscal losses	Consolidated spending with quasi-fiscal losses
Expenditures	19.5	6.5	1.1	3.4	0.3	4.3	35.2	6.0	41.2
of which:									
wages	9.5			1.0			10.5		10.5
pensions	0.9	4.9	0.0	0.1			5.9		5.9
capital spending	2.4	1.3	0.7	0.6	0.3		5.2		5.2
other	6.8	0.3	0.4	1.7		4.3	13.5	6.0	19.5

Arrears are also rarely covered in PERs. A review in 2021 of arrears during the COVID-19 pandemic revealed a significant spike, but also a significant dearth of data, an issues that needs to be addressed in future PERs (Figure 37). Tanzania (2018) offers one of best examples, with an entire PER dedicated to the subject. Good discussions are also presented in Ghana (2017), Laos (2018), North Macedonia (2018), Malawi (2019), Namibia (2020), Somalia (2021), and Zimbabwe (2017).

Figure 37: Stocks and change in levels of arrears during the COVID-19 pandemic



Source: M. Mastruzzi, E. Tanjiki, R. Utzi, and F. Vazquez-Ahmed, "An Overview of the Potential Impact of the COVID-19 Crisis on the Accumulation of Government Expenditure Arrears" (World Bank, Washington, DC, 2021).

Box 4 presents a list of the best practices of PERs in which some of the thematic areas mentioned in this section have been addressed. For further access to underlying analysis, consult the PER repository under the PER Programmatic Support Platform website.

Box 4: Best practice examples on the treatment of fiscal risks from quasi-fiscal activities

- Comoros (2016): A financial model of the electricity sector is developed to analyze fiscal aspects related to cost recovery.
- Indonesia (2020) and Sri Lanka (2020): The fiscal impact of guarantees and contingent liabilities are extensively analyzed.
- Malawi (2019), Mexico (2016), Tunisia (2019), and Zimbabwe (2017): Contingent liabilities are discussed.
- Angola (2018; Social Protection): Projections are made of the fiscal sustainability of the pension system at various levels.
- Tanzania (2019): PEFA is used to assess arrears and budget credibility.
- Myanmar (2017) and Uzbekistan (2020): Deep dives on SOEs are included in the analyses.
- Mongolia (2019): The debt burden from extra budget funding of capital investments is considered.
- Uzbekistan (2020): The quasi-fiscal deficit in major sectors is estimated.
- Various INFRA-SAP reports: Extensive discussions of PPP frameworks and options for maximizing funding for development are included. For instance, Morocco (2020) discusses legal framework issues, cost recovery, and subsidies in the water and roads sector. Indonesia (2020) analyzes subsidies and the legal and regulatory framework governing water utilities, as well as the regulations covering toll roads. Mexico (2016) discusses the regulatory framework for the water sector. Zimbabwe (2017) discusses subsidies and cost recovery for the water sector.

5. Budgetary Institutions

Budgetary institutions are a critical building block of PER policy analysis. Integrated assessment of institutions and policies involves evaluating how the institutional framework and incentive structure deliver aggregate fiscal discipline and a good allocation of resources, as well as efficiency and equity in the use of budgeted resources. For the purpose of the present stocktaking exercise, this thematic area covers a broad range of public finance issues (such as budget planning, preparation, and execution; accounting and reporting; and institutional accountability and transparency), in addition to budget procedures and laws more broadly. Strengths and weaknesses of budgetary systems and processes are important drivers of fiscal outcomes. For example, low levels of operational spending on service delivery may be caused by service delivery facilities not having spending authority to process transactions. Geographical inequities in spending on services may not be driven by an unwillingness to allocate money to underserved areas, but an inability to attract staff to remote areas in a country and HR practices may exacerbate this pattern.

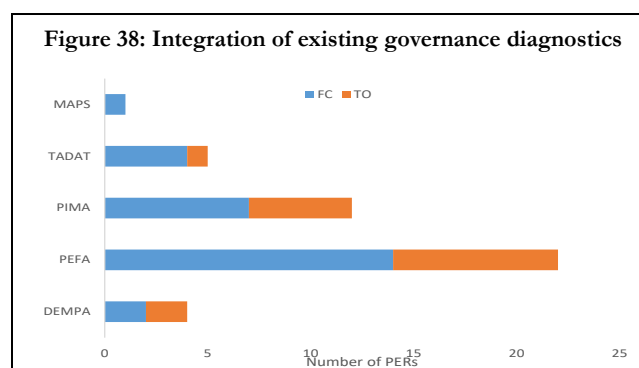
Within this thematic area, several subtopics are salient:

1. Fiscal decentralization and subnational fiscal frameworks.
2. TSAs, accounting, and budget classification.
3. Budget preparation and MTEFs/MTFFs/Medium Term Budget Frameworks.
4. Budget execution, IFMIS, and budget credibility.
5. Audit, internal/external controls, and oversight.
6. Public Investment Management (PIM) systems and practices.
7. Procurement systems and practices.
8. Other key issues, like corruption, political economy, digital governance.

To look into the questions of budgetary institutions PERs may:

1. Review legislative and regulatory frameworks (such as laws and procedures).
2. Consider the policy framework (such as the national development plans).
3. Assess the institutional framework (such as for the relevant ministries, agencies, and departments).
4. Analyze the general government sector accounting practices and their consistency with budget classifications.
5. Examine the quality of budgetary and expenditure control systems.
6. Review TSA functionality and IFMIS comprehensiveness.
7. Consider controls and audit and oversight practices.
8. Analyze intergovernmental fiscal transfers.
9. Review public investment and procurement management practices.
10. Use existing governance diagnostics such as the Methodology for Assessing Procurement Systems (MAPS), PEFA, PIMA, MAPS, DEMPA, and TADAT.

Coverage of budgetary institutions continues to expand, but the stocktaking found there is still room for improvement. About 50 percent of FC PERs provided an extensive treatment of PFM considerations, up from 16 percent in 2015, although at varying levels of granularity and depth. The most covered topic was fiscal decentralization (18 reports), followed by discussion of PFM aspects of budget execution and credibility (23), PIM (20), budget preparation (15), audit and oversight (9), procurement (16), and IFMIS (5). Box 5 lists some examples of good practices on the integration of budgetary institutions analysis in PERs for each of the topics mentioned. Integration of governance diagnostics proved to be a challenge, with only a handful of PERs incorporating feedback from these reports



into their analyses, particularly for TO PERs. The limited use of PEFA, in particular, was surprising given the broad availability of such reports for most developing countries and their holistic approach to PFM.

Most PERs under review have room to expand further the scope and coverage of PFM in their analyses. Challenges were detected particularly in the integration of existing diagnostics (Figure 38) and in-depth coverage of PFM issues with expenditure analysis. Best practice examples in justice sector PERs are Liberia (2019), Moldova (2019) and Ukraine (2017), which took a comprehensive and structured approach to PFM—looking at the entire budget cycle as well as budgeting practices, HRM, and ICT—along with a granular expenditure analysis. As already discussed in the macroeconomic section, Ghana (2017) and Lesotho (2018) provide balanced coverage of fiscal and PFM considerations in their macro chapters, but extensive discussion of PFM challenges that hinder fiscal discipline remains far from the norm in PERs.

Box 5: Best practice example on integrating budgetary institutions analysis in PERs

- Tanzania (2019) notes PEFA findings to review arrears and budget credibility.
- The Gambia (2020), Malawi (2018), and Nigeria (2020) connect the discussion of PFM with findings from existing PEFA, PIMA, MAPS, and TADAT diagnostics.
- Liberia (20) provides a comprehensive and structured approach to PFM and expenditure analysis in sectors.
- Chad (2019) uses PEFA to assess determinants of poor budget execution outcomes.
- Kenya (2020) reviews current procurement and PIM practices to identify options for fiscal savings.
- Comoros (2016) and Guinea Bissau (2018) investigate PFM aspects that undermine budget deviation analysis.
- Argentina (2018) and Comoros (2016) analyze options to improve the quality of intergovernmental transfers.
- The Gambia (2020) and Kazakhstan (2018) offer comprehensive assessments of the procurement framework.
- Cambodia (2016), The Gambia (2020), and Lesotho (2018) offer comprehensive assessments of the public investment framework.
- Lesotho (2018) integrates PEFA findings into an analysis of the macroeconomic context to explain the determinants of volatility and the lack of fiscal discipline.
- Cambodia (2018) documents progress in implementing a medium-term budget framework and remaining challenges.
- Somalia (2020) discusses challenges in commitment and payroll control.
- Ghana (2017) provides a discussion of political economy considerations in pay reform.

a. Budget cycle

Budget preparation is typically examined by focusing on the quality of budget planning and forecasting, as well as the quality of the medium-term framework to guide expenditure allocations. Guinea Bissau (2018), for example, documents a top-down approach with little involvement of the sector ministries in the preparation process, a practice which results in a disconnect between sectoral plans and overall expenditure decision-making. Kenya (2019) documents the need to further improve revenue modeling and forecasting tools, including the re-estimation of key parameters, such as elasticities. Malawi (2019) highlights how systematic manifestations of poor revenue forecasting lead to unpredictable releases and substantial budget deviations. Mali (2019) identifies the lack of a multiyear framework as a key shortcoming in budget preparation and improved alignment across government tiers.

Budget credibility is well analyzed with a PFM lens. Chad (2019) makes good use of its PEFA report for explaining the determinants of weak budget credibility, including poor planning, limited reporting tools, and lack of internal and external controls (in addition to compliance issues), leading to an accumulation of arrears. Comoros (2016) identifies bottlenecks in the budget execution chain to explain the main three findings of its budget deviation analysis (poor revenue forecasting, high volume of expenditure reallocation, and large over-execution of the wage bill). Bottlenecks include poor integration of sector plans, weak budget classification, and significant discretion in reallocation procedures. PEFA is also used in Guinea Bissau (2018) to document PFM issues underlying low budget credibility—including cash management, procurement practices, and IFMIS integration. The Gambia (2020), Lesotho (2018), and Malawi (2019) document weak legal frameworks that

allow in-year reallocations and administrative transfers of budgetary funds that are the main cause of budget variations. Kenya (2019) relates low budget execution to weaknesses in capacity, project appraisal and planning, and sequencing of capital project implementation, while Mali (2019) identifies streamlining the expenditure chain as critical to improving budget credibility and the overall efficiency of spending at the subnational level.

Budget execution analysis can be strengthened by examining budgetary and expenditure control systems. Unpredictability in budget execution can be traced to PFM flaws such as unreliable revenue forecasts that lead to reduced exchequer releases (Tanzania 2017), unrealistic budget assumptions and costing projections and limited IFMIS utilization (Kenya 2020), poor cash management and lack of a TSA (Lesotho 2018), and arrears accumulation (Tanzania 2018), among other examples. Assessing IFMIS coverage and utilization can help explain bottlenecks that hinder better expenditure controls because the larger the number and volume of transactions that occur outside IFMIS, the harder it will be to get a full picture of budget execution. As documented in Kenya (2020), about 20 percent of transactions occur offline, weakening overall expenditure control. Guinea-Bissau (2018) provides an extensive discussion of IFMIS functionalities in improving cash management functions and expenditure control, while recognizing the need for further reforms to improve budget realism and predictability. Tanzania (2018) discusses the cascading effects of unreliable revenue forecasts and poor revenue performance that lead to the implementation of a monthly system of “Exchequer releases,” which in turn results in systematic under-execution of the budget and persistent arrears accumulation outside of the IFMIS.

Analysis of audit, controls, and oversight performance is rarely presented. Analysis of the adequacy of budget classification and broader accounting and financial reporting functions could help surface factors hindering better oversight, as undertaken in a few PERs, such as Comoros (2016) and Guinea-Bissau (2018). The effectiveness of the audit function can also be examined through a review of the broader legal framework, as well as regulatory, institutional, or capacity constraints. Guinea-Bissau (2018) finds that the lack of legal foundations and standard protocols, and limited IT and broader coordination are the main impediments for to stronger audits by the Inspector General.

b. Procurement

Despite the relevance and size of expenditures involved, analysis of the quality of procurement systems is significantly underrepresented in PERs. In most cases, procurement systems are briefly discussed within broader assessments of budget execution and public investment management. Only 12 PERs present in-depth discussions of procurement systems with a view to identifying measures to realize efficiency savings, greater value for money, and better transparency and accountability.

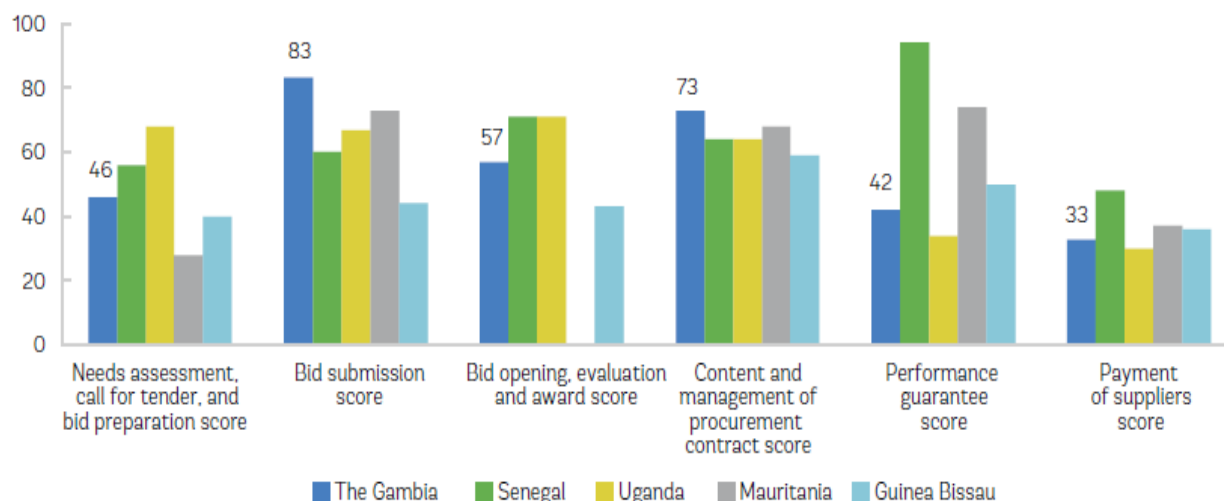
Procurement is typically analyzed through a review of legal and regulatory frameworks, highlighting bottlenecks that hinder efficiency and value for money. Various diagnostics have been implemented to facilitate this analysis, including PEFA, the Methodology for Assessing Procurement Systems (MAPS), and the 2017 Benchmarking Public Procurement report.³⁹ The use and integration of these diagnostic instruments, however, has been limited across PERs. When one assesses procurement systems, it is important to recognize the importance of assessing functions. This work entails looking beyond existing laws and frameworks to assess de facto practices on the ground. One of the most practical ways entails identifying the prevalence of a preferred method of open tendering versus other less competitive approaches (tackling bid rigging in public procurement could generate significant cost savings). The Gambia (2020), for instance, documents how only 9 percent of large bids were subject to open tendering, despite being legally required. Another comprehensive assessment of the public procurement framework and system is presented in Kazakhstan (2018).

The 2017 *Benchmarking Public Procurement* report provides an important tool to measure and benchmark country performance across eight pillars that capture the critical stages of the procurement process. The initiative collects data on a number of critical dimensions underpinning the performance of

³⁹ World Bank, *Benchmarking Public Procurement 2017: Assessing Public Procurement Regulatory Systems in 180 Economies* (Washington, DC: World Bank, 2016).

procurement systems across the full procurement cycle (from need identification to contract execution) and spanning 180 countries. Application of this framework in The Gambia (2020) reveals both strengths and gaps in relation to peers (Figure 39), particularly in terms of need assessment and procurement plans, performance guarantees, and payment processing.

Figure 39: Benchmarking Public Procurement Toolkit, The Gambia 2020



MAPS is another important tool that can help PERs assess public procurement systems. Implemented in over 20 countries, the MAPS framework assesses the quality of procurement systems by reviewing the legislative, regulatory, and institutional framework to identify the quality and functioning of the formal rules and procedures that govern public procurement. In addition to discussing factors that influence operational efficiency, MAPS addresses the accountability, integrity, and transparency of public procurement systems. This resource, however, is rarely used, with Malawi (2019) being the only study leveraging the diagnostic to document deficiencies in terms of the competitiveness and transparency of the procurement process (with 80 percent of the total value of procurement awarded through restricted tenders).

Strategic sourcing methodologies are applied in a few PERs to help identify operational efficiency and cost-saving measures, as discussed in section 3.2. The use of this approach in the Costa Rica (2018) is one of the most comprehensive. Using transaction-level data collected from the integrated procurement system, the report first categorizes procurement bids across quadrants using the supply position (Boston) matrix and then identifies customized strategies (demand consolidations, minimization of noncompetitive bidding, and so on) for each quadrant to maximize efficiency gains, estimated at more than 20 percent of current total spending via micro-level price estimation. The analysis also reveals important predictors of efficient performance, including completion times and rates, market characteristics, organizational capacity, and the level of competition.

c. Public investment management

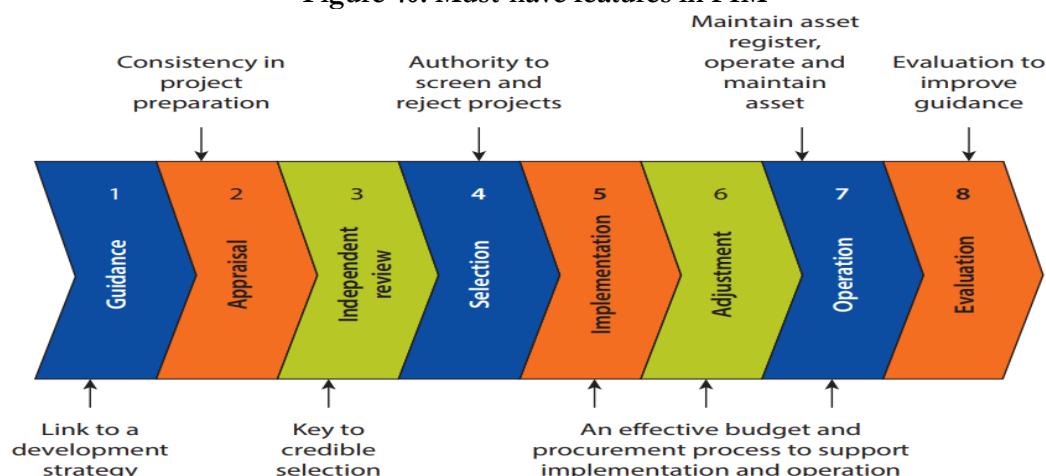
PIM represents one of the greatest sources of inefficiency. There are many drivers of inefficiency, including poor selection and appraisal practices, low implementation capacity, lack of multiyear capital budget planning, procurement gaps, poor monitoring and evaluation practices, weak interagency coordination and political economy considerations, among others. All of these issues can result in costly time delays, budget over-runs, dormant projects, and ineffective investment due to poor screening and prioritization.

Discussion of public investment management typically revolves around the must-have features of a good-quality PIM framework (Figure 40). Following the approach described by Rajaram et al. (2014),⁴⁰

⁴⁰ A. Rajaram, L. Minh, K. Kaiser, J. Kim, and F. Jonas, “The Power of Public Investment Management: Transforming Resources into Assets for Growth” (Directions in Development—Public Sector Governance, World Bank Group, Washington, DC, 2014).

these features include activities related to planning, appraisal, execution, and post evaluation. Common issues among developing countries include: (1) limited integration of development plans into the budget process; (2) lack of a mechanism to prioritize projects on the basis of their economic rates of return; (3) weak capacity to appraise projects and independently review appraised projects; (4) inadequate funding for O&M; (5) implementation gaps and poor monitoring; and (6) no ex-post evaluations.

Figure 40: Must-have features in PIM



Source: Rajaram et al, 2014.

The framework is well applied in several PERs. Romania (2017), for instance, undertakes PIM gap analysis to highlight the key drivers of underperformance of public investment projects, including a lack of strategic planning that inhibits effective screening; no single oversight institution for approving projects; limited use of appraisal methods in selecting investment projects; absence of a functioning independent review procedure; poor monitoring and prioritization; and budget execution challenges and delays. The Gambia (2020), Kenya (2020), Lesotho (2018), and Myanmar (2017) provide similar gap analyses of PIM systems and practices. Inclusion of an analysis of the impact of extensive donor funding can also strengthen the overall narrative. As documented in Cabo Verde (2019), for instance, inefficiencies in the management of donor aid have resulted in significantly lower execution rates than domestic-funded capital projects.

PIMA reports—when available—also provide extensive insights to help inform procurement reforms. PIMAs are typically led by the IMF and evaluate the procedures, tools, decision-making, and monitoring processes used by governments to provide infrastructure assets and services. They can be impactful in helping identify reform priorities and devise practical steps for their implementation. Using a PIMA report, Ghana 2017 describes challenges in public investment management: capacity is weak and reforms are needed at all stages of the project cycle. Similarly, The Gambia (2020) references a recently implemented PIMA to discuss two main weaknesses—the inadequacy of budget nomenclature to properly identify capital projects and heavy reliance on donors, both of which lead to fragmentation.

The analysis can be complemented by the development of ad hoc capital expenditure modules. As discussed in Section 3.2., collecting and analyzing disbursement and administrative data at the project-level can help document the extent of cost overruns, time delays, and other portfolio assessments as presented in Bangladesh (2015), Kenya (2020), and Myanmar (2017). This approach in turn can support the overall narrative by helping strengthen the link between form and function.

d. Intergovernmental transfers and relations

The issue of intergovernmental transfers and relations is typically addressed by reviewing the quality of the subnational fiscal and institutional framework and its ability to ensure the efficient and effective use of resources. This task typically comprises assessing challenges to the intergovernmental finance system,

including vertical and horizontal imbalances, subnational revenue autonomy, the predictability and transparency of funding, key features of the intergovernmental transfer system,⁴¹ and clarity on tax and expenditure assignments.

The stocktaking finds that 17 PERs look these issues. Comoros (2016) examines the ability of the fiscal framework to deliver services in a decentralized context by reviewing options to improve the current quota system and ensure more efficient allocation of resources at the subnational level (dominated by the wage bill). Somalia (2020) explores options to improve the current intergovernmental framework to resolve functional ambiguities and develop suitable fiscal arrangements to promote greater devolution to states. Tanzania (2017) looks at options to reduce fiscal inequities, while Uzbekistan (2020) discusses challenges to the intergovernmental finance system mainly due to large vertical imbalances, limited autonomy, low predictability, and little clarity on functional assignments. Three reports (Albania 2019, Myanmar 2018 and Peru 2017) are fully devoted to reviewing subnational fiscal and institutional frameworks, while the Mali 2019 decentralization report estimates the efficiency of spending at the subnational level by conducting an SFA in a two-step approach that identifies key PFM challenges as main determinants of inefficiency.

One of the most comprehensive discussions of intergovernmental transfers is in Argentina (2020). The report applies fiscal gap methodology to consider improvements to the design of equalization transfers,⁴² with aiming of making distributions of resources across provinces more equitable and helping address fiscal capacity and economic needs (Figure 41). The estimation of economic needs is made by calculating a multidimensional index (Table 8). Fiscal capacity is proxied by estimating tax potential in each province using typical econometric analysis. The report then compares identified parameters with those currently used in the existing outdated system to highlight the greater progressivity of the proposed new formula with a potential phased in transition to soften the effects of the changes on the “losing provinces.” The report then discusses potential arrangements for the establishment of a risk-sharing fund that could help mitigate the fiscal and output effect of asymmetric regional shocks in Argentina’s highly procyclical spending environment. By helping to decrease total revenue volatility, the proposed risk-sharing model helps the provinces adopt a more countercyclical spending stance. Similar compelling analysis is presented in Peru (2017), among other PERs.

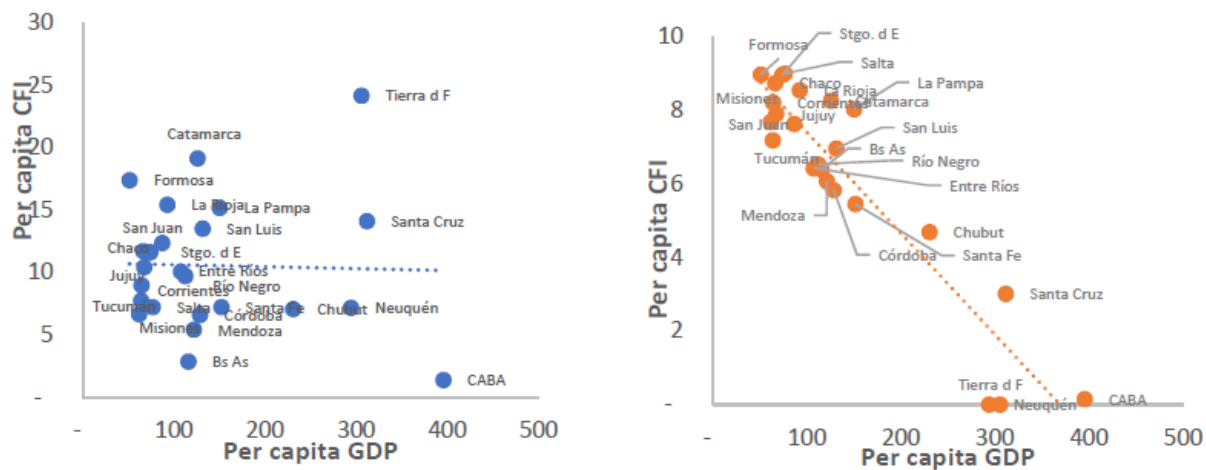
⁴¹ See also R. Bahl, J. Boex, and J. and Martinez, “The Design and Implementation of Intergovernmental Fiscal Transfers,” 2001.

⁴² One of the most common approaches is the one discussed in J. Martinez-Vazquez and C. Sepulveda, “A Theoretical Rationale for the Fiscal Gap Model of Equalization Transfers,” Andrew Young School of Policy Studies, Georgia State University, 2018.

Table 8: Indicators used to estimate spending needs

Function	Indicators and weights
Government Adm	Population (4/5) & Area (1/5)
Security	Population (4/5) & Area (1/5)
Social Services	
Education	School aged population (5-19)
Science	School aged population (5-19)
Health	Population (1/3) & UBN (1/3) & Poverty (1/3)
Water and sewage	Population (1/3) & UBN (1/3) & Poverty (1/3)
Housing	Population
Social Assistance	UBN (1/2) & Poverty (1/2)
Labor	UBN (1/2) & Poverty (1/2)
Other urban Service	Urban Population
Economic Services	GDP (1/2) & Population (1/2)

Figure 41: Impact of current (left) and proposed system based on fiscal gap methodology (right)



Source: Argentina (2020).

e. Other PFM and governance areas in need of expanded coverage

Several other governance-related areas warrant greater coverage. They include the treatment of corruption and its macro-criticality, governance of SOEs (as in Myanmar 2017), political economy analysis, budget classification, and digital governance (covered only in Myanmar 2019). Few PERs look in-depth at the challenges of transitioning to International Public Sector Accounting Standards (Kazakhstan 2018), enhancing the quality of IFMIS systems (Guinea Bissau 2018, Kenya 2020, and Tanzania 2018, which documents persistent expenditure arrears outside of the IFMIS), ensuring TSA functionalities (Lesotho 2018), and addressing broader budget classification issues.

Only a handful of PERs present some discussion of political economy issues. Mongolia (2018), for example, discusses the extent of political influence on budget outcomes, and Ghana (2017) and Somalia (2020)

elaborate on the impact of a strong labor movement and institutional fragmentation in wage setting, which act as impediments to ambitious wage bill reform (as discussed above). Other examples include the Philippines Education PEIR (2015) that diagrams the influence of agents on the budget, and Indonesia (2020).

Finally, corruption is largely unaddressed, with the exception of few references to VAT and procurement fraud. Tanzania (2017), for example, references recent studies to document how extensive fraud in the road sector undermines value for money of these investments by affecting project performance through several channels, including delays, cost escalations, and lower quality, leading to weaker project outcomes and higher costs in O&M. North Macedonia (2018) draws on existing surveys to highlight that a large percentage of households report having to make informal payments in order to access health services.

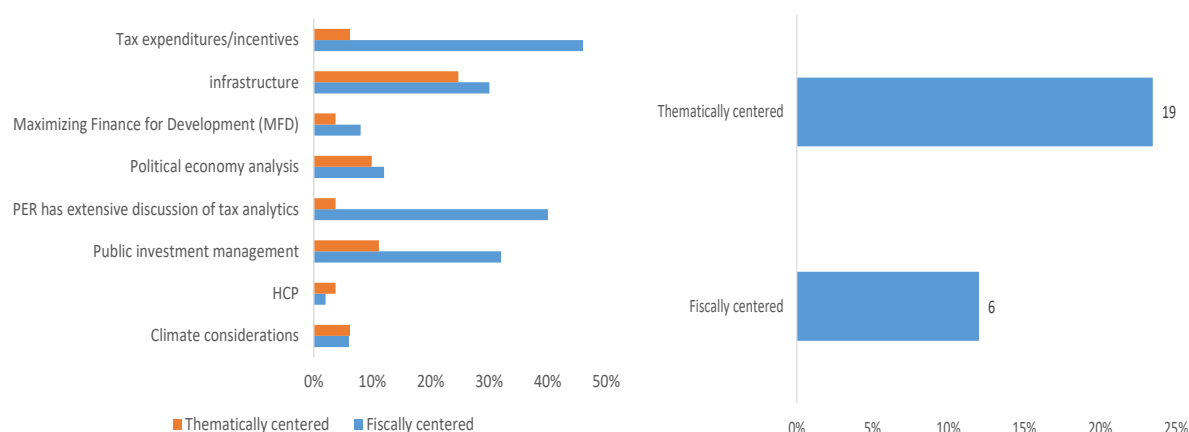
6. Thematic/sectoral areas

The stocktaking documented the extent to which PERs have tackled a range of thematic and sectoral areas. The treatment of thematic / sectoral areas is mixed, and in many contexts increased coverage could be considered of emerging themes, like climate change and pandemic preparedness. While tax expenditures and more general tax analytics have witnessed significantly expanded coverage and sophistication, gaps remain in other areas:

- Only 5 percent of reviewed reports have some coverage of climate considerations. With the development and dissemination of new tools and analytical approaches, and the importance of climate change on the international development agenda, treatment is expected to expand sharply.
- Integrated HCP analyses are nascent in PERs.
- Very few PERs include gender analyses.
- Discussion of MFD is limited, with fewer than 5 percent of PERs addressing the topic.
- Political economy issues are central to making progress on policy and institutional reforms, but less than 10 percent of PERs include related analysis.
- Coverage of SDGs was also limited, particularly among fiscally centered PERs.

The analysis revealed that most of the above thematic areas are heavily under-represented. As Figure 42 documents, the most covered area is tax expenditures with almost half of FC PERs presenting some analysis (but rare in TO PERs). Infrastructure has decent coverage but very different analysis between FC and TO (INFRASAP) while the least represented area is HCP which is a recently introduced approach. Discussion of MFD is also limited with less than 5 percent coverage of reports. Coverage of political economy aspects is also very limited with around 10 percent coverage, with good examples from Ghana 2017 (pay reform), Somalia 2020 and Indonesia 2020. Coverage of SDGs was also limited particularly among FC PERs. Best examples include water PERs in Bangladesh, Myanmar and Nepal. Finally, less than 5 percent of reviewed reports have some coverage of climate considerations. Good examples include Mali SP, Morocco 2020 and St Maarten 2020.

Figure 42: Breakdown of PERs by new frontier area and type



Source: 2020 Stocktaking, World Bank's calculations.

In an effort to guide and inspire future efforts, this section provides links to main resources as well as more detail on select sectors and thematic areas. Section 6.1 provides a summary view of typical analytical perspective in key sectors, drawing from both the stocktaking and existing guidelines. Section 6.2 provides a discussion on how climate-informed aspects can be mainstreamed in core diagnostics, amid a growing recognition about the central role of fiscal policy on the challenging climate agenda. Section 6.3 presents a discussion on the integration of human capital perspectives in PERs, with a guidance note under way, while Section 6.4 provides guidance on the integration of gender considerations in PERs, consistent with IDA

commitments. Deeper dive guidance notes, to be prepared together with colleagues in regions, across GPs and the Development Research Group, are envisaged.

6.1 Sectoral perspectives

In this section we summarize typical analytical approaches and perspectives in key sectors, drawing from both the stocktaking and existing guidelines. In particular, we examine analytical approaches in health, education, agriculture and infrastructure sectors along with references to new emerging areas such as judiciary and security.

a. Education

Education is the most covered sector in PERs. Within the universe of reviewed PERs, 25 percent of FC PERs had a dedicated chapter on education along with 26 percent of TO PERs. While differing in scope and content, most analysis revolved around issues of equity and efficiency complemented by a sector overview and a discussion of main PFM challenges.

The 2017 World Bank report, “Education Public Expenditure Review Guidelines” contains a comprehensive discussion on how to conduct a PER in education. The report presents a range of steps involved in the implementation of an education PER from preparation to dissemination of results, along with main analytical approaches complemented by country examples⁴³. The foundational approach to education should envision assessing the country’s education finance system, financial accountability, adequacy, sustainability, efficiency and effectiveness, and equity of education financing. The recently completed 2021 Iraq HD PER provide a good example of the implementation of this approach, analyzing the current context of education sector with a view toward adequacy, equity, and efficiency of public and private expenditures on education, as well as the institutional and public financial-management aspects of the education sector.

In particular we emphasize the following guidelines for each pillar:

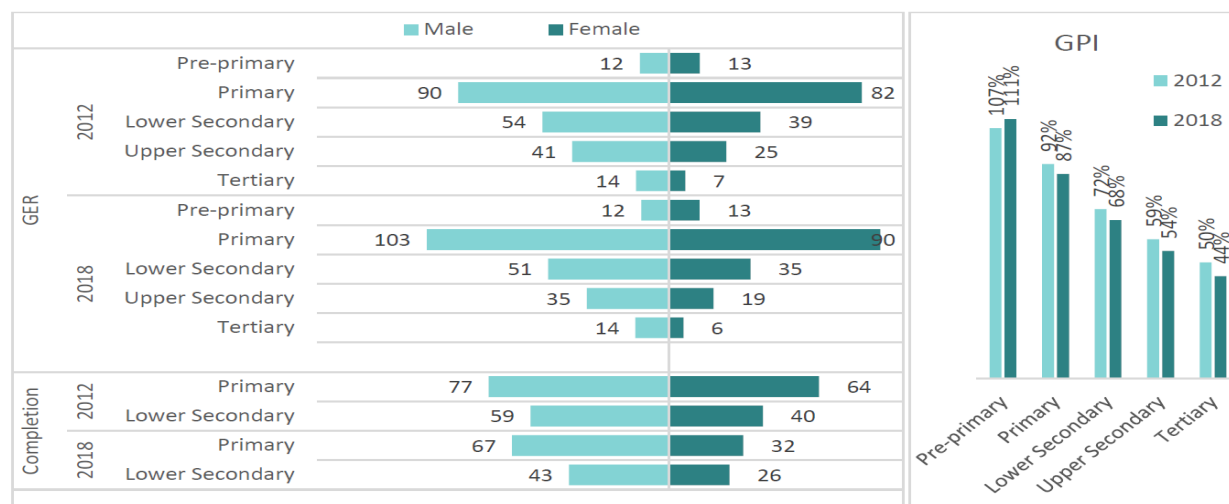
Sector overview should discuss existing strategic framework and objectives along with key achievements and challenges. Guinea (2021) for instance documents main challenges of the sector including limited access to post-primary education; large numbers of out-of-school children, mostly from poorer families, rural areas, and girls; large numbers of youth who do not have any formal education, representing a direct concern for the future human capital stock of the country and their potential livelihoods; high disparities in access to education across geographic areas and socioeconomic groups; inadequate and inefficient financial resources; and low learning outcomes, especially for children from disadvantaged backgrounds. The section should also provide basic education statistics – ideally in comparable perspective – such as number of schools by level, location, and type; number of students in school by level, gender, school location, and school type; net or gross enrollment rates by level; number of out-of-school children, dropout and repetition rates by grade, ideally disaggregated across student population characteristics (gender, income, etc.). Gender parity for instance was discussed in Guinea 2021 PER documenting growing disparities in enrollment rates (Figure 45) In addition, it should explore key financing and spending patterns, identifying who finances education and how funds are channeled and disaggregate public spending by functional, economic, administrative and program classifications, when available.

Financial accountability analysis should assess how effective public financial management systems are in contributing to providing better inputs and improving accountability. Widely available reports such as PEFA, SABER School Finance, Public Expenditure Tracking Surveys (PETS), Quantitative Service Delivery Surveys (QSDS), and Service Delivery Indicators (SDI) surveys provide ample documentation on PFM challenges in the sector. This analysis can reveal important insights into waste, leakages and ineffectiveness

⁴³ Report available at <https://openknowledge.worldbank.org/handle/10986/27264>

of public spending in education as discussed for instance in St Maarten 2010 documenting teachers and student absenteeism rates, along with other internal efficiency indicators.

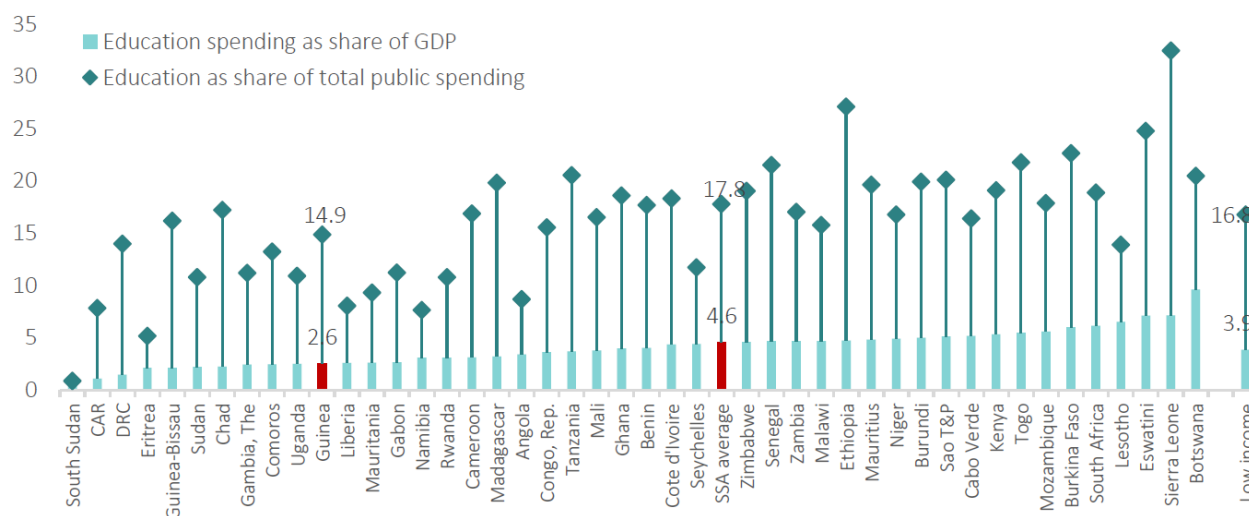
Figure 43: Guinea (2022) - Gender disparities in access to education



Source: Guinea PER 2022.

Adequacy and sustainability analysis help assess how much assistance is needed now and what can be afforded in the medium and long term. Beyond using existing international benchmarks, the issue of adequacy can be addressed by looking at unit costs per student, teacher compensation, quality of infrastructure, and the ratios of students to teachers (including by gender), to classrooms, or to textbooks, including examining both international and subnational benchmarking. Comparing the projected costs of education with the anticipated size of the resource envelope can shed light on whether planned spending levels are realistic. A comparison of Guinea's spending share in education against regional peers and international standards reveals substantial underfunding in the sector (Figure 44)

Figure 44: Guinea (2022) Share of education spending to total expenditures

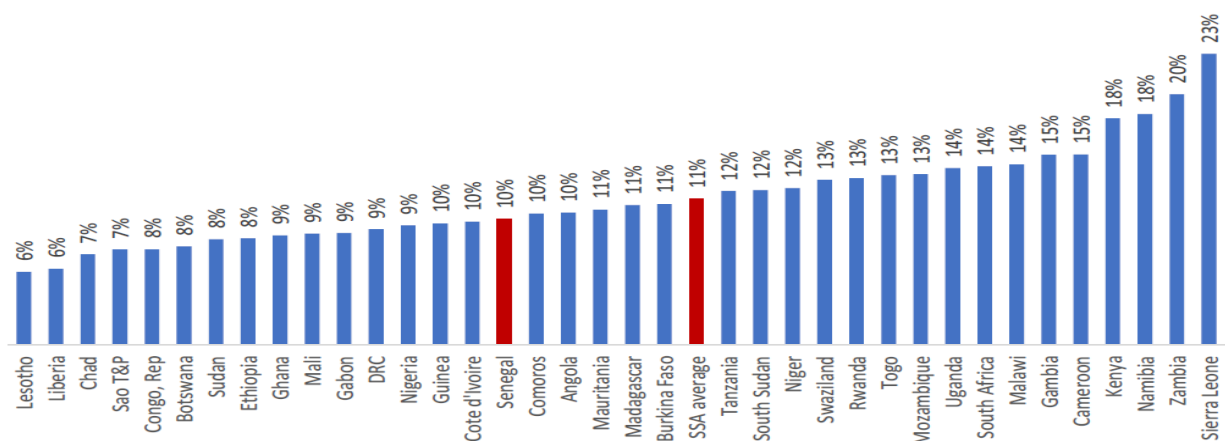


Source: Guinea PER 2022.

Efficiency analysis implies looking at issues of allocative efficiency, technical efficiency, internal efficiency, and external efficiency. Allocative efficiency examines whether the sector is allocating its

resources among subsectors so as to maximize social welfare and can be assessed by comparing de facto indicators with country standards and within region discrepancies. This might include the inefficient composition of spending with too much spending on wages crowding out other expenditures (Egypt 2019, Zimbabwe 2017), inefficient allocation of resources driven by incrementalism rather than economic and demographic needs (Kosovo 2015), inadequate funding of primary and secondary versus tertiary education and options for school network optimization (Trinidad and Tobago 2019, Ukraine 2017). Technical efficiency looks at unit costs and the relation between inputs and outputs through DEA and SFA approaches. Examples of this approach include Costa Rica 2018, Iraq 2021, North Macedonia 2019, and Mali 2019, among others. Internal efficiency looks at issues of drop out and repetition rates to examine the level of spending inefficiency in getting children to complete their educational cycle. Senegal 2021 uses this analysis to estimate the cost of school repetition and dropout rates equivalent to losses to the public sector and households of CFAF 17.3 billion and 19.9 billion, respectively. Similar analysis was implemented in Guinea 2022, Guinea Bissau 2018, Malawi 2020, and Niger 2021. Finally, external efficiency measures the returns to individuals, employers, and the country of public and private investments in education. As a practical application of this analysis, in Senegal the rate of return associated with an additional one year of education is shown to be below the Sub-Saharan Africa average (Figure 45), while Iraq 2021 compares country performance in terms of per pupil expenditure per actual learning-adjusted school years (Figure 46). Similar analysis was presented in Guinea 2022.

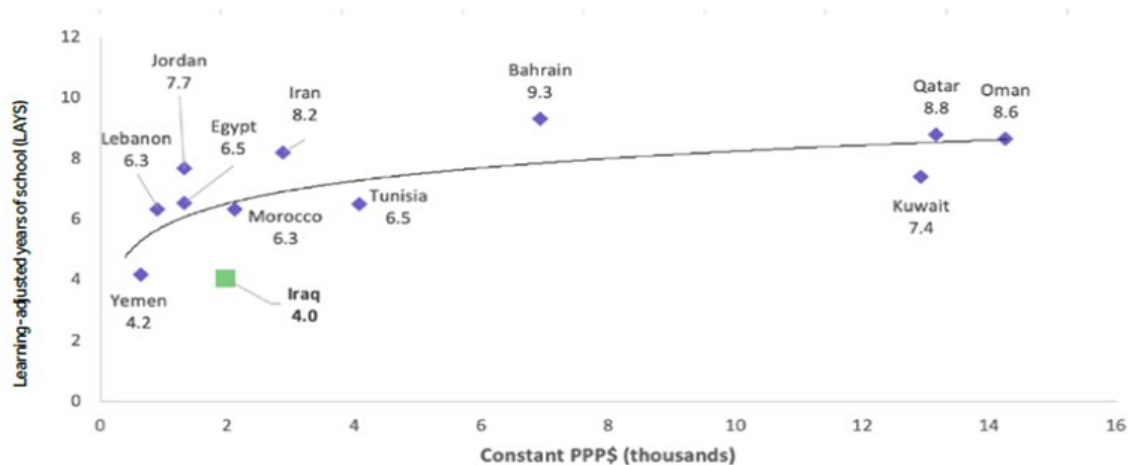
Figure 45: Average incremental earning associated to additional year of schooling in Sub-Saharan African countries



Source: Senegal (2021).

Note: SSA = Sub-Saharan Africa.

Figure 46: Per pupil expenditure and learning adjusted years of schools, Iraq 2021



Source: Iraq (2021).

Finally, equity analysis assesses disparities in access to education, completion, and learning achievements across different student groups to indicate existing problems of inequity in education. Equity in education financing can be assessed in terms of two main principles: horizontal (that is, ensuring similar per pupil spending across regions, schools, or both) and vertical equity via targeted support programs for the most vulnerable students. Examples of these approaches include use of benefit incidence analysis (Indonesia 2020, Malawi 2020, Senegal 2021), CEQ analysis (Egypt 2019, Montenegro 2019 Sri Lanka 2019), or discussion of regional disparities and per pupil allocations.

b. Health

Health is the second most covered sector in PERs. Since 2015, 21 percent of FC PERs had a dedicated chapter on health, as did 16 percent of TO PERs. While differing in scope and content, most analysis of health has been similar to that of education and revolved around issues of equity and efficiency complemented by an overview of sector context and performance and a discussion of the main PFM challenges.

Unlike for the education sector, existing guidelines on analytical approaches in health are quite outdated.⁴⁴ Typically, PERs with a health lens have focused on the following pillars: (1) analysis of sector context presenting an overview of the health system, along with discussion of main outcomes and challenges, often in a comparative perspective; (2) examination of the service delivery structure; (3) assessment of the financing context with a focus on discussing the level and composition of government expenditures versus household OOP expenditures as well as the role of donors; and d) equity and efficiency considerations. Each of these are briefly discussed in this section with references to analysis undertaken in select PERs.

The sector overview should discuss the existing strategic framework and objectives along with an illustration of the main achievements and challenges. The section should also provide basic health statistics such as child and maternal mortality rates, immunization rates, and life expectancy, ideally in a comparable perspective.

Service delivery analysis should address a range of inputs, including distribution of health care facilities, infrastructure quality, capacity, access, utilization rates, and quality of care. This analysis should be complemented by an analysis of governance and PFM bottlenecks along with human resources management considerations. Guinea 2022, for instance, reveals inadequate staffing levels when compared to regional peers (Figure 47), while analysis in Saudi Arabia 2019 reveals significant idle bed capacity across hospitals that results in overstaffing per occupied bed and, therefore, operations at low efficiency levels. Budget planning and execution challenges—particularly procurement of pharmaceuticals—are often analyzed to identify drivers of inefficiency in the sector. as documented in Guinea 2022 and Iraq 2021.

Figure 47: Guinea (2022) - Availability of health personnel

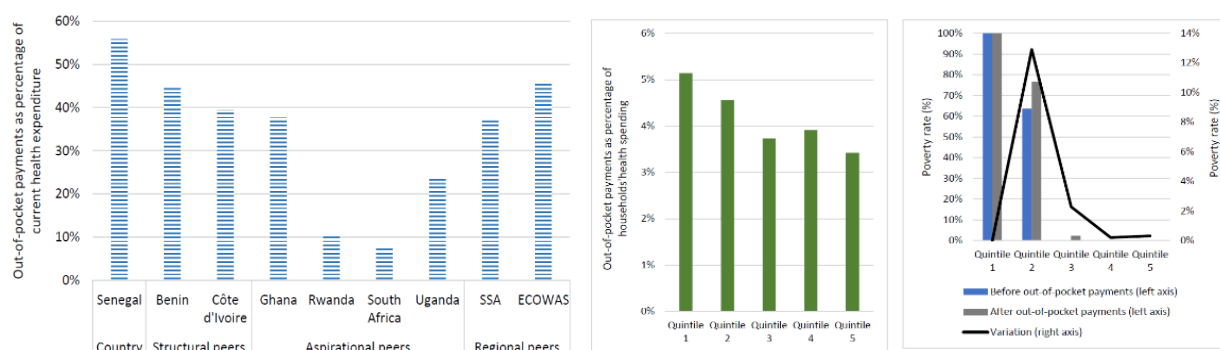


Source: Guinea (2022).

⁴⁴ New guidelines are currently under way and will be published on the PER support platform resource page once available.

Financing analysis should explore key spending patterns, identifying trends, profile, and source of funding. As is commonly found across developing countries, the lack of health insurance schemes and typical underfunding by the government lead to high OOP expenditures that often account for most than 50 percent of total health spending, rendering the system highly inequitable and ineffective. This is the case, for instance, in Senegal 2021, which documents how current allocations (accounting for 5 percent of total expenditures) for the sector remain well below the Abuja Declaration commitment targets of 2001 (15 percent), resulting in the highest OOP payments in the region (60 percent of total health spending), which are highly regressive as documented in Figure 48. Similarly, Iraq 2021 demonstrates the way that the absence of contributory schemes results in a high share of OOP expenditures, with the potential launch of financial protection schemes expected to change this landscape.

Figure 48: Levels and regressivity of OOP expenditures



Source: Senegal (2021).

Efficiency analysis implies looking at issues of allocative efficiency, technical efficiency, internal efficiency, and external efficiency. Allocative efficiency examines whether the sector is allocating its resources among subsectors so as to maximize social welfare; it can be assessed by comparing de facto indicators with country standards and within-region discrepancies. This might include inefficient composition of spending with too much spending on wages crowding out other expenditures, or inefficient allocation of resources driven by incrementalism rather than by economic and demographic needs. For example, Sri Lanka 2020 documents that health resource allocations are input based, which implies that funding is not based on need but rather on staff location. Allocative efficiency can also be examined by assessing the level of funding of primary and secondary (preventive) health services versus tertiary and quaternary (curative) health services. Senegal 2021 estimates funding to tertiary and quaternary to exceed 70 percent of total spending with less than 6 percent devolved to primary and secondary functions; similar findings are presented in Cameroon 2018, Ghana 2017, and Iraq 2021, among others. Technical efficiency looks at unit costs and the relation between inputs and outputs through DEA and stochastic frontier approaches, with Guinea 2022 offering a good example of both of these approaches. When extended to the facility level, this analysis has helped identify options for hospital network optimization as in Belarus 2021.

Examining spending efficiency in health subsectors would help rationalize spending as necessary. In this respect, internal efficiency analysis helps with evaluating the efficiency of the subfunction within the sector, while external efficiency analysis focuses on the performance of the sector within the public sector and the economy. Internal and external efficiency analysis should be carefully balanced. For instance, internal efficiency rates are much higher in hospitalization than in primary care, but overall costs are also higher. While internal efficiency considerations would lead to increased spending on curative services, the resulting external efficiency of the entire sector may then be lower because poor preventive medicine efficiency and increased hospitalizations mean higher costs for the sector, increased absenteeism in the economy, and poor productivity due to morbidity and disease outbreaks.

Finally, equity analysis assesses disparities in access to health services across different income groups and other characteristics (such as gender). Equity can be assessed by looking at health-related resource

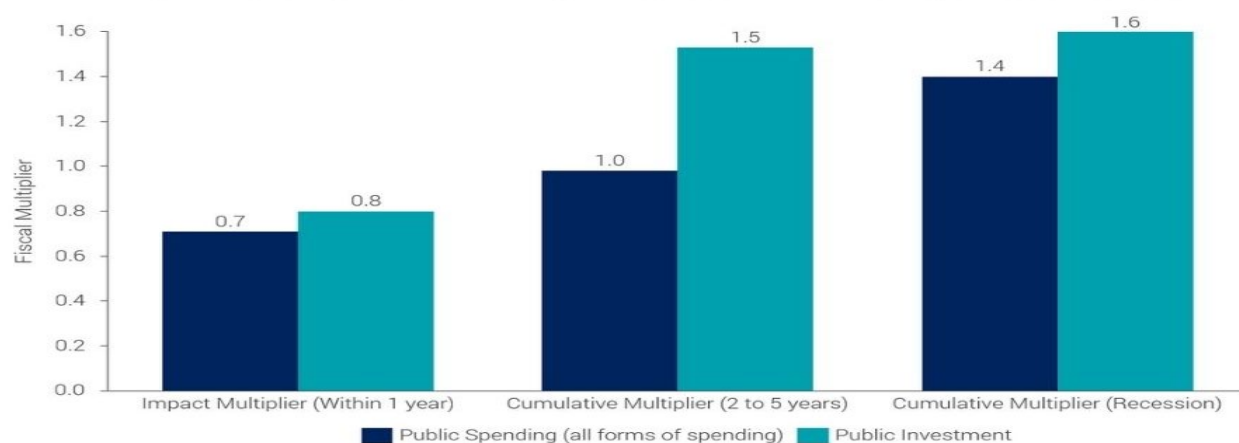
allocations across geographic areas. As documented in Senegal 2021, for example, the density of medical personnel and universal health insurance financing is unevenly distributed across regions and heavily favors the capital, leading in turn to similar uneven distribution of outpatient care. Equity can also be assessed by quantifying the impact of health OOP expenditures on household income (Senegal 2021), or by using benefit incidence or CEQ analysis.

c. Infrastructure

Investments in infrastructure are critical for economic growth and improvement in service delivery.

As evidenced in several studies and in Figure 49, infrastructure spending exhibits higher economic multipliers than other types of spending do. Beyond economic growth, infrastructure investments have also been found to be critical to the attainment of SDG goals (Thacker et al. 2019) and poverty alleviation (Brenneman and Kerf 2002, Songco 2002). Investments in infrastructure in low-income and middle-income countries, however, are plagued by inadequate levels and procyclicality (Foster et al. 2022), poor planning and implementation processes leading to cost overruns and time delays and limiting their overall impact, thus requiring careful examination of these dynamics. Similarly, the large role played by quasi-fiscal activities and off-budget operations (PPPs, SOEs, and so on) make this analysis more challenging given the paucity of available data.

Figure 49: Estimates of fiscal multipliers for public investments vs. other types of spending



Source: Global Infrastructure Hub, <https://www.gihub.org/infrastructure-monitor/insights/fiscal-multiplier-effect-of-infrastructure-investment/>.

Despite its relevance, coverage of infrastructure in traditional PERs is quite limited. Less than 10 percent of PERs present a dedicated chapter while an additional 20 percent include a sufficient discussion of infrastructure's dynamics, often in the context of a broader analysis of the efficiency of capital investments. INFRA-SAP appears to be the preferred instrument of choice to examine the quality of public finance in infrastructure with over 30 reports already produced on the basis of this novel diagnostic and many pilots under way using the recently released INFRA-SAP 2.0 methodology⁴⁵ to integrate public finance, governance, and connectivity considerations, as described in more detail as follows:

- **Connectivity:** This pillar examines the physical performance of the infrastructure sectors in delivering the services needed to meet a country's social and economic goals. Separate connectivity modules have been developed for each infrastructure sector (energy, digital, transport), but broadly they all consider different dimensions of infrastructure performance, encompassing accessibility, affordability, quality, and sustainability.
- **Finance:** This pillar examines how infrastructure is funded and financed, including both public and private sources of finance, at both the domestic and international level.

⁴⁵ Available internally at <https://worldbankgroup.sharepoint.com/sites/ppp/isap/Pages/index.aspx>.

- **Governance:** This pillar examines the institutional and regulatory organization of the sector, distinguishing between the governance framework surrounding the project cycle for infrastructure investment and the governance surrounding service delivery. The latter is affected by market structure and competitive dynamics, as well as governance arrangements for SOEs where these play a role as service providers.

Additional resources in support of infrastructure analysis include the Infrastructure Global Indicators Dashboard and the Global Infrastructure Map. The former resource is a comprehensive compilation of more than 250 infrastructure indicators that allows for rapid identification of country performance indicators and trends to facilitate benchmarking against different peer groups. The latter resource is a web platform that provides a growing number of geospatial information layers that permit customized visualizations of a wide variety of infrastructure characteristics and related geographical contextual factors. For more information, consult the INFRA-SAP 2.0 methodological note.

In addition to INFRA-SAP, additional guidelines have been developed to look specifically at the water sector comprising water/sanitation, irrigation, and hydropower.⁴⁶ The guidelines reflect recent efforts to elevate the profile of public finance dynamics in the water sector, drawing from a growing list of PERs with a focus on water and sanitation, irrigation and hydropower. The guidance note is available on the PER support platform resource page.

d. Agriculture

Agriculture is the third most covered sector in PERs. Within the universe of reviewed PERs, 5 percent of FC PERs have a dedicated chapter on agriculture along with 12 percent of TO PERs. Within these, coverage typically includes crops, livestock, forestry/hunting, fisheries, and irrigation.

A 2011 World Bank Report⁴⁷ provides a toolkit for analyzing public expenditures in agriculture. More specifically, the toolkit has two goals: to provide checklists for practitioners conducting various kinds of agriculture public expenditure analyses, and to provide selected examples on aspects of the checklist to help guide analysis. Coverage of the agricultural sector in PERs would typically include an overview of sector objectives, strategy, characteristics, and trends; analysis of level, composition, execution rates, and adequacy of budgetary allocations along with sources of funding; efficiency, value for money, and equity considerations including subsidies; and impact of climate change, particularly on land degradation.

The sector overview typically discusses existing policies, institutional arrangements, and the strategic framework underpinning sector performance. It also should present an overview of recent efforts and reforms along with recent developments, including a discussion of key drivers (resource versus productivity gains) and impediments to growth. Myanmar 2017 agriculture PER, for instance, documents the ways that the country underperforms in creating an enabling environment for business in agriculture by underproviding public goods critical to the sector.

The financing analysis should explore key spending patterns and identify trends, profiles, and sources of funding. Common patterns identified across developing countries include wide fluctuations in recurrent and capital expenditures along with low execution rates, often due to the prevalence of donor funding in the sector and leading to inadequate and unpredictable levels of funding. This was the case discussed in Mozambique 2017, which documents misalignment between donor and government procurement procedures that results in higher budget unpredictability. In this respect, a World Bank review of several PERs in the Sub-

⁴⁶ In addition, a guidance note was issued in 2012: [Public Expenditure Review from the Perspective of the Water and Sanitation Sector \(World Bank, 2012\)](#).

⁴⁷ Practitioners' Toolkit for Agriculture Public Expenditure Analysis, World Bank, 2011.

Saharan Africa region⁴⁸ revealed persistent challenges in meeting the 10 percent target of public spending dedicated to agriculture set through the Malabo/Maputo Declaration.

Efficiency in agriculture can be measured through a variety of approaches. Allocative efficiency can be measured by assessing whether resources are allocated to the programs with the highest payoffs and with the right mix of inputs or by comparing the alignment of spending with government policies. The analysis could also look at the extent to which public funds are devolved to private sector financing (for fertilizers and other farm inputs) as opposed to provision of public goods such as extension services. Myanmar 2017 agriculture PER, for example, documents current allocations that fail to stimulate economic growth, particularly with underfunding of agriculture research and climate-smart agricultural programs and inadequate funding of O&M. Under-spending in O&M is a typical shortcoming in agriculture as also noted in Mink 2015 in a review of Sub-Saharan Africa PERs. Technical efficiency can be measured by using instruments such as PETS to measure the extent of public resources reaching beneficiaries or examining scope for reducing the unit costs of agriculture services.

Analysis of subsidies carry important ramification in terms of equity, effectiveness, and value for money considerations. As documented in the agriculture toolkit, significant efficiency gains can be extracted by switching focus of public expenditures away from subsidies to private sector and toward public goods. In many low-income countries, the size of subsidies remains elevated; for instance, Sierra Leone 2021 assesses subsidies to account for almost 50 percent of the total agriculture budget. Furthermore, subsidies are often regressive as they tend to support the largest producers and are often vulnerable to poor targeting. Senegal 2021 documents the challenges faced in the distribution of subsidized inputs in terms of targeting and eligibility criteria along with weak budget discipline and significant overruns that lead to reduced effectiveness of such interventions.

Analysis of revenue generation might be appropriate to assess the sustainability and affordability of agriculture financing. This is particularly relevant in the case of irrigation, where user fees are applied to cover government support but are typically insufficient to cover even basic operational costs. As documented in Myanmar 2017 agriculture PER, irrigation has relatively low fiscal revenue generation capacity, insufficient to support needed investments in the sector, which increases the country's reliance on foreign assistance.

Finally, an analysis of agricultural productivity cannot neglect an assessment of climate change impact on land degradation and its effect on future growth. In this respect, Senegal 2021 estimates the annual cost of land degradation at 9 percent of GDP, while Myanmar 2017 agriculture PER focuses extensively on climate-smart agriculture and the vulnerability to climate disasters as measured by the global climate risk index. Vulnerability to climate-induced natural disasters is further analyzed in Myanmar 2020 water PER, which documents how the irrigation sector is a high recipient of contingency funds, typically for flood management purposes.

e. Social protection

An analysis of social protection spending is a critical building block to fulfill a PER core objective of examining the equity and effectiveness of public spending. Its analysis is particularly relevant to assess the degree to which the government is achieving goals of equity, inclusion, poverty alleviation, and shared prosperity, while protecting pro-poor spending. Social protection can be broadly defined as the set of programs that provide financial or in-kind assistance to individuals and households, especially poor and vulnerable people. These in turn can be broadly categorized as contributory social insurance programs (pensions and unemployment and health insurance), noncontributory social assistance/safety net programs (cash transfers, scholarships, health subsidies, and school feeding), and labor market schemes (skills training, economic inclusion, and employment support programs). Given the diversity and range of approaches across countries, it is important for a PER to identify relevant categories along with individual programs and activities within

⁴⁸ S. Mink, "Synthesis Review of Agricultural Public Expenditure Reviews Conducted in African Countries" (World Bank, Washington, DC, 2015).

them. Around 15 percent of all PERs reviewed from 2015 onward have a dedicated chapter (or full PER) on social protection, with additional PERs offering some references on levels and targeting of such spending within broader discussions of expenditure trends and composition.

A 2001 World Bank seminal paper⁴⁹ presents the most comprehensive toolkit to date on analytical guidelines on social protection. The aim of this paper is to provide some guidance on how to carry out a social protection (SP) expenditure, performance, and finance review, measuring level, accuracy, targeting, adequacy, sustainability, efficiency, and equity while disaggregating analysis by main categories (disability, labor, pensions social insurance, safety nets, social funds, and so on).

Identification of expenditures can be challenging. Social protection schemes are typically administered by a range of administrative units which might vary considerably across countries. Even when available, social protection functions might not properly include all social protection schemes as social protection programs (for example, school meals and labor market programs). As documented in Nepal 2021 social protection PER, there could be significant differences in total amounts depending on whether economic or functional classification is used to identify total spending.

Fiscal analysis focuses on trends and profiles of social protection. At a minimum, this would envision examining the evolution of total spending over time disaggregated across main categories and financing sources. The analysis should be complemented by deep dives across each category and individual programs, along with identification of beneficiaries and the poverty impact of each program. Sierra Leone 2021, for instance, documents recent efforts by the government to increase coverage of its core social assistance program, which doubled in size although still with limited resources, while also emphasizing heavy dependence on foreign aid. The analysis would also benefit from an examination of budget execution rates with a view toward identifying bottlenecks to program implementation.

Social insurance typically captures the majority of social protection, primarily dominated by large pension payments. As documented in several PERs, pensions tend to dominate spending in social protection while typically crowding out other pro-poor expenditures. In Nepal for instance, pensions account for the majority of total spending in SP while benefiting only 1 percent of the population, mostly in urban areas (Figure 52). Analysis of pensions can also look at the fiscal sustainability of existing schemes and potential measures to redress systemic imbalances through PROST and other simulation tools. Analysis of existing social health insurance should discuss targeting, equity adequacy, and sustainability as the absence of these schemes in low-income countries, combined with low per capita public spending levels, often leads to high OOP household expenditures, particularly in instances of catastrophic events. In this respect, Iraq 2021 discusses key reform to scale up well-targeted financial protection schemes in order to ensure health care affordability with concerns around its feasibility. Finally, the existence and funding of unemployment insurance schemes is rare, even though, as demonstrated in higher-income countries, their implementation might act as important automatic stabilizers to ensure a better countercyclical fiscal stance in times of need.

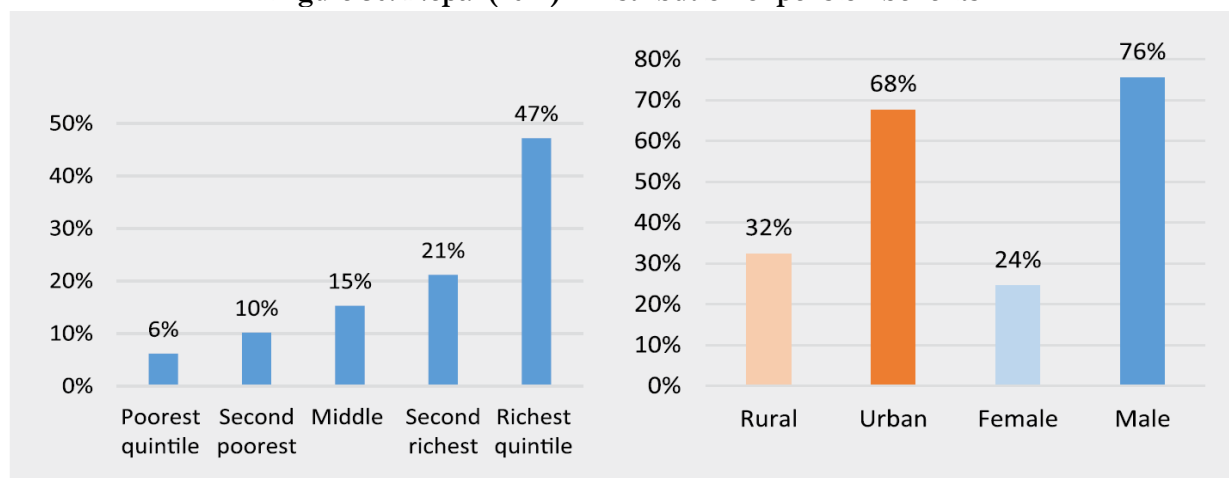
Analysis of safety nets/noncontributory social assistance should center around three core pillars. First, it should document relative spending levels for each component. Second, it should examine coverage of each scheme by identifying beneficiary profiles and the overall equity impact of each scheme. Finally, the analysis should include an examination of the adequacy of benefits along with potential measures to increase the impact and equity of these interventions. Indonesia 2020 documents how several rounds of reduction in expensive and untargeted subsidies (from 16 percent to 6 percent of total spending) were implemented, with some of the expenditure savings reallocated to means-tested social assistance programs, increasing targeting and the allocative efficiency of social assistance spending. Similarly, Belarus 2021 offers an analysis of poor targeting of the social safety net, indicating a need to improve formulas and move away from universal subsidies. A range

⁴⁹ Roy S Canagarajah, Karin Hitzmann, and Robert Holzmann, “Guidance for the Conduct of a Social Protection Expenditure, Performance and Finance Review (English)” (Social Protection discussion paper series, World Bank Group, Washington, DC; 2001).

of other PERs (Cameroon 2019, Moldova 2016, Montenegro 2019, Trinidad and Tobago 2019, and Zimbabwe 2017, among others) also document how the targeting of most social assistance programs is not progressive.

Finally, the analysis would include an examination of labor market support policies. These programs might include unemployment support, training activities, and special programs for entrepreneurs. As documented in various PERs (Nepal 2021), these schemes typically represent only a trivial share of social protection spending despite their obvious merit in encouraging employment and entrepreneurship.

Figure 50: Nepal (2021) - Distribution of pension benefits



Source: Nepal Social protection PER (2021).

A critical analytical perspective revolves around the assessment of impact/effectiveness of social protection spending. This can be measured by the extent to which SP programs effectively target most vulnerable households, and pro-poor spending is protected in cyclical downturns. In doing so it is important to consider multidimensional poverty rates to account not only for monetary poverty but also deprivation in dimensions such as health, education, or living standards. In this regard, use of means-tested methodologies is more likely to be effective than untargeted universal subsidies which are often regressive in nature, as discussed in Indonesia 2020, Ukraine 2019 and Belarus 2021 among others. Political economy considerations might also play an important role, as discussed in Kosovo 2015 documenting how social protection schemes driven by policy choices rather than economic needs. Effectiveness can also be examined by estimating the impact of specific social protection schemes. Nepal 2021 PER for instance documented limited evidence of a correlation between the scholarships and midday meals program budget and the performance of students and schools. On the other hand, Sierra Leone 2021 PER documented how the core cash transfer program is pro-poor and has a significant impact on lowering extreme poverty rates, using simulations from recent household survey.

Unlike other sectors, technical efficiency analysis is less relevant for social protection. Allocative efficiency considerations however can be explored to measure the responsiveness of per capita budget allocations to economic needs across the range of multi-dimensional deprivation indices and social protection schemes. Regression analysis conducted to estimate the determinants of per capita program budget allocations in Nepal for instance revealed a pattern of favoritism towards better (and wealthier) performing geographic areas

Finally, assessing the policy and institutional framework for design and delivery of social protection programs can offer important insights into determinants of equity and efficiency outcomes. The analysis should examine the quality of budget processes highlighting main challenges and bottlenecks as related to institutions, systems, and capacities. Of particular relevance is the issue of institutional fragmentation and reliable social registry for proper targeting along with quality of payment systems. In decentralized contexts, the analysis should also assess the quality of underlying policy and legal framework to clarify roles and responsibilities across tiers, as discussed in Nepal 2021 PER, as well as assess the capacity of subnational entities

in providing adequate financing, when applicable. Several PERs (Egypt 2019, Gabon 2018, Haiti 2015) document significant regional disparities in distribution of social assistance programs, covering only a small fraction of the poor and bias towards the capital and fragmentation in their design and implementation.

f. Other sectors with limited coverage in PERs

While still limited in scale, the issue of public safety warrants increased attention in many cases, especially in contexts of fragility, conflict and violence. Around 8 percent of PERs included some analysis of public finance considerations vis-a-vis security with additional 2 percent focused on the judiciary sector. To increase the profile of public finance in this critical sector, new guidelines have recently been published, highlighting the role played by public finance in the delivery of security and criminal justice services.⁵⁰ The note seeks to strengthen policy and operational dialogue on security sector issues by providing national and international stakeholders with key information on security expenditure policy and management. This has resulted in an increasing number of PER covering the issues of efficiency in public safety (police), such as The Gambia (2020). At the same time, a number of PERs – such as Liberia (2019), Moldova and Ukraine JSPEIRs – have implemented a comprehensive and structured approach to PFM and expenditure analysis in the judiciary sector.

Finally, analysis of nutrition issues in PERs is very limited. Less than 5 percent of PERs included any analysis of public finance considerations vis-a-vis nutrition. Access to spending and performance data on nutrition is also quite limited often lacking inter-operability⁵¹. To increase the profile of public finance in this critical sector, new guidelines have recently been published.⁵² The guidance note establishes key elements of a nutrition PER including profile and trend analysis as well as equity, efficiency and effectiveness considerations.

6.2 Towards Climate-Informed PERs

PERs need to engage with climate change because of mounting fiscal impacts from climate change, and the critical role of fiscal policy in supporting mitigation, enhancing adaptation, and building resilience, all while promoting robust development trajectories. As noted above, only a few PERs to date have included a core focus on climate change, but with increasingly salient motivation, there is a growing recognition about the central role of fiscal policy on the challenging climate agenda, and PERs need to be fully climate informed, based on sound evidence and impactful tools.

a. Highlights of climate change in PERs to date

As noted above, the focus on climate change in PERs has generally been limited. PERs that have placed climate change at the center of the analysis include Vietnam 2015 and Indonesia 2017, while others, such as the Dominican Republic 2019 and Niger 2020, note climate change issues as part of their wider narrative.

The pathbreaking Vietnam 2015 climate change PER is titled: “Financing Vietnam’s Response to Climate Change: Smart Investment for a Sustainable Future - Laying the foundation for resilient low-carbon development through the Climate Public Expenditure and Investment Review (CPEIR).” It was built on strong client demand from the Government of Vietnam (GoV) and prepared in collaboration with the United Nations Development Program. Recognizing that climate impacts are intensifying and that on the existing trajectory Vietnam would become a major greenhouse gas emitter, the GoV felt the time was right to

⁵⁰ Harborne, Bernard; Dorotinsky, William; Bisca, Paul M.. 2017. Securing Development: Public Finance and the Security Sector. Washington, DC: World Bank. © World Bank.

⁵¹ A recent reform in Indonesia focuses on connecting various financial and performance monitoring data systems to make it easier to simultaneously track spending and outputs, enabling stock taking and quick evaluation by line ministries.

⁵² Wang, Huihui; Shibata Okamura, Kyoko; Subandoro, Ali Winoto; Hoberg, Yurie Tanimichi; Qureshy, Lubina; Ghimire, Mamata. 2021. A Guiding Framework for Nutrition Public Expenditure Reviews. World Bank, Washington, DC. © World Bank. <https://openknowledge.worldbank.org/handle/10986/36907> License: CC BY 3.0 IGO.”

undertake a review to protect development gains. The effort drew upon strong client commitment to addressing climate change, as reflected in extensive National Climate Change Action Plans, Green Growth Action Plans and other strategic planning initiatives, along with the establishment of a strong coordinating body in 2012.

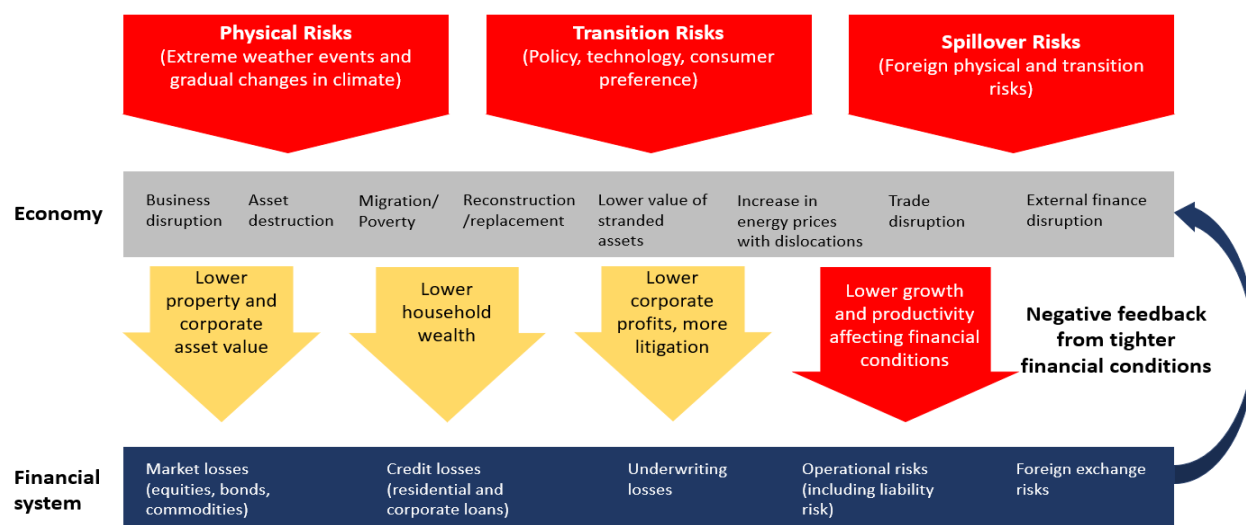
The Vietnam CPEIR developed a typology of climate change response expenditure (TCCRE) to provide a unifying framework to analyze spending and assess contributions to adaptation and mitigation. The CPEIR highlights significant progress in enhancing climate change response (with most related spending going to investment and adaptation), while noting scope for improvement in aligning the policy objectives across strategic planning documents, strengthening monitoring and reporting, and enhancing recurrent spending for mitigation as well as capacity and policy activities. The CPEIR articulates key short-, medium-, and long-term recommendations centered on: (a) developing a climate mainstreaming plan, (b) using the TCCRE to underpin monitoring, budgeting and planning, (c) strengthening the role of the central coordinating body, (d) enhancing the climate change adaptation, planning, financing, and policy implementation processes, and (e) boosting climate finance. It also includes a detailed “Results Framework for a Climate Budget and Financing Action Plan: Key Activities, Objectives and Risks.” Overall, the Vietnam CPEIR carries lessons for other PERs seeking to strengthen the focus on climate change.

The Niger PER 2020 reflects on climate change as part of the analysis of the agriculture sector, noting exposure to adverse impacts and the need for increased spending on climate smart agriculture research. The Niger PER also emphasizes protection of vulnerable groups and the imperative to enhance resilience. The Dominican Republic PER 2019 highlights the importance of generating adequate fiscal space to address vulnerability to natural disasters (especially intensifying hurricanes that are becoming more frequent), which historically have wrought damages equivalent 0.7 percent of GDP.

b. Fiscal risks and impacts of climate change

Climate change brings significant physical, transition and spillover risks that need to be addressed in PERs. Physical risk pertains to the increasing frequency and intensity of extreme weather events (droughts, floods, and tropical cyclones), higher average temperature levels (affecting wildfires and productivity, among others), and sea-level rise. Transition risks arise from sudden changes in climate related policies, consumer preferences, and technology. Spillover risks are associated with policy changes in other countries, such as a reduction in financing flows as pledges are made to cease financing fossil fuels, the establishment of carbon border adjustments in trading partners, or deepening private sector sustainability commitments. Together, these risks can manifest in numerous direct and indirect impacts on public finance (Figure 51). Only a few PERs include fiscal risk assessments and scenario analysis to climate shocks, as well as the role of climate informed fiscal policies for macro-stability, which are becoming increasingly salient.

Figure 51: Climate risks can negatively impact the economy with potential amplification effects



Source: Adapted from Grippa et al. (2019).

c. Climate-fiscal policies can be critical drivers of development outcomes

Growing evidence highlights how climate smart fiscal policies are key drivers germane to PERs in helping promote better development outcomes. These include: (i) development co-benefits from eliminating inefficient fossil fuel subsidies and getting prices right, (ii) employment, growth and equity effects, and (iii) scope for raising revenue.

A core topic of PER analysis is efficiency of public spending against clearly articulated public policy goals. Eliminating fossil fuel subsidies is a key step that also brings significant development co-benefits and helps address climate change.

The fiscal gap for fuels remains huge and is widening, driven by continued direct subsidies, as well as implicit subsidies (Figure 52). Estimates for efficient gasoline and coal prices indicate wide gaps across all regions (Figure 53, Panels A&B). While eliminating subsidies poses political economy challenges, they remain a substantial fiscal burden in many countries, in addition to increasing distortions and providing incentives for excessive fossil fuel consumption.

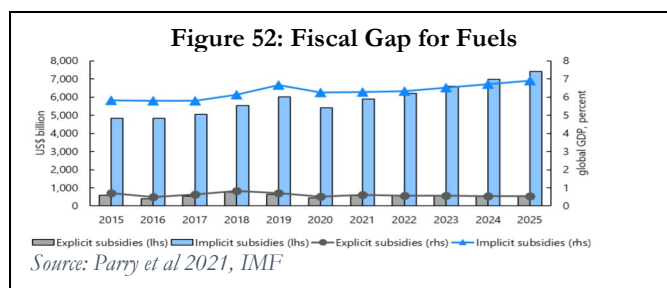
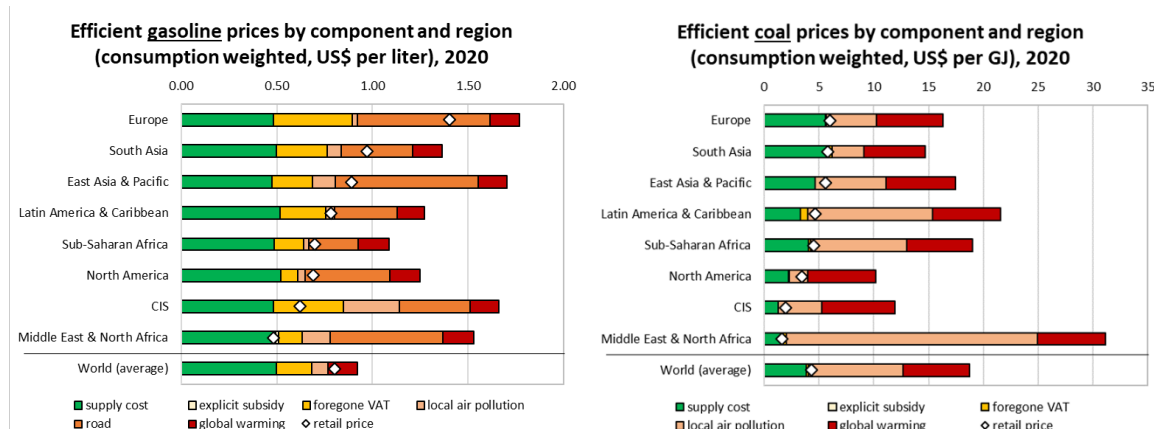


Figure 53. Efficient Fossil Fuel Energy Prices

Panel A. Gasoline

Panel B. Coal



Key development co-benefits from reducing the use of fossil fuels are lower air pollution and traffic congestion / fatalities. About 9 million people die prematurely due to pollution, of which over 90 percent are in developing countries. Morbidity in terms of respiratory problems is also severe in many countries. Decreasing the use of fossil fuels lowers associated health costs by more than the cost of mitigation in many contexts. Mounting evidence also shows that road fatalities and congestion costs also fall with higher fuel costs.

Environmental taxes, such upstream carbon taxes, support better employment, growth, and equity outcomes, compared to labor taxes. Sound tax analysis as part of PER preparation considers the direct and indirect impacts of the full range of different taxes, including environmental taxes compared with alternatives. Evidence suggests that the effect of environmental taxes on employment and growth are more benign than taxes on labor, in part because low-carbon production tends to be more labor intensive, and environmental taxes brings only a transitory contraction of growth (Figure 54 and Figure 55). A key feature of upstream taxation is that it also covers the informal sector, typically missed by other tax measures, broadening the tax base, and narrowing the formal-informal tax wedge. Fuel price reforms can generate significant and relatively stable revenue, especially important in countries seeking to rebuild fiscal buffers in the aftermath of the COVID-19 pandemic. These are central issues to be addressed in PERs.

Figure 54: Employment effects of environmental (Panel A) versus labor taxes (Panel B)

Panel A. Environmental taxes

Panel B. Labor taxes

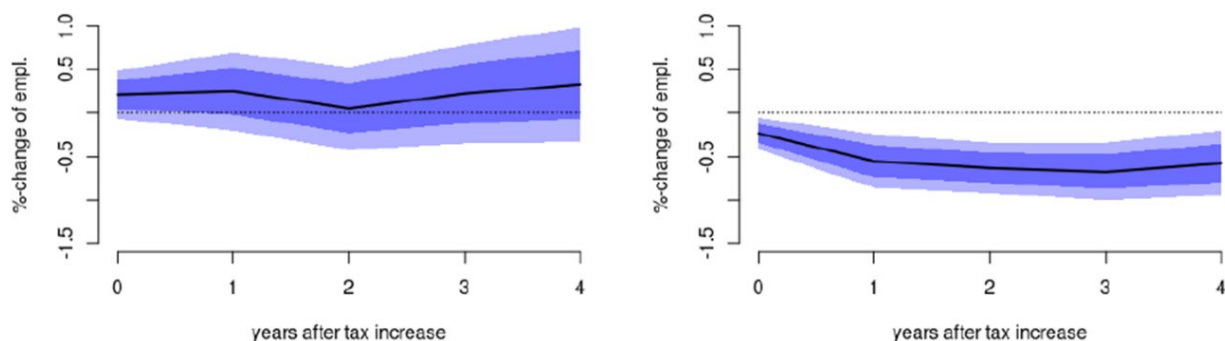
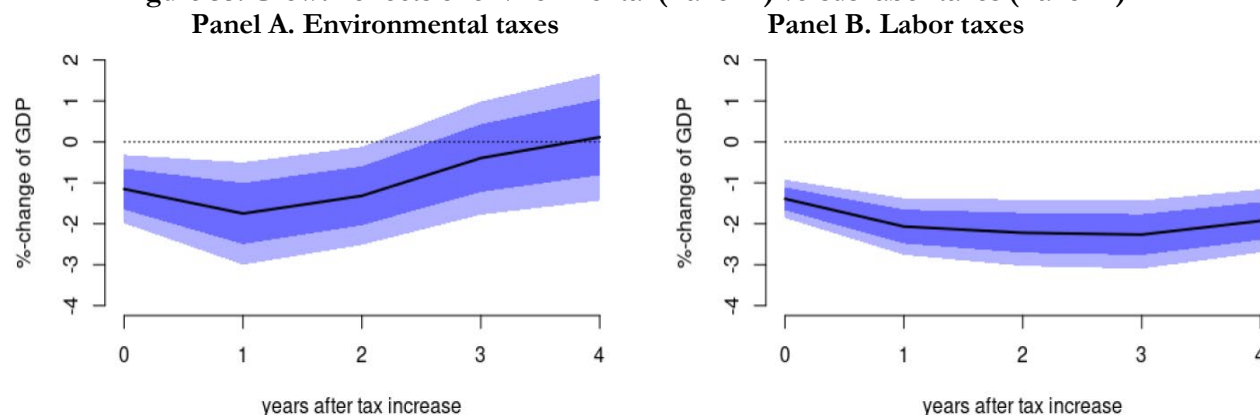


Figure 55: Growth effects of environmental (Panel A) versus labor taxes (Panel B)



d. Fiscal policies to mitigate climate change

Addressing climate change has moved to the top of the policy agenda, and fiscal policy has a central role to play in cost-effectively reducing greenhouse gas emissions. In many ways, fiscal policy sets the incentive structures that determine the trajectory of carbon emissions and development. The recognition of the role of public finance is reflected also in the establishment the Coalition of Finance Ministers for Climate Action. The Coalition’s work is organized around the six Helsinki Principles, covering many of the key topic areas of PERs, including working towards carbon pricing, and taking climate into account in macroeconomic policy, fiscal planning, budgeting, public investment management, and procurement practices. The World Bank’s Climate Change Action Plan 2021-2025 also notes the importance of taking a ‘whole of economy’ approach towards climate action, including mainstreaming carbon pricing in fiscal policy.

e. Development of tools to underpin climate informed PERs

In order for PERs to engage effectively with climate change, analytical tools need to be developed and deployed. Two key tools are being rolled out to underpin strengthened analysis: the Climate Pricing Assessment Tool (CPAT), and the Multi-Regional Input-Output (MRIO) model.

CPAT is a joint effort with the IMF and enables the estimation of multiple economic and non-economic effects of carbon pricing and fossil fuel subsidy reform. Key features include estimates of macroeconomic impacts (GDP, revenues, balance of payments), energy and emissions (energy consumption, global and local pollutants), distribution and incidence (effects of tax and expenditures across the income distribution, as well as urban versus rural), and development co-benefits (reductions in mortality and morbidity from improvements in air quality, as well as road safety and reduced congestion).

The MRIO is a forward-looking model to help quantify the employment and competitiveness impacts of climate change mitigation policies.⁵³ The structural shifts induced by tax, subsidy, and public investment changes, as well as other exogenous structural changes that arise from countries implementing their Nationally Determined Contributions, will heterogeneous effects across countries, regions, sectors, and groups of workers, depending on sectoral composition of the economy and integration in global value chains, among other factors. The tool will facilitate analysis and consideration of various policy options as part of PER engagements with the client.

6.3 Integrating Human Capital Approaches in PERs

While it is recognized that human capital accounts for about two-thirds of total wealth and is essential for inclusive growth, it has been not featured in most PERs as an integrated driver of policy. Building

⁵³ The development and roll-out of the MRIO is being undertaken as part of a sub-task of the PER Programmatic Support Platform.

and deploying human capital in conjunction with other types of capital investments, as well as leveraging a sound business climate and good governance, can help countries accelerate their economic growth, improve equity, and reduce poverty. Realizing the full potential of human capital, however, hinges on strategic public investment in cohesive human capital-building services, comprising select spending in health, sanitation, water, nutrition, education, skills development, labor markets, and social protection. It will also require considering the interplay between the megatrends (demography, climate change, and digitalization) and human capital, given that government expenditures need to impact structural transformation. PERs are an important avenue of support to achieving these goals, and integrated efforts are needed.

a. Recognizing the role of human capital

Several PERs emphasize the importance of human capital as part of the narrative, but more is needed to develop an integrated analysis. Several PERs make reference to the Human Capital Index (HCI) as part of the motivation of seeking to strengthen public spending in education, health and social protection (Guinea (2022) and most recently, Nigeria and Uzbekistan (drafts for Decision Review, 2022). The Senegal (2021) PER highlights the HCI, including across regions as part of the discussion on spending and outcome disparities in education. The subsequent discussion then focuses on key sector components contributing to human capital.

b. The role of public finance to build, protect and utilize human capital

Public human capital expenditures are substantial and difficult to manage. They account for a significant portion of government spending, but are not as visible as physical infrastructure, for example, and are often considered current spending rather than capital investments. The Kyrgyz PER 2021 notes substantial education spending, higher than the average for lower middle-income countries, that has greatly expanded access, but calls for an increased focus on quality to improve education outcomes. Human capital finance has to navigate the complexities of public administration, decentralized service delivery, procurement and distribution arrangements (such as for medical supplies), as well as ensuring integrated public investments (such as energy and digital infrastructure for schools, social protection, and health systems), to realize the benefits of human capital investments.

To contain human capital losses and support recovery from the COVID19 crisis, it is critical to ensure the sufficiency, efficiency, and long-term sustainability of public investment underpinning human capital outcomes. COVID-19 has caused massive human capital losses as well as a devastating recession, according to the joint HD-EFI 2021 report "Investing in Human Capital for a Resilient Recovery: The Role of Public Finance," with recovery prospects reliant on the virus's containment and the availability of adequate fiscal support. To prevent long-term human capital losses and promote human capital for economic recovery, high-priority strategic investments must be maintained. This includes boosting health care services for all, protecting children from hunger (which supports their cognitive development) and returning them to school, and enabling the working age population to avail of income earning opportunities as key priorities.

The COVID-19 crisis and the tightened fiscal conditions in many countries heightens the importance of ensuring adequate financing for human capital expenditures, as well as boosting efficiency and efficacy. Over 50 nations are likely to see a reduction in borrowing capacity between now and 2026, affecting 900 million people, the majority of whom live in low-income countries (LICs). As public spending capacity declines and budgets are modified, it will be critical to protect the high-impact human capital investment that generates high growth dividends and development results. Because of the tightened budgetary conditions caused by the economic fallout from the COVID19 epidemic, a persistent effort to improve public expenditure management, as well as both allocative and technical efficiency, is critical. For example, the Kyrgyz PER (2021) emphasizes the importance of containing / reducing inefficient tax expenditures to ensure adequate fiscal space for human capital spending. Landscape analysis of public expenditures that links to key human capital outcomes will support reforms and strengthen the results focus of human capital public finance.

To achieve the objectives of the IDA20 Special Theme: Human Capital, it will be critical to strengthen public finance for human capital via policy and institutional reforms. Key initiatives include: resource

availability, efficiency of expenditure management, and results-orientation. This approach necessitates more analytical effort and proactive client involvement to inform effective public finance reforms for human capital. DPOs can help with policy and institutional changes that focus on: strengthening PFM, increasing DRM, enhancing public workforce management, and placing human capital investment within a broader medium-term fiscal framework.

c. Supporting improved human capital outcomes with PERs

PERs are a key instrument for making progress on the human capital financing agenda. Building on the HD-EFI 2021 report noted above, the HCP and PER Programmatic Support Platform teams are collaborating to support front-line WBG teams and their government counterparts on four components:

- o Component 1: Building the analytical base on human capital finance and sound public expenditure management. This includes preparing a good practice ‘guidance note’ with methodologies for conducting PER analysis of human capital related expenditure and fiscal policy, building and maintaining a database of PERs, and surfacing good practices in terms of process, design, methodology and outcomes.
- o Component 2: Developing and improving tools, including for better measurement and monitoring of the macro-fiscal context (such as indicators of debt distress) to help inform budget priorities through the development of a “Policy Priority Inference Tool”. Efforts will also support continued improvements in BOOST systems to track human capital investments.
- o Component 3: Sharing knowledge and experience on good practice internally and externally.
- o Component 4: Convening stakeholders and strengthening relations with external partners to advance the agenda on a global level.

The team expects to deliver on the guidance note and Policy Priority Inference Tool in the first half of FY23, and is considering pilots in select countries undertaking a PER.

6.4 Integrating Gender considerations in PERs

Well-planned fiscal policy and budget systems are central for achieving strategic priorities such as gender equality. Fiscal policy is a powerful tool to change incentives and correct market failures such as unequal economic opportunities. Fiscal policy choices that address gender equality—such as investing in education or infrastructure, developing better sanitation facilities, implementing individual-based tax regimes and other reforms, or subsidizing childcare —create more economic opportunities for women, increase growth, and reduce poverty and inequality. To this end, the World Bank agreed under the IDA20 Gender and Development Special Theme to Policy Commitment 8:

***Implementing fiscal policy and budget systems to close gender gaps:** Support at least 10 IDA countries to make their fiscal policy and budget systems more inclusive and gender responsive by, for example, budget reforms, removing discriminatory provisions from tax legislation and/or monitoring the effectiveness of public spending, including where appropriate through fiscal incidence analysis for equality policies.*

Systematic treatment of gender issues in PERs to date is limited. The PER stocktaking exercise for the repository includes tagging on the cross-cutting SDG pertaining to gender, as well as techniques for gender disaggregated data and analysis. Of the 140 PERs reviewed, 3 had extensive coverage on gender, 8 had sufficient coverage, and five included a basic reference. Seven PERs considered gender informed recommendations and other analysis.

- **Extensive coverage:** Niger Social Protection (2019), Philippines GOV (2015), Philippines MTI (2021).
- **Sufficient coverage:** Botswana Education PER (2019), Burundi GOV PER (2017, 2019), DRC Education (2015), Georgia MTI (2017), Lebanon (2016), Mali Education (2019), Nepal Health (2020) - sufficiently; Guinea Education (2015) (including a recommendation to: “Tackle disparities across regions, gender, and income group at all levels”).

- **Basic reference:** Guinea-Bissau GOV (2018), Mali Social Protection (2018), Mali MTI (2019), Romania MTI (2016) - basic reference.
- **Other:** The Gambia Education (2017) includes: “Benefit incidence analysis - Use of HIS for equity/efficiency/gender analysis”; Indonesia Social Protection (2020) includes: “Deep dive on Employment, insurance and pension schemes and gender considerations”; Lebanon, Education (2016) includes: Gender analysis/Gender Parity Index; North Macedonia MTI (2018), Papua New Guinea Education (2015) Equity of School Access Between Genders, Sudan Education (2021) recommendations including: “Use special transfers (2 percent fund) to support targeted interventions to eliminate or reduce disparities between states, including on gender disparities, nomadic tribes, and conflict areas, and to increase internal efficiency to increase survival rates, especially in rural areas.” Ukraine, GOV (2017) data on legal challenges disaggregated by gender.

Comprehensive effort is needed to ensure PERs incorporate a more thorough consideration of key gender gaps within and across sectors, including those beyond education, health and social protection. The MTI and Gender teams will collaborate to help meet Policy Commitment 8 by developing appropriate tools/guidance for country economists on bringing a gender-aware perspective into tax policy reform dialog and design, as well as conducting training on these tools. This entails:

- a. Producing a series of policy notes for Bank economists distilling the literature on gender, taxation, and fiscal policy more broadly.
- b. Developing a guidance note with inputs from GOV and other relevant GPs that provides methodological guidance for teams to integrate gender analysis and access relevant data and evidence (as was done earlier for poverty diagnostics and SCDs for growth and shared prosperity). This module would contain information and resources on where to find data and analyses, existing work by World Bank economists and others on how to assess the costs of gender disparities and examples of good diagnostics.
- c. Convening a workshop for EFI teams focused on the Bank’s approach to incorporating gender analysis into PERs prior to finalizing the draft.
- d. Soliciting interest from country teams to pilot at least one PER for more hands-on support to incorporate a gender lens. This effort would be led by MTI, with cross-GP participation (POV, GOV, the Gender Group, and others as relevant).
- e. Organizing a session during MTI learning week and the MTI bootcamp on how to use these tools in policy dialog and operations.

Delivering on these elements would help underpin deepened engagement with clients, including through DPOs. Those in the pipeline that are focused on tax policy, tax administration, and fiscal decentralization, among others, would be good candidates for enhanced upstream support to reflect prior actions and triggers to close relevant gender gaps or enhance women’s voice and agency. This could include providing gender tag training to EFI teams in each region, liaising with the IMF on potential joint work, including joint analyses, technical assistance in selected countries, and events, among others, and collaborating with the ADB and OECD to field a survey in Asia-Pacific and selected low-income countries, that adapts and extends the survey on gender and taxation conducted by the OECD for the G20.

7. Lessons Learned and Key Considerations

The stocktaking highlights substantial scope for elevating the quality and effectiveness of PERs, as well as surfacing good practices in terms of both substance and process that contribute to impactful PERs. These include both opportunities to enhance consistency in the depth of analysis and implement several critical features of best practice PERs, including deep client engagement, strong leverage with other analytical work, expanded dissemination and related training, and tailored recommendations. While this section offers broad guidance building on the stocktaking exercise and feedback from PER TTLs, the final scope and structure of individual PERs (and the selection of the task team to deliver) will need to be tailored to the client context, policy dialogue, reform needs, and other country specific circumstances.

a. Enhancing consistency in macro-fiscal chapters

Understanding the macroeconomic stance is critical for sound diagnostics and effective policy interventions. It helps identify the nature of existing macroeconomic challenges, the vulnerability to shocks, and the scope for government intervention through fiscal and monetary policy measures to generate fiscal space, boost growth, support fiscal consolidation and/or improve long-term financing prospects. In this respect, the stocktaking revealed significant variation in the scope, content and approach of the analyses. It is critical that all PERs provide well-crafted coverage of the macroeconomic context to ensure that findings and recommendations are feasible and consistent.

Drawing from best practices identified during the stocktaking, the following components should be considered for inclusion in a macro fiscal chapter (as applicable):

1. **Reviewing macro-fiscal developments**, including key revenue and expenditure perspectives, off budget operations (Uzbekistan 2019) and debt composition and management considerations (Belarus 2020 and Myanmar 2017). This analysis could also connect macro-fiscal challenges with broader development goals, as articulated in national development plans and the Sustainable Development Goals (Sierra Leone 2021 and Costa Rica 2019).
2. **Discussing fiscal risks**, both in terms of short-term fiscal pressures (Egypt 2019 and Nepal 2021) and the impact of long-term structural issues on fiscal stability (Argentina 2020, Sri Lanka 2020 and St. Maarten 2020). This could include assessing fiscal policy resilience to shocks in terms of expenditure rigidity (Dominican Republic 2019 and Argentina 2020), revenue volatility (Chad 2019), cyclicalities of fiscal stance (Costa Rica 2018) and existence and effectiveness of fiscal rules (Papua New Guinea 2021) for stabilization of macroeconomic imbalances.
3. **Assessing medium-term fiscal space.** Identification of efficiency savings through expenditure rationalization and increased revenue mobilization are critical to identification of options for fiscal space considerations. Senegal (2021) offers a compelling analysis of tax expenditure rationalization, while Costa Rica (2019), Peru (2017) and Ecuador (2019), among others, offer best practice approaches to efficiency savings on the expenditure side.
4. **Integrating PFM considerations affecting fiscal stability** to examine the role and impact of public financial management practices in underpinning fiscal discipline and macroeconomic stability (Lesotho 2019 and Ecuador 2019). This can readily be undertaken based on existing PEFA diagnostics, among others.
5. **Implementing macro-scenario analysis** to inform fiscal consolidation efforts (Costa Rica 2019), fiscal space (Kenya 2020) and debt sustainability perspectives (Cabo Verde 2018).

b. Scope for more structured analysis in PERs

The stocktaking notes significant variation in scope, content and level of sophistication in PER analyses. In general, revenue dynamics were more consistent, due to the application of a common analytical framework (TPAF), uniform metrics (collection, productivity, effort and efficiency), and similar dynamics driving performance (policy and compliance gaps). On the other hand, expenditure analysis in PERs typically follows a less structured approach, leading to variation of outputs that are difficult to compare and often differ

in quality and level of sophistication. As a result, it is important to develop an ecosystem of resources that can help PER teams adopt a more structured approach to expenditure analysis to enhance the overall effectiveness of PERs. These are further elaborated below.

c. Enhancing cross-cutting analysis of public expenditures

Fuller assessment of underlying, system-wide determinants of efficiency, equity and effectiveness in public spending is needed, underpinning the articulation of recommendations. Currently, the majority of PERs continue to be structured by sector when examining quality of expenditure policy, resulting in missed opportunities for system-wide reform in support of fiscal consolidation and greater value for money. In this respect, it is critical that PERs sharpen their focus on cross-cutting dynamics to help ensure rigorous analysis of the drivers of the quality of spending and identify opportunities for efficiency savings. A few PERs have excelled in implementing this approach, resulting in strengthened policy dialogue around reforms. In particular, the following pillars are key:

1. **Discussing the overall expenditure framework**, assessing the economic and functional composition of public spending, considering adequacy, equity and efficiency considerations and including international benchmarks to regional and aspirational peers.
2. **Undertaking wage bill analysis**, as it accounts for an average of over 25 percent of total spending and can reveal important insights about sustainability, efficiency and productivity of public spending. Best practice approaches implement a holistic analysis that decomposes the wage bill and looks at options for allowance rationalization (Kenya 2020), identifies drivers of wage bill growth (Lesotho, 2019), estimates fiscal sustainability and impact of alternative wage bill policies (Costa Rica, 2018 and Guinea Bissau 2018), assesses crowding out of other expenditures, discusses the wage premium (Uzbekistan 2018 and Ecuador 2018), examines the structure and compression of pay grades (Cameroon 2018 and Ghana 2017), and reviews per worker productivity. The analysis can cover HRM, quality of establishment control (Lesotho 2018) and related institutional and political economy considerations (Costa Rica 2018).
3. **Reviewing subsidies**, which on average accounted for about 10 percent of total expenditures over the 2006-21 period (including to SOEs, the private sector or individuals). As done in several PERs, a review of subsidy schemes can help identify options for rationalization to maximize equity and efficiency. A review of 70 EFI-led PERs revealed that untargeted (and typically regressive) subsidies account for up to 80 percent of the total, and there is substantial scope for improvement by moving towards targeted, means-tested schemes. Value for money and efficiency analysis of subsidies can be undertaken in relation to SOEs/utility companies (Comoros 2016), including below-the-line support (Cabo Verde 2019).
4. **Assessing pension liabilities**, which represent a sizeable share of public expenditures, and often present a fiscal sustainability challenge, particularly in countries with rapidly aging populations. Typical coverage of pensions would look at sustainability through PROST, as well as exploring adequacy and equity perspectives.
5. **Analyzing procurement**, seeking opportunities for greater efficiency and value for money. Costa Rica (2018), Ecuador (2020), The Gambia (2020), and Kenya (2020) present deep dives on procurement using strategic sourcing methodology and other diagnostics, resulting in the identification of significant efficiency savings of up to 20 percent of total spending. On average, procurement spending accounted for 17 percent of total expenditures during 2006-2021.
6. **Assessing the quality of public investment**, which represents one of the greatest sources of inefficiency. A good practice PER seeks to identify the bottlenecks to more efficient public investment using the World Bank PIM framework (Romania 2017) or PIMA reports (Ghana 2012 and The Gambia 2020), complemented by data deep dives to document the extent of cost over-runs, time delays and other portfolio assessments as presented in Kenya (2020), Myanmar (2017) and Bangladesh (2015). On average, capital spending accounts for over 15 percent of total expenditures.

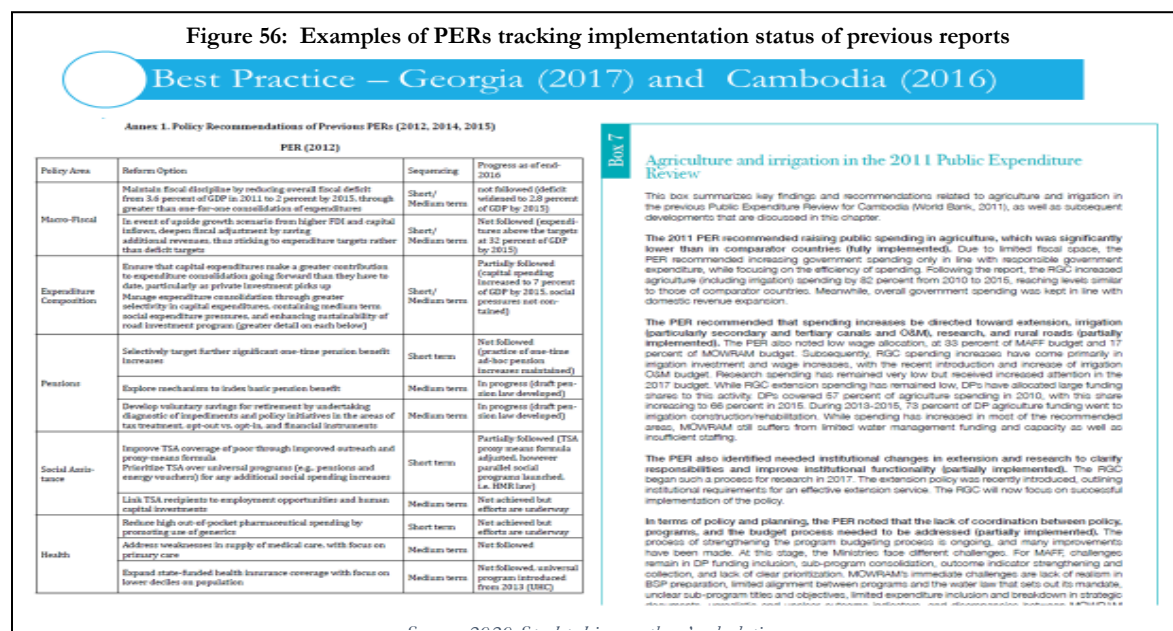
d. Leveraging existing knowledge

PERs are one of most important WB instruments to underpin policy dialogue and shape the operational context. Comprehensive coverage of topical issues, analytical rigor and innovative, sophisticated analysis can help sharpen messaging and provide a roadmap for reforms. For this reason, and in conjunction with limited available resources, it is important for PERs to take stock of all available information, integrating content as applicable.

Integrating existing diagnostics can yield important insights without the need to undertake new analysis. For instance, less than 20 percent of the PERs reviewed as part of the stocktaking noted PEFA findings, despite the broad coverage and availability of PEFAs. Similarly, the expanded implementation of diagnostics such as TADAT and DIAMOND has not been reflected in greater use and integration in PERs. This is a missed opportunity for deepening the knowledge base of tax administration aspects without deployment of additional resources, as successfully undertaken in The Gambia (2020), Kyrgyz (2021), Maldives (2022), and Congo (2022), among others.

Given the ambitious nature of PERs and limited resources, it is essential to build on previous PERs and seek synergies with other existing analyses, but the record is mixed. While around 30 percent of PERs made some references to previous efforts, only 10 percent looked at the implementation status of previous recommendations, including Georgia (2017) and Cambodia (2016) (Figure 56). Integration of existing diagnostics such as PEFA, PIM, TADAT has expanded compared to the stocktaking of 2015, but there is scope for improvement.

More needs to be done to cross-reference EFI-led and sector-led PERs in order to avoid duplication and ensure consistency. While more than half of EFI-led PERs cover education and health, other sectors are underrepresented, and it is important to strengthen connections, including with related diagnostics like INFRA-SAPs. Similarly, sector PERs often do not include analysis of the macro-fiscal context within which sector policy reforms should be framed, and they are even less likely to draw upon existing diagnostics such as PEFAs and PIMAs. Impact assessments conducted as part of the stocktaking reveal that sector PERs are strong analytical vehicles to connect to sector operations, but are less likely to be incorporated in DPOs or contribute to reforms of the budget process or legal framework.⁵⁴



⁵⁴ This may be changing, however, including through climate-informed DPOs.

e. Ensuring greater consistency in data gathering and dissemination

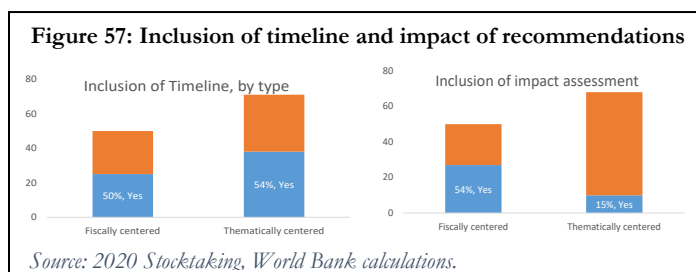
Data collection continues to be a challenge in most PERs. There is room to introduce greater consistency in these efforts by defining a common set of tables that should be included in appendices. Based on best practices identified as part of the stocktaking, several key tables and summary findings are recommended for inclusion in PERs. These cover trends in public expenditures and revenues, execution rates, comparative / benchmarking indicators, and extra budgetary, quasi-fiscal and off-budget activities. Further details are presented in Annex 5. A compilation of key findings from all existing diagnostics (PEFA, PIMA, TADAT, MAPS) is also recommended to be included in an Appendix to ensure that all PERs are based on the most recent and available information, integrating findings in the main text to strengthen the overall narrative.

f. Ensuring strong client engagement

Extensive engagement with the client is essential to tailoring PERs and ensuring relevance. The appropriate scope of PERs depends on consultations with government authorities and other stakeholders to identify their interest in the analysis and topics to be covered in each chapter, build ownership, and enable testing of some hypotheses, raise awareness, and contribute to consensus building for enhanced impact.⁵⁵ For example, Belarus (2021) presented initial findings as part of consultations, affording opportunities to receive early feedback from counterparts and fostering intense policy dialogue that was critical in driving the focus of the work. The strength of the engagement during the PER preparation process is important also to the ownership of the findings and likely follow-through on policy recommendations that emerge from the analysis. Finally, the engagement needs to involve sector colleagues to ensure consistency.

g. Tailoring recommendations

The stocktaking highlights substantial scope for improvement on how PERs formulate policy recommendations. Just over half of PERs include timelines and sequencing of proposed recommendations, and about a third estimate their potential fiscal and/or social impact, (Figure 57). Examples of the latter include discussion of fiscal gains and uses (The Gambia, 2020), estimates of equity and efficiency gains (Chile, 2017, Mexico, 2016 and Peru 2017), and assessment of policy coherence (Lesotho, 2019 and Uganda, 2019).⁵⁶ FC PERs were more likely than TO PERs to include estimates on the impact of policy recommendations, while little difference was detected when it came to the inclusion of timelines. Identifying fiscal impact of reforms is arguably the most important determinant of uptake of recommendations or DPO actions.



Recommendations that are either very general or not linked to the analysis are less likely to be impactful. Some examples from the stocktaking that are hard to implement and operationalize include:

1. **“Allocate greater resources to social and economic sectors”** – the recommendation is very general.
2. **“Contain purchases of goods and services and wage bill”** – the PER did not have specific analysis on these categories making it hard to understand cost drivers and areas for potential savings.
3. **“Reduce the number of public sector employees; reinforce procurement practices; better regulate public investments”** – while consistent with analysis performed in respective PERs, how and where to proceed was not articulated.

⁵⁵ Custer, Samantha, Zachary Rice, Takaaki Masaki, Rebecca Latourell and Bradley Parks. 2015. “Listening to Leaders: Which Development Partners Do They Prefer and Why?” Williamsburg, VA: AidData. Accessed at <http://aiddata.org/listening-to-leaders>.

⁵⁶ Annex 2 presents an illustration of these approaches.

4. **“Reduce tax expenditures to increase the fiscal space”** – The recommendation is in line with the overall narrative, but the PER did not elaborate on sequencing and prioritization.

Recommendations that are actionable and closely linked to sound analysis that is well adapted to the local context are more likely to gain traction. Similarly, articulation of the timeline, sequencing and expected fiscal impact are important determinants of client uptake and potential inclusion in DPOs. PERs should avoid being either too general or not linked to the analysis, which might make it harder to operationalize. Options should be provided on how and where to implement proposed recommendations and strive to elaborate on sequencing and prioritization. Annex 2 presents an example of good practice drawn from The Gambia (2020).

Estimating the fiscal impact and other important dimensions of proposed recommendations is essential. About 40 percent of PERs implemented between 2016 and 2020 include estimates of the anticipated fiscal impact for each proposed recommendation, typically estimating efficiency savings in percent of GDP. Additional dimensions might include technical feasibility (Guinea Bissau, 2018), efficiency and equity effects (Chile, 2017, Mexico, 2016, and Peru, 2017), policy coherence (Uganda, 2019), and potential uses (The Gambia, 2020). Annex 2 presents an example of good practice drawn from The Gambia (2020).

It is useful to summarize recommendations in the overview, as well as at the end of each chapter. Recommendations are often scattered across several sections, including within a chapter. It is important that they are presented in tables at the end of each chapter and – at least in a reduced form capturing main recommendations – in a single table in the introductory chapter for ease of reference.

b. Expanding dissemination and related training activities

Dissemination is essential for impact. For example, the Serbia (2016) team conducted wide-ranging dissemination events beyond the capital Belgrade to help build consensus for proposed reform measures. Laos (2018) shows that breaking a PER down into smaller targeted policy notes can facilitate client uptake. As part of the preparation of Timor Leste (2017), the PER team carried out a series of capacity-building events, including working together with the client on datasets, presenting on key methodologies (such as medium-term expenditure frameworks, and classification of the functions of government), and collaborating through workshops and brainstorming sessions.

i. Proposed next steps

To help move the agenda forward and raise the bar on PERs, we propose the following actions:

- **Continuous identification and dissemination of best practices and innovation** – Building on the PER stocktaking and repository, continued efforts will be made to capture innovation and best practices, as well as underpinning just in time advice to PER teams.
- **Technical guidance notes** – Targeted guidance notes are envisaged to support enhanced analytical endeavors across a range of thematic areas, including efficiency, wage bill and procurement analysis.
- **Scaled up staff training** – A fiscal analysis academy is being developed with new material on the expenditure side, complementing existing resources on revenue.
- **Development of tools and benchmark data sets** – Data resources, such as BOOST, are already widely available to PER teams, but there is scope for further advances in terms of inter-operability between financial and non-financial indicators for greater depth of analysis and comparability. A cadre of STCs with significant expertise in specific sectors and thematic areas is being developed to support PER TTLs.

- **Development of a framework on core elements of a PER** – PERs can benefit from more consistent coverage of key fiscal policy issues while striving to adopt rigorous analytical approaches and innovations.
- **Enhanced review process** – Contestability is an important pillar safeguarding the integrity and quality of PERs. Global MTI is well positioned to provide an overarching perspective on the proposed design and implementation of PERs, enhancing the overall review process.
- **Community of Practice on Fiscal Policy and Quality of Expenditure** – The platform was established in June 2022, and Global MTI is putting mechanisms in place to ensure robust and sustained engagement.
- **Fiscal Policy BBL series** – Efforts are underway to attract renowned public finance specialists and speakers to support continued thinking and innovation.
- **PER help desk** – Global MTI is establishing a one-stop-shop of support for PER TTLs and teams to seek help in their endeavors.

Appendix A: List of PERs reviewed

Over 160 reports were reviewed according to the framework. Below is the list along with basic classifiers.
Links to individual PER reports can be found in the PER repository.

Country	FY	GP	Type	Project	Report title
Afghanistan	2020	MTI	Other	P157287	Afghanistan: Public Expenditure Update
Albania	2019	Agriculture	PER	P169730	Albania - Agriculture Potential and Public Support: A Public Expenditure and Technical Efficiency Review (English)
Albania	2018	MTI	PFR	P156233	Albania Subnational Public Finance and Institutional Review
Angola	2018	SP/Jobs	PER	P162124	Angola Social Protection Public Expenditure Review
Angola	2020	Transport	PER	P170237	Angola Road Sector Public Expenditure Review
Argentina	2020	MTI	PER	P168968	Argentina Public Expenditure Review
Bangladesh	2015	MTI	PER	P132921	Bangladesh Public Expenditure Review Update. An update of fiscal trends based on the 2010 Public Expenditure Review
Bangladesh	2020	Water	PER	P171452	Public Expenditure Reviews of the Water Sector of Countries in Asia (Bangladesh, Nepal and Sri Lanka) - Bangladesh
Bangladesh	2020	INF	other	P167117	Bangladesh INFRASAP
Belarus	2021	MTI	PER	P169790	Belarus Public Expenditure Review: Strengthening the Equity, Efficiency and Effectiveness of Social Expenditures in Belarus
Belarus	2021	Transport	other	P169218	Belarus INFRASAP report
Botswana	2019	Education	PER	P168023	Basic Education sector in Botswana
Bulgaria	2016	GOV	PEIR	P156952	Bulgaria Judicial Performance, Caseload and Expenditure Review (2008-2014)
Bulgaria	2018	MTI	PEIR	P161180	Spending review: improving efficiency and effectiveness of waste management spending
Burundi	2019	GOV	PER	P164387	Burundi: Addressing the Challenges and Opportunities in Basic Education
Burundi	2017	GOV	PER	P151575	Burundi: Delivering Health Services Under Fiscal Stress
Cabo Verde	2019	MTI	PER	P164637	Revisiting the Efficiency of Public Spending to Reduce Debt and Improve Education and Health Outcomes
Cambodia	2018	MTI	PER	P157949	Improving the effectiveness of Public Finance
Cameroon	2018	MTI	PER	P161120	Aligning Public Expenditures with the Goals of Vision 2035
Chad	2019	MTI	PER	P168773	Chad Public Expenditure Analysis: Fiscal Space for Productive Social Sectors Expenditure
Chile	2017	MTI	PER	P158693	Chile Public Expenditure Review
China	2019	MTI	Other	P164116	Public Expenditures Under the 2011-2020: Poverty Reduction Strategy in China
Comoros	2016	MTI	PER	P151770	Comoros - Public expenditure and fiscal management review
Congo, DRC	2015	Education	PER	P147553	Democratic Republic of Congo: Education Sector Public Expenditure Review
Costa Rica	2018	MTI	PFR	P162556	Costa Rica Public Finance Review
Costa Rica	2015	SP/Jobs	PEIR	P150371	Central America social expenditures and institutional review: Costa Rica
Dominican Republic	2019	MTI	PER	P164680	Dominican Republic Public Expenditure Review, 2012-18
Ecuador	2020	MTI	PFR	P164213	Ecuador Public Finance Review Phase II
Egypt	2019	MTI	PER	P161888	Making Egypt a Better Fiscal Space
El Salvador	2015	SP/Jobs	PEIR	P133348	El Salvador - Central America social expenditures and institutional review
Ethiopia	2015	MTI	PER	P148336	Ethiopia Public Expenditure Review (P148336)
Gabon	2018	MTI	PER	P159858	Improving public spending quality to foster inclusive growth
Gambia, The	2018	Education	PER	P153483	The Gambia: Education Sector Public Expenditure Review - An Efficiency, Effectiveness, Equity, Adequacy, and Sustainability Analysis
Gambia, The	2018	GOV	PER	P164522	The Gambia Security Sector Public Expenditure Review
Gambia, The	2020	MTI	PER	P169825	The Gambia PER: Creating fiscal space
Georgia	2015	MTI	PER	P151523	Georgia - Public expenditure review: selected fiscal issues (English)
Georgia	2017	MTI	PER	P156724	Georgia - Public expenditure review: building a sustainable future
Ghana	2016	POV	PER	P156462	Ghana - Social protection assessment and public expenditure review (English)
Ghana	2017	GOV	PER	P151519	Fiscal consolidation to accelerate growth and support inclusive development: Ghana public expenditure review

Guinea	2015	Education	PER	P155946	Republic of Guinea Public Expenditure Review: Volume I: Education
Guinea	2022	MTI	PER	P174834	Guinea PER
Guinea-Bissau	2018	GOV	PER	P164582	Guinea-Bissau - Public Expenditure Review: Managing Public Finance for Development
Haiti	2015	MTI	PER	P132356	Towards Greater Fiscal Sustainability and Equity: A Discussion of Public Finance in Haiti
Indonesia	2020	SP/Jobs	other	P167664	Investing in People: Social Protection for Indonesia's 2045 Vision (Vol. 1)
Indonesia	2020	MTI	PER	P165627	Indonesia Public Expenditure Review: Spending for Better Results
Iraq	2021	MTI	PER	P171782	Iraq Public Expenditure Review for Human Development Sectors
Jordan	2016	Education	PER	P148394	Hashemite Kingdom of Jordan - Education Sector Public Expenditure Review
Kazakhstan	2018	MTI	PFR	P162003	Kazakhstan - Enhancing the fiscal framework to support economic transformation: public finance review (English)
Kenya	2020	Agriculture	PER	P165731	Improving the Quality of Public Expenditures on Agriculture
Kenya	2019	SP/Jobs	PER	P168878	Kenya Social Protection and Jobs Programs Public Expenditure Review
Kenya	2019	MTI	other	P168277	Kenya Public Expenditure Analysis 2019: Creating Fiscal Space to Deliver the Big Four while Undertaking a Needed Fiscal Consolidation
Kenya	2020	MTI	PER	P172158	Kenya Public Expenditure Review 2020: Options for Fiscal Consolidation after the COVID-19 Crisis
Kosovo	2014	MTI	PFR	P133100	Kosovo Public Finance Review: Fiscal Policies for a Young Nation
Laos	2018	MTI	PER	P158831	Education and Health Sectors - Public Expenditure Arrangements
Lebanon	2016	Education	PER	P152803	Lebanon: Education Public Expenditure Review 2016
Lesotho	2019	GOV	PER	P160784	Improving expenditure efficiency for inclusive development and growth
Lesotho	2017	Health	other	P153463	Lesotho Public Expenditure Tracking Survey and Service Delivery in the Health Sector
Lesotho	2019	Agriculture	PER	P168208	Kingdom of Lesotho: Agriculture Public Expenditure Review (Vol. 2)
Liberia	2019	Justice			
Liberia	2020	MTI	PEIR	P168219	Towards A More Secure and Justice Liberia
Liberia	2019	Agriculture	PER	P165734	Liberia: Analysis of Public Expenditure in Agriculture
North Macedonia	2016	MTI	PER	P147200	FYR Macedonia Public Expenditure Review Fiscal Policy for Growth
North Macedonia	2018	MTI	PFR	P165084	Republic of North Macedonia Public Finance Review - Sowing the Seeds of a Sustainable Future
Malawi	2020	MTI	PER	P171768	Malawi Public Expenditure Review 2020: Strengthening Expenditure for Human Capital
Malawi	2019	MTI	PER	P164759	Malawi Public Expenditure Review 2019: Putting Fiscal Policy on a Sustainable Path
Mali	2019	Education	PER	P151537	Better Governance for Better Education Quality and Equity
Mali	2019	MTI	PER	P168677	Mali Fiscal decentralization follow up: A First Approach to Assess the Efficiency of Subnational Government Expenditures
Mali	2018	SP/Jobs	other	P162072	Social Protection Financing Diagnostics for Mali
Mali	2021	MTI	PER	P171692	Mali Public Expenditure Review
Mexico	2016	MTI	PER	P150646	Mexico Public Expenditure Review
Moldova	2019	GOV	PEIR	P158938	Moldova - Judicial Reform: From Resources to Results - A Court User Survey: Measuring and Monitoring Justice Sector Performance
Moldova	2015	MTI	PER	P151612	Moldova PER: Agriculture
Moldova	2018	Education	PER	P161676	Moldova: Education Sector Public Expenditure Review Selected Issues
Moldova	2017	MTI	PFR	P157689	Republic of Moldova Public Finance Review Towards More Efficient and More Sustainable Public Finances
Moldova	2018	Health	PER	P165679	Moldova - Public expenditure review: improving public health expenditure efficiency
Mongolia	2018	MTI	PER	P161057	Growing Without Undue Borrowing. Enhancing efficiency of spending and revenue
Montenegro	2019	MTI	PFR	P167665	Montenegro - Second Fiscal and Financial Sector Resilience Policy-Based Guarantees Program (English)
Morocco	2020	Water	PER	P162398	Morocco Infrastructure Review (English)
Morocco	2016	Transport	PEIR	P152499	Morocco Road sector PEIR

Mozambique	2019	Agriculture	PER	P168732	Republic of Mozambique Agriculture Public Expenditure Review: Assessment and Result-Focused Expenditure Management
Myanmar	2020	Water	PER	P169593	Myanmar Water Sector Public Expenditure Review
Myanmar	2020	GOV	PER	P166313	Subnational Public Expenditure Review 2019 Fostering Decentralization in Myanmar
Myanmar	2017	MTI	PER	P159067	Myanmar PER 2017 Fiscal Space for Economic Growth
Myanmar	2019	Agriculture	PER	P158765	Increasing the Impact of Public Spending on Agricultural Growth - Myanmar agricultural public expenditure review
Namibia	2020	Health	PER	P168550	Namibia - Health Sector Public Expenditure Review (English)
Nepal	2019	Water	PER	P168191	Reviews of the Water Sector in select countries in Asia (Bangladesh, Nepal and Sri Lanka)
Nepal	2020	Health	PER	P168568	Assessing Public Financing for Nutrition in Nepal (2011-17)
Nepal	2021	SP/Jobs	PER	P169010	Nepal Social protection PER
Nepal	2021	MTI	PER	P172086	Nepal PER
Niger	2021	MTI	PER	P171693	Niger PER
Niger	2019	SP/Jobs	PER	P166144	Review of the Public Expenditures in Social Protection in Niger
Nigeria	2020	MTI	other	P161752	Nigeria Fiscal Review
Oman	2017	Education	PER	P157882	Oman Education PER
Panama	2017	Education	PEIR	P146907	Panama - Central America social expenditures and institutional review (English)
Papua New Guinea	2015	Education	PER	P129764	Papua New Guinea - From expanded access to better quality: a call for consolidation of education investment - a PER of the education sector
Papua New Guinea	2021	MTI	PFR	P171817	Papua New Guinea Public Finance Review : Resuming Fiscal Consolidation while Improving Public Service Delivery
Paraguay	2020	Water	PER	P169997	Servicio de Agua y Saneamiento en Paraguay: Aspectos Institucionales, de Gobernanza y el Gasto Público
Paraguay	2020	Transport	PER	P169044	Paraguay Roads Sector Public Expenditure Review
Paraguay	2019	SP/Jobs	PER	P168287	Paraguay Invertir en Capital Humano: Una revisión del gasto público y de la gestión en los sectores sociales
Peru	2017	MTI	PER	P158618	Peru Public Expenditure Review
Philippines	2015	GOV	PEIR	P144796	Philippines Autonomous Muslim Region of Mindanao (PH-ARMM) Public Expenditure and Institutional Review
Philippines	2021	MTI	other	P166521	The Philippines' Fiscal Policy – Recent Developments
Romania	2018	Education	PFR	P151510	Romania - Public finance review: enhancing the efficiency of public spending in pre-university education (English)
Romania	2017	Health	PFR	P159644	Romania public finance review: enhancing the efficiency and effectiveness of public spending in the health sector
Romania	2017	MTI	PFR	P159659	Romania Public Finance Review Consolidating public finances, improving tax and spending efficiency
Saudi Arabia	2019	MTI	PER	P165032	Kingdom of Saudi Arabia PER: Strengthening Fiscal and Spending Performance to Support Vision 2030
Senegal	2021	MTI	PER	P170349	Senegal Public Expenditure Review
Serbia	2016	MTI	PFR	P151518	Serbia - Public finance review: toward a sustainable and efficient fiscal policy
Serbia	2019	FCI	PER	P158246	Public Expenditure Review of Small and Medium Enterprise and Competitiveness Programs in Serbia
Sierra Leone	2016	Agriculture	PER	P146159	Basic Agricultural Public Expenditure Diagnostic Review (2003-12)
Somalia	2021	MTI	PER	P160458	Somalia Programmatic Public Expenditure Review: Overview Module
Sri Lanka	2020	Water	PER	P171604	Reviews of the Water Sector in select countries in Asia (Bangladesh, Nepal and Sri Lanka)
Sri Lanka	2020	MTI	PER	P163965	Sri Lanka Public Expenditure Review
Sri Lanka	2021	Health	PER	P168570	PER for Nutrition in Sri Lanka Assessing Public Financing for Nutrition in Sri Lanka
St. Maarten	2020	MTI	PER	P170066	St Maarten - Public Expenditure Review (English)
Sudan	2018	Agriculture	PER	P162413	Sudan Agriculture Public Expenditures An Initial Overview
Sudan	2021	Education	PER	P168442	Financing Education in Sudan: Strengthening the Efficiency and Equity of Public Spending in Education
Tanzania	2021	Health	PER	P162456	Tanzania Health Sector Public Expenditure Review 2020

Tanzania	2019	MTI	PER	P168175	Public Expenditure Review FY19: Are Resources and Policies Aligned?
Tanzania	2020	MTI	PER	P171715	Public Expenditure Review FY20 Policy options for improving DRM
Tanzania	2019	SP/Jobs	other	P161653	Financing Social Protection in Tanzania
Tanzania	2017	MTI	PER	P162759	Tanzania Public Expenditure Review 2017
Tanzania	2018	MTI	PER	P164502	Tanzania Public Expenditure Review 2018
Timor Leste	2018	MTI	other	P165836	Timor-leste - TL public expenditure analysis: a joint ministry of finance and World Bank analysis on public spending 2008-2016
Timor Leste	2016	MTI	PER	P133282	Democratic Republic of Timor-Leste Public Expenditure Review: Infrastructure
Togo	2021	MTI	PER	P171688	Togo Public Expenditure Review
Trinidad & Tobago	2019	MTI	PER	P162267	Productivity gains to recover fiscal balance in Trinidad and Tobago
Tunisia	2019	MTI	PER	P158118	Tunisia Public Expenditure Review
Uganda	2020	GOV	PER	P171711	Judiciary of The Republic of Uganda: Rapid Institutional and Economic Assessment
Uganda	2019	Agriculture	PER	P165598	Uganda - Agriculture Sector Public Expenditure Review (English)
Uganda	2019	SP/Jobs	PER	p169097	Uganda - Social Protection Public Expenditure Review (English)
Ukraine	2017	GOV	PEIR	P159960	Ukraine. Improving Courts and Enforcement. From Resources to Performance
Ukraine	2017	MTI	PFR	P155716	Ukraine Public Finance Review
Ukraine	2019	MTI	PFR	P166106	Towards a More Effective and Fiscally Sustainable Public Sector for Ukraine
Uzbekistan	2020	MTI	PER	P168168	Uzbekistan Public Expenditure Review
Vietnam	2015	ENV	PER	P144625	The Vietnam Climate Public Expenditure and Investment Review (CPEIR)
Vietnam	2017	MTI	PER	P158393	Fiscal Policies towards Sustainability, Efficiency, and Equity
Vietnam	2018	Energy	other	P165729	Vietnam Energy INFRASAP
Zambia	2016	Education	PER	P149455	Zambia Education Public Expenditure Review
Zambia	2019	Health	PER	P162287	Zambia Health Sector Public Expenditure Review
Zimbabwe	2017	MTI	PER	P146472	Zimbabwe Public Expenditure Review

Appendix B: Background Resources by Select Topics

Agriculture in PERs:

PERs: Albania 2019, Cambodia 2019, Ghana 2017, Kenya 2020, Lesotho 2019, Liberia 2019, North Macedonia 2018, Mali 2021, Mexico 2016, Moldova 2015, Mozambique 2019, Myanmar 2019, Niger 2021, Sierra Leone 2016, Sri Lanka 2020, Sudan 2018, Uganda 2019, Uzbekistan 2020, Vietnam 2015, Zimbabwe 2017

Resources:

Gautam, Madhur; Laborde, David; Mamun, Abdullah; Martin, Will; Pineiro, Valeria; Vos, Rob. 2022. Repurposing Agricultural Policies and Support; Options to Transform Agriculture and Food Systems to Better Serve the Health of People, Economies, and the Planet. World Bank, Washington, DC. © World Bank.

Goyal, Aparajita; Nash, John. 2017. Reaping Richer Returns; Public Spending Priorities for African Agriculture Productivity Growth. Africa Development Forum. Washington, DC: World Bank and Agence Francaise de Developpement. © World Bank.

Mink, S., Synthesis Review of Agricultural Public Expenditure Reviews Conducted in African Countries, World Bank 2015

Practitioners' Toolkit For Agriculture Public Expenditure Analysis, Report No. 60015-Glb World Bank 2011

Climate-informed analysis in PERs:

PERs: Vietnam climate 2015 PER, Dominican Republic 2019, Malawi 2019, St. Maarten 2020, Kazakhstan 2018, Malawi 2019, Niger 2018 social protection PER

Resources:

Parry, I. Implementing the United States' Domestic and International Climate Mitigation Goals: A Supportive Fiscal Policy Approach, IMF Working Paper 21/57, 2021

Carney, Mark. 2015. "Breaking the Tragedy of the Horizon—Climate Change and Financial Stability." Speech delivered at Lloyd's of London, September 29.

Commitment to Equity

PERs: Egypt 2019, The Gambia 2020, Georgia 2015, Montenegro 2019, Sri Lanka 2020

Resources:

Lustig (2016) "Commitment to Equity Handbook. A guide to estimating the impact of fiscal policy on Inequality and Poverty", Tulane University

Lustig, N. 2016. "The impact of taxation and social spending on income distribution and poverty in Latin America: Argentina, Bolivia, Brazil, Chile, Colombia, Costa Rica, Ecuador, El Salvador, Guatemala, Honduras Mexico, Peru, and Uruguay: An application of the methodology framework of the Commitment to Equity (CEQ) project," Centre for Global Development, Washington, DC

Cyclical policy of fiscal

PERs: Argentina 2020, Belarus 2021, Cabo Verde 2019, Cambodia 2018, Chad 2019, Chile 2017, Costa Rica 2018, Dominican Republic 2019, Kazakhstan 2018, Kenya 2019, Kenya 2020, Lesotho 2018, Moldova 2017, Montenegro 2019, Trinidad and Tobago 2019, Ukraine 2017, Vietnam 2017

Resources:

Abbott, Andrew, R. Cabral, P. Jones, and R. Palacio. (2015). “Political pressure and procyclical expenditure: An analysis of the expenditures of state governments in Mexico.” *European Journal of Political Economy*

Abbott, A. and P. Jones (2012) “Intergovernmental transfers and procyclical public spending”. *Economics Letters* 115: 447-451.

Abdih, Y. Lopez-Murphy P., Roitman A., and Sahay R., The cyclical policy of Fiscal policy in the Middle East and Central Asia: Is current crisis different? IMF working paper, 2010

Akitoby, B., B. Clements, S. Gupta, and G. Inchauste (2006) “Public spending, voracity, and Wagner’s law in developing countries”. *European Journal of Political Economy*, 22:908-924.

Meloni, O. (2018) “Turning a Blind Eye to Policy Prescriptions: Exploring the Sources of Procyclical Fiscal Behavior at the Subnational Level”. *Public Finance and Management*. Vol. 18(3-4): 265-284.

Vegh, Carlos; Lederman, Daniel; Bennett, Federico R.. 2017. *Leaning Against the Wind Fiscal Policy in Latin America and the Caribbean in a Historical Perspective*. LAC Semiannual Report; April 2017. World Bank, DC.

DEA and stochastic frontier

PERs: Belarus 2021, Cabo Verde 2019, Chile 2017, Costa Rica 2018, Dominican Republic 2019, Ecuador 2020, Egypt 2019, The Gambia 2020, Ghana 2017, North Macedonia 2018, Mali 2019, Moldova 2017, Romania 2017, Saudi Arabia 2019, Trinidad and Tobago 2019, Zimbabwe 2017

Resources:

Battese G.E. and Coelli, T.J. (1988): “Prediction of Firm-level Technical Efficiencies: With a Generalized Frontier Production Function and Panel Data”, *Journal of Econometrics*, 38, pp. 387-399.

Farrell, M. (1957): “The measurement of productive efficiency,” *Journal of the Royal Statistical Society, Series A*. Vol. 120, No. 3 (253-290).

Grigoli, F. and J. Kapsoli (2013) *Waste not, want not: the efficiency of health expenditure in emerging and developing economies*. IMF Working paper WP/13/187.

Grigoli, F. (2014) *A hybrid approach to estimating the efficiency of public spending on education in emerging and developing economies*. IMF Working Paper WP/14/19.

Gupta, S. and Verhoeven, M. (2001): “The Efficiency of Government Expenditure. Experiences from Africa” *Journal of Policy Modelling*, 23, pp. 433–467.

Herrera, S. and Pang, G. (2005): “Efficiency of Public Spending in Developing Countries: An Efficiency Frontier Approach” *World Bank Policy Research Working Paper*, 3645.

Herrera, S. and A. Ouedraogo (2018): *Efficiency of Public Spending in Education, Health, and Infrastructure: An International Benchmarking Exercise*. Policy Research Working Paper 8586.

Jondrow, J., Lovell, C.A.K., Materov, I.S. and Schmidt, P. (1982): "On the Estimation of Technical Inefficiency in the Stochastic Frontier Production Function Model", *Journal of Econometrics*, 19, pp. 233-238

Fiscal consolidation and multipliers

Fiscal multiplier PERs: Cabo Verde 2019, Dominican Republic 2019, Ethiopia 2015, Moldova 2017, Timor Leste 2021

Fiscal consolidation PERs: Argentina 2020, Cabo Verde 2019, Cambodia 2018, Cameroon 2018, Chad 2019, Chile 2017, Comoros 2016, Costa Rica 2018, Dominican Republic 2019, Ecuador 2019, Egypt 2019, Gabon 2018, The Gambia 2020, Georgia 2017, Ghana 2017, Kazakhstan 2018, Kenya 2019, Kenya 2020, Lesotho 2018, North Macedonia 2016, Mexico 2016, Moldova 2017, Montenegro 2019, Myanmar 2017, Romania 2016, Saudi Arabia 2019, Serbia 2016, Sri Lanka 2020, St. Maarten 2020, Tunisia 2019, Ukraine 2017, Ukraine 2019, Vietnam 2017

Resources:

Alesina, A., C. Favero, and F. Giavazzi. 2015. "The Output Effect of Fiscal Consolidation" *Journal of International Economics*, Vol. 96, pp..S19–S42.

Alesina, Alberto, Barbiero, Omar, Favero, Carlo A. Giavazzi, Francesco, and Paradisi, Matteo. 2017. "The Effects of Fiscal Consolidations: Theory and Evidence," CEPR Discussion Papers 12016, C.E.P.R. Discussion Papers

Estevao, M and Samake, I. 2013. "The Economic Effects of Fiscal Consolidation with Debt Feedback", IMF WP 13/136, DC.

Ilzetzki, E, Mendoza E.G., Vegh, C. 2013. "How Big (Small?) Are Fiscal Multipliers?" *JME*, Vol. 60,. 239-54.

Molnar, M. 2012. What factors determine the success of consolidation efforts?

OECD. 2012. Fiscal Consolidation: Part 1. How Much Is Needed and How to Reduce Debt to a Prudent Level? OECD Economics Department Working Papers, No. 932, OECD Publishing, Paris.

Vegh, Carlos; Lederman, Daniel; Bennett, Federico R.. 2017. *Leaning Against the Wind: Fiscal Policy in Latin America and the Caribbean in a Historical Perspective*. LAC Semiannual Report; April 2017. World Bank.

Force, Momentum and/or Rigidity Analysis

PERs: Cabo Verde 2019, Argentina 2020, Belarus 2021, Chad 2019, Chile 2017, Comoros 2016, Dominican Republic 2019, Ecuador 2020, The Gambia 2020, Ghana 2017, Kenya 2019, Kenya 2020, North Macedonia 2018, Malawi 2019, Mexico 2016

Resources:

Alier M. "Measuring Budget Rigidities in Latin America, IMF WP, Washington DC 2006

Cetrángolo, Oscar; Jiménez, Juan Pablo; Ruiz del Castillo, Ramiro. 2010. Rigidities and Fiscal Space in Latin America: A Comparative Case Study. *Macroeconomía del Desarrollo Serie*. Santiago, Chile April 2010

Merotto et al, *Dismal Science, Accounting and Newton's Second Law Identifying Force and Rigidity in Public Expenditure Analysis*, World Bank 2015.

Fiscal gap methodology

PERs: Argentina 2020 PER

Resources:

Martinez-Vazquez, Jorge and Sepulveda, Cristian F., A Theoretical Rationale for the Fiscal Gap Model of Equalization Transfers (2018). International Center for Public Policy, Working Paper 18-23, November 2018, Andrew Young School of Policy Studies Research Paper Series No. 19-12

Martinez-Vazquez, Jorge and Sepulveda, Cristian F., "Intergovernmental Transfers in Latin America: A Policy Reform Perspective" (2011). *ICEPP Working Papers*. 88.

Infrastructure

PERs: Angola 2020, Bulgaria 2018, Cameroon 2018, Comoros 2016, Dominican Republic 2019, Egypt 2019, Haiti 2015, Indonesia 2020, North Macedonia 2016, Moldova 2019, Morocco 2016, Morocco 2020, Myanmar 2020, Nepal 2019, Paraguay 2020, Peru 2017, Tanzania 2017, Timor Leste 2016, Tunisia 2019, Vietnam 2015

Resources:

Brenneman A. and Kerf M., Infrastructure & Poverty Linkages A Literature Review, World Bank 2002

Foster V. Rana A.Gotgulu N., Understanding Public Spending Trends for Infrastructure in Developing Countries, WB policy research number 9903, World Bank 2022

Songco, Jocelyn A.. 2002. Do Rural Infrastructure Investments Benefit the Poor? Evaluating Linkages--A Global View, A Focus on Vietnam. Policy Research Working Paper No. 2796. World Bank, Washington, DC.

Thacker S. Adshead, D., Fay, M. et al., 2019 Infrastructure for sustainable development. Nat Sustain 2,

Procurement – strategic sourcing methodology and similar approaches

PERs: Kenya 2020 PER, Chile 2017 PER, Ecuador 2018 PFR, Costa Rica 2018 PFR, Haiti 2015 PER, Malawi 2019 PER, Vietnam 2017 PER

Resources:

Talluri, S. and Narasimhan, R., 2004. A methodology for strategic sourcing. European journal of operational research,154(1), pp.236-250.

Shadow/Informal economy

PERs: Chad 2019, Comoros 2016, Ecuador 2020, Kosovo 2015, North Macedonia 2018, Serbia 2016, Tanzania 2019, Tanzania 2020

Resources:

Awasthi, Rajul; Engelschalk, Michael. 2018. Taxation and the Shadow Economy: How the Tax System Can Stimulate and Enforce the Formalization of Business Activities. Policy Research Working Paper; No. 8391. World Bank, Washington, DC. © World Bank

Leandro Medina, Andrew Jonelis, and Mehmet Cangul, The Informal Economy in Sub-Saharan Africa: Size and Determinants, IMF Working Paper 17/156, July 2017.

Bruhn, M. and J. Loeprick (2014), “Small Business Tax Policy, Informality, and Tax Evasion Evidence from Georgia”, World Bank Policy Research Working Paper 7010, The World Bank, Washington, D.C.

Tax effort/potential and gap analysis

PERs: Cabo Verde 2019, Chad 2019, The Gambia 2020, Guinea-Bissau 2018, Tanzania 2020

Resources:

Battese G.E. and Coelli, T.J. (1988): “Prediction of Firm-level Technical Efficiencies: With a Generalized Frontier Production Function and Panel Data”, Journal of Econometrics, 38, pp. 387-399.

Hutton, M. E. (2017). The Revenue Administration–Gap Analysis Program: Model and Methodology for Value-Added Tax Gap Estimation. International Monetary Fund.

Khwaja, M & Lyer I: Revenue potential, tax space and tax gap, a comparative analysis, WP6868, World Bank

Le, Tuan Minh and Moreno-Dodson, Blanca and Bayraktar, Nihal, Tax Capacity and Tax Effort: Extended Cross-Country Analysis from 1994 to 2009 (October 1, 2012). World Bank Policy Research Working Paper No. 6252, Available at SSRN: <https://ssrn.com/abstract=2167186>

Tax expenditures

PERs: Cabo Verde 2019, Cambodia 2018, Chad 2019, Costa Rica 2018, Dominican Republic 2019, Ecuador 2020, The Gambia 2020, Georgia 2017, Ghana 2017, Guinea-Bissau 2018, Haiti 2015, Kazakhstan 2018, Kenya 2019, Kenya 2020, Lesotho 2018, North Macedonia 2018, Moldova 2017, Mongolia 2018, Myanmar 2017, Romania 2017, Tanzania 2020, Ukraine 2017, Uzbekistan 2020, Vietnam 2017

Resources:

Burton, M. and K. Sadiq. 2013. “Tax Expenditure Management – A Critical Assessment”. Cambridge Tax Law Series. Cambridge: Cambridge University Press

Hutton, E. 2010. “A Basic Framework for Evaluating Value Added Tax Expenditures”, Inter-American Development Bank Discussion Paper IDB-DB-138.

James, S. 2014. “Effectiveness of Tax and Non-Tax Incentives – Evidence and Policy Implications.” World Bank Policy Paper

VAT in-depth analysis

PERs: Costa Rica 2018, Ecuador 2020, The Gambia 2020, Ghana 2017, Kazakhstan 2018, Lesotho 2018, North Macedonia 2018, Sri Lanka 2020, Tanzania 2020

Resources:

Pierre Bachas, Lucie Gadenne and Anders Jensen (2020), Informality, Consumption Taxes, and Redistribution, Policy Research Working Paper No. 9267. World Bank, Washington, DC.

Paolo Dudines and Joao Tovar Jalles (2017), How Buoyant Is the Tax System? New Evidence from a Large Heterogeneous Panel, IMF Working Paper 17/4, International Monetary Fund.

Liam Ebrill, Michael Keen, Jean-Paul Bodin, and Victoria Summers (2001), The Modern VAT, (Washington, D.C.: International Monetary Fund), chapter 4.

Eric Hutton (2017), "The Revenue Administration – Gap Analysis Program: Model and Methodology for Value-Added Tax Gap Estimation," Technical Notes and Manuals TNM/17/04, Fiscal Affairs Department (Washington, DC: International Monetary Fund).

Michael Keen (2013), The Anatomy of the VAT, WP/13/111, Washington, D.C.: International Monetary Fund, 2013. Okello, M. Pessoa, A. Swistak, *Managing VAT Refunds; IMF Guidance Note*

Wage Bill Analysis

PERs: Belarus 2021, Cabo Verde 2019, Cameroon 2018, Chad 2019, Costa Rica 2018, Ecuador 2020, Gabon 2018, Ghana 2017, Guinea-Bissau 2018, Kenya 2020, Laos 2018, Lesotho 2018, Liberia 2020, Mexico 2016, Mongolia 2018, Saudi Arabia 2019, Serbia 2016, Somalia 2021, Timor Leste 2016, Tunisia 2019, Uzbekistan 2020, Vietnam 2017, Zimbabwe 2017

Resources:

Acker, W. and Hasnain, Z. (2019): “Wage Bill and Pay Reform: A First Glance and a Look Forward”, World Bank Governance Note No. 16

Barna, N. and J. Orac. 2014. “Tailoring Civil Service Pay Analysis and Advice to Context.” Policy Research Working Paper WSP6744, The World Bank, January 2014.

Dorotinsky, B., Manning, N. and Rinne, J. (2009): “Options for Restraining the Wage Bill (While Preserving Essential Service Delivery)”, World Bank GET Note.

Formi, L. and N. Novta. 2014. “Public Employment Compensation Reform during times of Fiscal Consolidation.” IMF Working Paper WP12/192. Washington, DC: IMF.

Gindling, T., Hasnain, Z., Newhouse, D. and Shi, R. (2019): “*Are Public Sector Workers in Developing Countries Overpaid? Evidence from a New Global Data Set*”, World Bank Policy Research Working Paper No. 8754.

IMF. (2016). Managing government compensation and employment institutions, policies, and reform challenges. Washington, D.C.: International Monetary Fund.

Tamirisa, N. and Duenwald, C. (2018): “*Public Wage Bills in the Middle East and Central Asia*”, IMF Fiscal Affairs Departmental Paper.

World Bank. 2021. Public Sector Employment and Compensation: An Assessment Framework. Equitable Growth, Finance and Institutions Insight, World Bank, Washington, DC. © World Bank.

Appendix C: Best Practices in Reporting Impact of Policy Recommendations

Best practice: The Gambia 2020 PER

Policy actions	Time horizon	
	ST	MT
1. Revenue mobilization		
Value-added tax		
R1. Consider mainstreaming the statutory VAT rate to the range of 16–18%, along country practices in SSA.		
R2. Increase the threshold to focus on larger taxpayers, in combination with improved taxation of SMEs in the informal sector tax scheme.		
Excises		
R3. Adjust key excise rates.		
R4. Expand the domestic excise base.		
R5. Build tax capacity to administer and control excises.		
Tax expenditures		
R6. Improve the efficiency of tax expenditures at the border.		
R7. Enhance the efficiency of domestic VAT by phasing out exemptions.		
R8. Rationalize corporate tax incentives.		
R9. Strengthen the fiscal oversight of tax expenditures.		
Corporate income tax		
R10. Limit further reductions in the statutory tax rate without a parallel broadening of the CIT tax base.		
Personal income tax		
R11. Revert the PIT rate structure to earlier practice in the Gambia, aligned more closely to country practices in SSA.		
R12. Improve taxation of self-employed professionals such as lawyers, doctors and accountants.		
Institutional and capacity development		
R13. Create a Tax Policy Unit (TPU) in MOFIA with a mandate to forecast revenue and prepare tax policy initiatives.		
R14. Enhance tax administration capacity by modernizing business processes and IT systems.		
2. Spending		
2.1. Sector		
(i) Education		
R1. Increase student-teacher ratio (STR) at lower basic schools without compromising learning outcomes and considering the varying context across regions.		
R2. Base teacher staffing on a predetermined set of criteria including the STR, classrooms, school size, subjects taught, and facilities available at the school level.		
R3. Increase non-salary school spending on inputs and learning materials.		
R4. Use unit cost as an instrument to prepare the primary school education budget, considering the numbers of out-of-school children.		

continues in the next page

Policy actions	Potential gains (percent of GDP)	Potential uses (percent of GDP)
1. Revenue mobilization		
R1. Improve the efficiency of tax expenditures at the border.	0.5	
R2. Enhance the efficiency of domestic VAT by phasing out exemptions.	0.05	
R3. Rationalize corporate tax incentives.	0.25	
R4. Revert the PIT rate structure to earlier practice in the Gambia, aligned more closely to country practices in SSA.	0.4	
R5. Adjust key excise rates.	0.3	
R6. Expand the domestic excise base.	0.3	
R7. Enhance tax administration capacity by modernizing business processes and IT systems.	1.5	
2. Spending		
2.1. Sector		
(i) Education		
R1. Increase student-teacher ratio at Lower Basic Schools from 36.8:1 to the SSA average of 38.4:1 without compromising learning outcomes and considering the varying context across regions.	0.02	-0.02
R2. Increase non-salary school spending on inputs and learning materials.	0.71	
R3. Use unit cost as an instrument in preparing the primary school education budget.		-0.71
R4. Bring out-of-school children back into school.		
(ii) Security		
R5. Reduce or reallocate police officers and soldiers in the GPF and GNA based on criteria such as crime or conflict levels or the working environment.	0.68	
R6. Reduce or reallocate vehicles to efficient stations/regions.	0.1	
(iii) Health		
R7. Implement a disaggregated health expenditure information system to monitor spending on inputs in health facilities to generate efficiency gains.	N/A	
R8. Rationalize the allocation of the health budget in favor of primary care and prioritize primary care in the essential health care package.		-0.84
2.2. Public financial management		
(i) Procurement		
R9. Promote more competitive procurement of standardized goods such as vehicles (efficiency gains generated on a small sample of 5 processes).	0.001	
(ii) Treasury single account		
R10. Improve TSA coverage from 30% to 100% that would result in interest savings and a reduction in the opportunity cost due to idle balances.	0.14	
Grand total:	4.87%	3.3%
- as share of required fiscal consolidation over 2020–2024 (3.7% of GDP)	132%	89%
- as a share of public expenditure analyzed in this PER (16.2% of GDP)	30%	20%

Chile (left) and North Macedonia (right)

Policy Context	Policy Action	Equity Impact	Potential Savings (CLP bn.)	Potential Savings (as % of GDP)
Improve the procurement of drugs in the public sector. Significant efficiency gains and savings can be obtained through better structured and more centralized procurement of drugs for public sector health services.	Reduce the volume of direct procurement of drugs and medical equipment by hospitals and municipalities.		26.1	0.017
	Implement multi-year framework contracts for high volume, high price drugs.		34.4	0.022
	Improve the management and cost control procedures of central pharmacies in public hospitals and expand the joint procurement of drugs with other Mercosur countries.		8.2	0.005
	Subtotal		68.7	0.044
	Reduce avoidable hospitalizations.		34.4	0.022
Improve hospital efficiency. Major savings can be derived from improvements in the clinical efficiency and effectiveness, as well as in the management of hospital care services.	Reduce Caesarean section rates.		1.4	0.001
	Reduce hospital readmission rates.		12.4	0.008
	Reduce hospital length of stay for some key conditions and procedures.		28.8	0.018
	Accelerate implementation of day surgery and other measures to reduce administrative costs and achieve operational efficiencies		26.1	0.017
	Subtotal		103.1	0.066
Total, health			171.8	0.11
Reduce inefficiencies and improve targeting in school feeding program at primary and secondary levels (PAE)	Introduce an on-demand modality for beneficiaries from the 3. quintile	Neutral	18.6	0.012
	Improve the monitoring of beneficiaries	Slightly positive	50.0	0.032
	Allow richer students to pay for food rations	Neutral	18.6	0.012
	Subtotal, school feeding		87.2	0.056
	Change technology to tablets (I)	Positive-Neutral	14.1	0.009
Reduce inefficiencies and improve targeting in laptop program (ENLACES-ICT Education).	Offer hardware only to the poorest households (II)	Slightly negative	19.1	0.012
	Offer tablets only to the poorest households (I and II combined)	Slightly negative	25.4	0.016
	Subtotal, laptop program		44.5	0.028
Total, education			141.8	0.093
Grand Total (Health and Education)			313.6	0.20
- as share of 2016 structural deficit adjustment target (0.25% of GDP)				72%
- as a share of public expenditure analyzed in this PER (2.6% of GDP)				7%

Annex 2. The Fiscal Impact of Reform Options

Reform Action	Priority	Cumulative Fiscal Impact (% of GDP unless otherwise indicated)
Chapter 2: Public Financial Management		
1. Draft and carry out a strategy for clearing arrears that is orderly, transparent, and credible.	S	-0.5–3.5
2. Update the treasury information system to (a) record the date an invoice is received and its due date(s), (b) monitor and report monthly on all accounts payable, and (c) record commitments before contracts are signed.	M	Neutral
3. Strengthen MoF capacity for macrofiscal forecasting.	M	Neutral
4. Draft a comprehensive medium-term budget framework linked to the overarching fiscal strategy.	M	Neutral
5. Strengthen assessment of fiscal risks and their monitoring, e.g., by integrating their oversight within a single coordinating body.	M	Neutral
6. Improve the transparency of SOEs by publishing all SOE financial reports.	M	Neutral
7. Enhance the monitoring of local government financial performance.	M	Neutral
Chapter 3: Fiscal Decentralization		
1. Clarify functional assignments by better defining the responsibilities of relevant bodies at each tier of government.	S	Neutral
2. Consider measures to increase local own revenue capacity, such as identifying new tax sources (e.g., an annual local vehicle registration tax) and measures to improve collection of all taxes (property tax), charges, and communal fees.	S-M	+0.5–3
3. Improve the effectiveness of conditional block grants by, e.g., making more transparent the calculation of transfers and selection of municipalities to receive transfers.	S	Neutral
4. Move to a more optimal size for municipalities by reducing the number of municipalities with fewer than 10,000 residents.	M	Positive
5. Improve the measurement of the spending needs related to functional assignments.	M	?
6. Stabilize LG revenues by raising the amount of PIT collections they receive.	M	Neutral
7. Enhance the equalization capacity of general grants by reforming the current formula.	M	Neutral
8. Restructure the capital grants system.	M	Neutral
Chapter 4: Tax Policy		
1. Remove the tax exemption on the sales of securities; add a higher PIT tax bracket with a top marginal tax rate above 15 percent; broaden the CIT base; and increase the PIT standard deduction.	S	1
2. Expand the VAT base by removing preferential treatment for items not in line with international standards.	S	> 1.5
3. Accelerate the increase in excise taxes on cigarettes and energy to meet EU requirements.	S	Positive
4. Review PIT, CIT, and VA tax expenditures and which tax incentives for corporations should be maintained and do a cost-benefit analysis of special economic zones.	S-L	Neutral
5. Review cadaster valuations and increase revenue from the property tax.	M	Positive
6. Move the CIT rate closer to 15 percent.	L	1.2 (CIT of 13 percent)

Peru

Table 1-5 Overview of policy options and their impact on efficiency, equity, and fiscal savings

Sector	Measure	Efficiency	Equity	Potential Savings (% of GDP)
Public Procurement	Reform of the procurement system, with the following options in terms of intensity:	+	Neutral	
	- Full reform of the procurement system (aggressive approach) *	+	Neutral	
	- Full reform of the procurement system (conservative approach) *	+	Neutral	0.4%
	- Reform of 9 low-value and low-risk categories (aggressive approach) *	+	Neutral	
Citizen Security	- Reform of 9 low-value and low-risk categories (conservative approach) *	+	Neutral	
	Optimization of human and physical resources at the district level	+	+	0.3%
	Creation of the resource allocation formula with efficiency measures	+	+	TBD
Water and Sanitation	Funding of programs for the construction of police infrastructure	+	Neutral	
	Reduce the undervaluation of water and sanitation services for EPS	+		0.1%
	Improve the EPS' collection efficiency	+		0.0%
	Reduce the losses of unaccounted-for water in the areas in which the EPS operate	+		0.0%
	Improve planning and coordination of the total investment and the investment by government level	+	+	0.1%-0.2%
	Evaluate the specific viability of every project as part of a greater investment program, which may help reduce the transaction costs for project preparation and implementation	+	+	
	Improve the structure incentive in the budget process: (i) decoupling the budget process and investment to enable multiannual planning of the investment and budget, and (ii) changing the rules that create adverse incentives.	+		TBD
Transport	Multiannual planning of transport works	+	+	
	Implement the multiannual budget with MTC as the unit coordinating different levels of government	+	Neutral	TBD
Health	Continue to expand the population coverage of insurance	Neutral	+	--
	Redefine the package of benefits, keeping in mind the fiscal impact and the impact in terms of equity	+	+	
	Adapt international payment systems for suppliers, incentivizing the efficient production of services	+	Neutral	
	Improve efficiency in the purchase and distribution of medicines	+	Neutral	TBD
	Build the institutional and operational foundations for the interoperability of insurance	+	Neutral	
	Reconfigure specialized hospital care programs	+	Neutral	
	Unify complementary services at the regional level	+	Neutral	
	Lower the administrative expenses of the health system	+	Neutral	0.1%-0.2%
	Remove the implicit subsidies for EsSalud	+	Neutral	0.1%
Education	Provide adequate financing of the current expenses for the provision and supervision of educational services in different contexts, integrating it into results-based budgeting	+	+	
	Allocate the investment budget in a multiannual manner, according to the priorities and strategies of the National Plan for Educational Infrastructure (PNI), budgeting also the necessary operating expenses	+	Neutral	
	Prioritize the expenses in highly effective and pro-poor interventions and policies	+	+	
	Implement an electronic government that connects schools and optimizes information flows, reduces transaction costs and speeds up administrative procedures	+	+	TBD
	Establish differentiated strategies (by region, university) and accredit capacities at the level of schools and education units in order to enhance budget responsibilities	+	+	
	Establish the conditions for allocating the subnational budget by means of student-based formulas that comprise equity criteria, starting with accredited schools and education units.	+	+	
	Link the budget to performance-driven incentives.	+	Neutral	
Risks	Contingencies: explicitly allocate public and private risks in new PPP contracts, as well as the instruments for mitigation	+	Neutral	TBD
Budget	Budget: reduce variability of the PIA versus the PIM, allocating a greater proportion of the budget at the beginning of the year	+	Neutral	TBD
TOTAL				1.0%-1.2%

Uganda (left) and Lesotho (right)

Table A1.1. Policy coherence assessment for selected PEAS indicators

Indicator	Observed	Target	Source of target	Gap	Recommendation
Level	3.6%	10%	Malabo (2014) Declarations, 2010 CAADP Compact, ASSP	-6%	Build up capacity to allow sector institutions to receive greater proportions of the national budget.
Administrative efficiency					
MAAIF and sub-branches	26%	41%	ASSP	-15%	Reconsider the role of NAADS in the sector's institutional framework and rebalance budgets accordingly.
NAADS	41%	23%	ASSP	17%	
NAIRO	11%	16%	ASSP	-5%	
Local governments	8%	12%	ASSP	-4%	Build up capacity at the decentralized level to increase district resources and gear up local service delivery.
Technical efficiency					
Execution rates	90%	No target, high value desirable			Maintain high execution rates but seek to increase the magnitude of PEAS.
Share of donor spending in PEAS	33%	No target, low value desirable	World Bank (2011)		Develop a strategy to phase out donor spending in the long run. Enhance expenditure monitoring systems for donor spending.
Economic efficiency					
Share of development spending in PEAS	66%	No target, high value desirable	World Bank (2011)		Refine classification systems of recurrent and development spending. Contemplate the reallocation of PEAS shares to long-term investments (capital spending).
Functional efficiency					
Share of PEAS allocated to specific commodities (including priority ones)	21%	53%	ASSP	-32%	Refine budget monitoring systems to better track and understand how PEAS interventions support specific commodities, including the priority commodities of the CBA.
Allocation of PEAS across commodity groups					Replenish livestock and fisheries, as well as food crops (even if they are poorly represented in the ASSP budget).
Livestock and dairy	20%	36%	ASSP	-16%	
Cash crops	41%	28%	ASSP	13%	
Crops, general	12%	3%	ASSP	9%	
Fisheries	6%	32%	ASSP	-26%	
Forestry	4%	NA	ASSP	NA	
Other	16%	NA	ASSP	NA	

ANNEX 2: Policy Coherence Analysis Summary Table

The table below summarizes all dimensions of PEA policy coherence.

Indicator	Observed	Target	Source of target	Gap	Recommendation
Level	3.3%	10%	Malabo (2014) Declarations, 2010 CAADP Compact	67%	
Technical efficiency					
Execution rates	79%	No target, high value desirable			
Economic efficiency					
Share of development spending in PEA	44.6%	No target, high value desirable			
Share of operating costs in PEA	43.9%	No target, high value desirable			
Functional efficiency					
Allocation of PEA across sectors					
Multi-sectoral	70%	58%	NSDP	-12	
Crop	25%	31%	NSDP	+6	
Livestock	4%	10%	NSDP	+6	
Forestry	1%	6%	NSDP	0	
Allocation of PEA across subfunctions					
Inputs/subsidies	0%	20%	NSDP	20	
Other infrastructure	6%	25%	NSDP	19	
Extension and advisory services	10%	28%	NSDP	17	
Capital subsidies	4%	5%	NSDP	0	
Services subsidies	1%	0%	NSDP	-1	
Training	9%	5%	NSDP	-4	
Inspection/quality control	4%	0%	NSDP	-4	
Research	7%	2%	NSDP	-5	
Irrigation	6%	0%	NSDP	-6	
Processing and marketing	12%	0%	NSDP	-12	
Administrative costs	39%	14%	NSDP	-25	
Share of private sector-related functions in PEA	42%	No target, low value desirable	World Bank (2011)		

Mexico (left) and Guinea-Bissau (right)

Table 1.3: Policy Actions: Efficiency, Equity and Fiscal Savings

Policy Context	Policy Action	Efficiency	Equity	Potential Savings (in % of GDP)
Preserving domestic fuel-price stability has functioned as a partial public revenue hedge against oil-price volatility, though at a high cost in terms of public revenue. The energy sector reform envisages a liberalization of the fuel distribution and retail market by 2018, including a liberalization of the current fixed final sales price. Fuel price liberalization would be consistent with a switch to a fixed per-unit excise tax (from the current variable excise tax). Current fuel price levels (domestic and international reference) imply a reasonable level of the fuel excise tax. Fixing the excise at its current level is viable as it does not lead to a direct final price shock.	Accelerate the fuel-price liberalization process by switching the excise tax on fuel from a variable rate to a fixed amount and allowing the final sales price to fluctuate in line with international prices, thereby locking in revenue at its current level of approximately 1 percent of GDP.	+	+	1% (by retaining what would otherwise be foregone revenue)
Subnational tax collection is low by international standards due to the centralization of major tax bases and weak subnational collection efficiency. States collect an average of just 62 percent of the potential revenue from the state payroll tax (<i>nomina</i>). Numerous exemptions and subsidies greatly reduce the revenue generated by the state tax on motor vehicle ownership and use (<i>tenencia</i>).	Enhance collection efficiency for <i>nomina</i> taxes, which could increase state revenue by approximately 20 percent. Reduce <i>tenencia</i> subsidies and deter tax competition among states, returning collection rates to the level that prevailed when the tax was administered by the federal government.	Neutral	Neutral	0.15-0.2%
Property taxes in Mexico represent 0.18 percent of GDP, far below the Latin American average of 0.6 percent. This is due primarily to issues with collection efficiency. Most municipalities have limited administrative and technological resources, making it difficult to maintain updated registries and accurately assess property values.	Incentivize tax administration agreements between state governments and small municipalities.	+	+	At least 0.05%
				0.3%

Policy Recommendations – Public Sector Wage Bill

	Technical feasibility	Timeframe policy impact
Public employment policy		
- Use the recent civil service census to carry out a comprehensive functional review to identify overlapping functions and redundancies.	Medium	Medium term
- Carry out a census of irregular public sector employees to inform future policy options aimed at regularizing their situation and/or reducing their numbers.	Low	Medium term
Wage setting and pay policy		
- Use hiring freezes selectively to reduce overstaffing in certain sectors.	High	Short term
- Freeze or limit the number of promotions to slow wage increases	Medium	Short term
- Carry out a comprehensive review of non-wage compensation to ensure that pay is competitive, equitable and transparent. Tax collection bonuses need to be a central part of this.	Medium	Medium term
HR payroll control		
- Complete organigram, including terms of reference for each position.	Medium	Short term
- Include the remaining civil servants not yet included the census, a total of 1,023 military staff.	High	Short term
- Initiate the process of removing civil servants from payroll when they reach retirement age.	High	Short term
- Implement a fully functional Human Resources Management System and link it to SIGFIP.	Low	Medium term
- Strengthen transaction controls, payroll reconciliations, and auditing.	Medium	Medium term
Accounting and reporting		
- Update the existing budget classification system to ensure that all employee compensation expenditures are appropriately accounted for.	High	Short term

Appendix D: Available Resources for TTLs

Tool / Resource	Description	Contact
PER Programmatic Support Platform	One-stop shop for PER teams to access data, tools and guidance notes to guide preparation and implementation.	Massimo Mastruzzi & Christian Eigen-Zucchi
PER Repository	List of PERs from 2015 onwards along with specific tags.	Massimo Mastruzzi
Expenditure analysis tools	Series of streamlined analytical modules.	Massimo Mastruzzi & Christian Eigen-Zucchi
Revenue Academy	A repository of data, studies and analytical techniques to learn fundamentals of tax policy, modeling and analysis.	Sebastian James
Cross country interface	This data module provides access to annual comparative fiscal statistics for over 80 countries across some 300 fiscal dimensions using existing BOOST platforms.	Massimo Mastruzzi
BOOST data	Access BOOST data presenting annual micro expenditure data for over 90 countries.	Massimo Mastruzzi
Climate fiscal policy hub	List of tools and resource papers on climate fiscal policy including CPAT.	Dirk Heine
<u>MTI benchmarking</u>	This new version includes about 30,000 variables from about 50 databases from various sources.	Julio Velasco
Fiscal data for PERs		Xun Yan
<u>MTI online tools</u>	This site offers access to a series of tools on jobs, growth debt, fiscal analysis and investment climate.	Mark Volk
<u>PER sector guidelines</u>	Link to most recent guidelines issues around analytical techniques across sectors.	Massimo Mastruzzi & Christian Eigen-Zucchi
GOV Diagnostics	<u>PEFA</u> , <u>TADAT</u> , <u>Diamond</u> , <u>PIMA</u>	Various
Business of the State	The database covers more than 80 countries and can help identify SOE footprint across sectors/markets, with performance indicators in terms of output, profits/revenues and employment.	Various
<u>IMF Fiscal Transparency Evaluation</u>	The IMF's <u>Fiscal Transparency Code</u> is the international standard for disclosure of information about public finances.	IMF

Links to all resource above are available at [PER support platform resource page](#).

Appendix E: Core Fiscal Tables for PERs

Table 1: Trends in public expenditures (% of GDP), by econ (a) and function (b)

BY MAIN ECON	2017	2018	2019	2020	2021	BY MAIN FUNC	2017	2018	2019	2020	2021
Total Expenditures	-	-	-	-	-	Total Expenditures	-	-	-	-	-
Total Expenditures (foreign funded)	-	-	-	-	-	701 General public services	-	-	-	-	-
Wage bill	-	-	-	-	-	702 Defense	-	-	-	-	-
Capital Expenditures	-	-	-	-	-	703 Public safety	-	-	-	-	-
Use of goods and Services	-	-	-	-	-	704 Economic relations	-	-	-	-	-
Interest on debt	-	-	-	-	-	705 Environment	-	-	-	-	-
Social benefits - (pensions)	-	-	-	-	-	706 Housing	-	-	-	-	-
Social benefits (other than pensions)	-	-	-	-	-	707 health	-	-	-	-	-
Inter-govt transfers	-	-	-	-	-	708 Recreation, culture and religion	-	-	-	-	-
Other grants and transfers	-	-	-	-	-	709 Education	-	-	-	-	-
Debt repayment	-	-	-	-	-	710 Social Protection	-	-	-	-	-

Table 1c: Trends in public expenditures, Consolidated public spending by institutional coverage

	Budget	Targeted funds	EBFs	Off-budget accounts	Foreign financed investment	Policy lending	Consolidated spending without quasi-fiscal losses	Quasi-fiscal losses	Consolidated spending with quasi-fiscal losses
Expenditures	19.5	6.5	1.1	3.4	0.3	4.3	35.2	6.0	41.2
<i>of which:</i>									
wages	9.5			1.0			10.5		10.5
pensions	0.9	4.9	0.0	0.1			5.9		5.9
capital spending	2.4	1.3	0.7	0.6	0.3		5.2		5.2
other	6.8	0.3	0.4	1.7		4.3	13.5	6.0	19.5

Source: Uzbekistan 2020 PER

Table 2a: Trends in revenues

	2017	2018	2019	2020	2021
Revenues: Total	-	-	-	-	-
Income tax	-	-	-	-	-
VAT	-	-	-	-	-
Customs	-	-	-	-	-
Excise	-	-	-	-	-
Royalties	-	-	-	-	-
Permits/Fees	-	-	-	-	-
Subnational own revenues	-	-	-	-	-
Sales of goods and services	-	-	-	-	-
Property taxes	-	-	-	-	-
Other	-	-	-	-	-

Table 2b: Additional Metrics

	Year
Tax Expenditures (% GDP)	-
VAT Compliance gap (% revenues)	-
VAT C-Efficiency	-
CIT productivity	-
PIT productivity	-
Tax effort	-
Tax potential	-
Tax capacity	-
Overall GAP	-
Informal economy	-

Table 3: Execution rates over time by econ (a) and function (b)

BY MAIN ECON						BY MAIN FUNC					
	2017	2018	2019	2020	2021		2017	2018	2019	2020	2021
Total Expenditures	-	-	-	-	-	Total Expenditures	-	-	-	-	-
Total Expenditures (foreign funded)	-	-	-	-	-	701 General public services	-	-	-	-	-
Wage bill	-	-	-	-	-	702 Defense	-	-	-	-	-
Capital Expenditures	-	-	-	-	-	703 Public safety	-	-	-	-	-
Use of goods and Services	-	-	-	-	-	704 Economic relations	-	-	-	-	-
Interest on debt	-	-	-	-	-	705 Environment	-	-	-	-	-
Social benefits - (pensions)	-	-	-	-	-	706 Housing	-	-	-	-	-
Social benefits (other than pensions)	-	-	-	-	-	707 health	-	-	-	-	-
Inter-govt transfers	-	-	-	-	-	708 Recreation, culture and religion	-	-	-	-	-
Other grants and transfers	-	-	-	-	-	709 Education	-	-	-	-	-
Debt repayment	-	-	-	-	-	710 Social Protection	-	-	-	-	-

Table 4a: Comparative tables across behavioral/aspirational peers by economic (% of GDP)

% of GDP	Total Expenditures			Donor funded			Wage bill			Capital Expenditures			Interest on debt			Social benefits		
	2019	2020	2021	2019	2020	2021	2019	2020	2021	2019	2020	2021	2019	2020	2021	2019	2020	2021
Country	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Peer 1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Peer 2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
...	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Peer X	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Table 4b: Comparative tables across behavioral/aspirational peers by function (% of GDP)

% of GDP	701 General public services			702 Defense			703 Public safety			704 Economic relations			705 Environment		
	2019	2020	2021	2019	2020	2021	2019	2020	2021	2019	2020	2021	2019	2020	2021
Country	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Peer 1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Peer 2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
...	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

% of GDP	706 Housing			707 health			708 Recreation, culture and religion			709 Education			710 Social Protection		
	2019	2020	2021	2019	2020	2021	2019	2020	2021	2019	2020	2021	2019	2020	2021
Country	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Peer 1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Peer 2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
...	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Peer X	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Table 5: Extra budgetary, quasi fiscal activities and off budget operations

General government, % of GDP	2015	2016	2017	2018	2019	2020	2021
Fiscal indicators							
Revenues	..	..	..	..	..	..	..
Expenditures	..	..	..	..	..	..	..
Fiscal balance	..	..	..	..	..	..	..
Cyclically adjusted balance	..	..	..	..	..	..	..
Gross debt	..	..	..	..	..	..	..
Foreign aid	..	..	..	..	..	..	..
Non fiscal indicators							
GDP (current prices)	..	..	..	..	..	..	..
GDP (annual average growth)	..	..	..	..	..	..	..
Exchange rate (CLP/USD, annual average)	..	..	..	..	..	..	..
Consumer price index (% , annual average)	..	..	..	..	..	..	..
FDI	..	..	..	..	..	..	..
Off budget items							
Tax expenditures	..	..	..	..	..	..	..
Arrears	..	..	..	..	..	..	..
SOE revenues	..	..	..	..	..	..	..
SOE expenditures	..	..	..	..	..	..	..
SOE fiscal balance	..	..	..	..	..	..	..
SOE gross debt	..	..	..	..	..	..	..
Activated guarantees	..	..	..	..	..	..	..